Chapter 2

# Dental Caries: Etiology, Clinical Characteristics, Risk Assessment and Management

*"There are no such things as incurables…*

*There are only things for which man has not yet found a cure…"*

—Bernard Baruch

This chapter presents basic definitions, terminologies and information on dental caries, clinical characteristics of the caries lesion, caries risk assessment and caries management in the context of clinical operative dentistry.

## Definition

*Dental caries is defined as a preventable, chronic and biofilm-mediated disease modulated by diet. This multifactorial, oral disease is caused primarily by an imbalance of the oral flora (biofilm) due to the presence of fermentable dietary carbohydrates on the tooth surface over time.*

## Demineralization–Remineralization Balance

Traditionally, the tooth–biofilm–carbohydrate interaction has been illustrated by the classical *Keyes–Jordan diagram*.[1] However, dental caries onset and activity are, in fact, much more complex than this three-way interaction, as not all persons with *teeth*, *biofilm* and *consuming carbohydrates* will have caries over time. Several modifying risk and protective factors influence the dental caries process, as will be discussed later in this chapter (Fig. 2.1).

At the tooth surface and sub-surface level, dental caries results from a dynamic process of attack (demineralization) (Figs. 2.2 and 2.3) and restitution (remineralization) of the tooth matter. This cycle is summarized in Box 2.1.

The balance between demineralization and remineralization has been illustrated in Fig. 2.4 in terms of

- Pathologic factors (i.e. those favouring demineralization) and
- Protective factors (i.e. those favouring remineralization).[3]

Individuals in whom the balance tilts predominantly towards protective factors (remineralization) are much less likely to develop dental caries than those in whom the balance is tilted towards pathologic factors (demineralization). It is essential to understand that caries lesions, or cavitations in teeth, are signs of an underlying condition, an imbalance between protective and pathologic factors favouring the latter. In clinical practice, it is very easy to lose sight of this fact and focus entirely on the restorative treatment of caries lesions, failing to treat the underlying cause of the disease. Although symptomatic treatment is important, failure to identify and treat the underlying causative factors allows the disease to continue and increases the chance of treatment failure. Caries management efforts must be directed not only at the tooth level (traditional or surgical treatment) but also at the total-patient level (caries management by risk assessment).

**Clinical Notes**

- Understanding the balance between demineralization and remineralization is the key to caries management.
- Restorative treatment does not cure the caries process. Instead, identifying and managing the risk factors for caries must be the primary focus, in addition to the restorative repair of damage caused by caries.

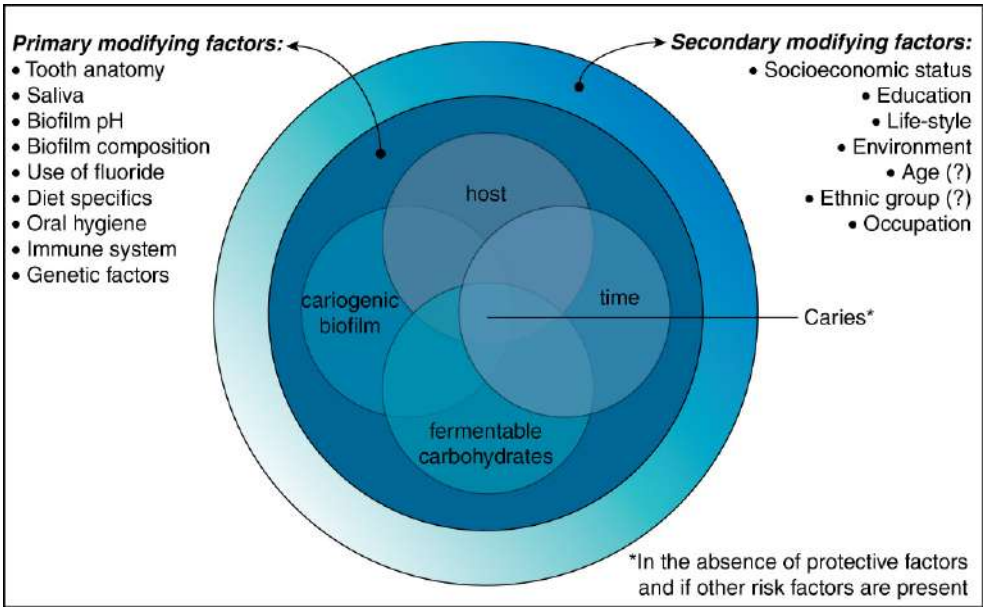



**Figure 2.1** Modified Keyes–Jordan diagram. As a simplified description, dental caries is a result of the interaction of cariogenic oral flora (biofilm) with fermentable dietary carbohydrates on the tooth surface (host) over time. However, dental caries onset and activity are, in fact, much more complex, as not all persons with teeth, biofilm, and who are consuming carbohydrates will have caries over time. Several modifying risk factors and protective factors influence the dental caries process.
(Modified from Keyes PH, Jordan HV: Factors influencing initiation, transmission and inhibition of dental caries. In Harris RJ, editor: *Mechanisms of hard tissue destruction*, New York, 1963, Academic Press.)

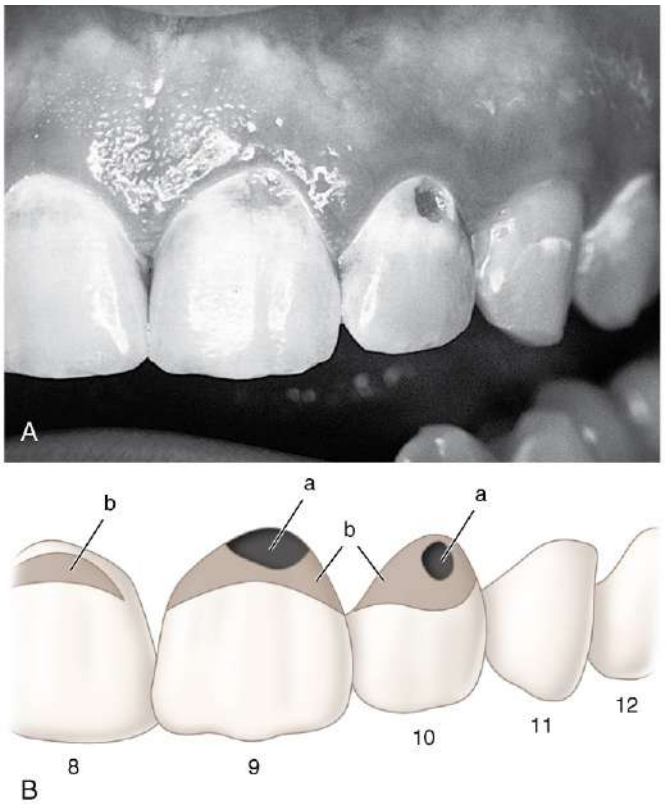



**Figure 2.2** (A) Young adult with multiple active initial and cavitated caries lesions involving teeth nos. 8–10 and (B) cavitated areas (*a*) are surrounded by areas of extensive demineralization that are chalky and opaque (*b*). Some areas of initial (non-cavitated) caries have superficial stain.

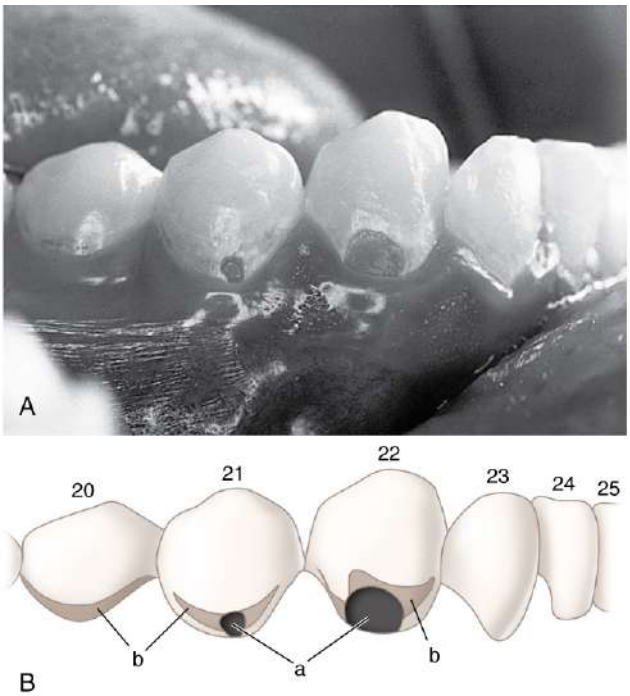



**Figure 2.3** Extensive active caries in a young adult (same patient as in Fig. 2.2). (A) Mirror view of teeth nos. 20–22 and (B) cavitated lesions (*a*) are surrounded by extensive areas of chalky, opaque demineralized areas (*b*). The presence of smooth-surface lesions such as these is associated with rampant caries. Occlusal and interproximal smooth-surface caries usually occur in advance of facial smooth-surface lesions. The presence of these types of lesions should alert the dentist to the possibility of a high caries risk patient and possibly extensive caries activity elsewhere in the mouth. The interproximal gingiva is swollen red and would bleed easily on probing. These gingival changes are the consequence of longstanding irritation from the biofilm adherent to the teeth.

**Box 2.1 Demineralization and remineralization cycle**

I. **Acid production:** Cariogenic bacteria in the biofilm metabolize refined carbohydrates for energy and produce *organic acid* by-products

II. **Critical biofilm pH:** These organic acids, if present in the biofilm ecosystem for *extended periods*, can lower the pH in the biofilm to below a *critical level* (5.5 for enamel, 6.2 for dentin). This low pH has effects both on the biofilm composition and at the tooth surface level (Fig. 2.12)

III. **Demineralization:** The *low pH* drives calcium and phosphate from the tooth to the biofilm in an attempt to reach equilibrium, hence resulting in a net loss of minerals by the tooth, or *demineralization*

IV. **Remineralization:** When the pH in the biofilm returns to *neutral* and the concentration of soluble calcium and phosphate is *supersaturated* relative to that in the tooth, mineral can then be added back to partially demineralized enamel, in a process called *remineralization*

V. **Demineralization and remineralization cycle:** This process takes place several times a day over the life of the tooth and are modulated by many factors, including:
   i. Number and type of microbial flora in the biofilm
   ii. Diet
   iii. Oral hygiene
   iv. Genetics
   v. Dental anatomy
   vi. Use of fluorides and other chemotherapeutic agents
   vii. Salivary composition, flow and buffering capacity
   viii. Inherent resistance of the tooth structure and composition that will differ from person to person, tooth to tooth and site to site

VI. **Progression of disease:** Repeated demineralization events may result from a predominantly pathologic environment causing the localized dissolution and destruction of the calcified dental tissues, evidenced as a caries lesion

Severe demineralization of enamel results in the formation of a cavitation in the enamel surface.

Subsequent demineralization of the inorganic phase and denaturation and degradation of the organic phase results in dentin cavitation.[2]

## Etiology of Dental Caries

Dental caries is a disease that is dependent on the complex interrelationships between the following four critical parameters:

I. Biofilm
II. Tooth habitat
III. Diet
IV. Saliva

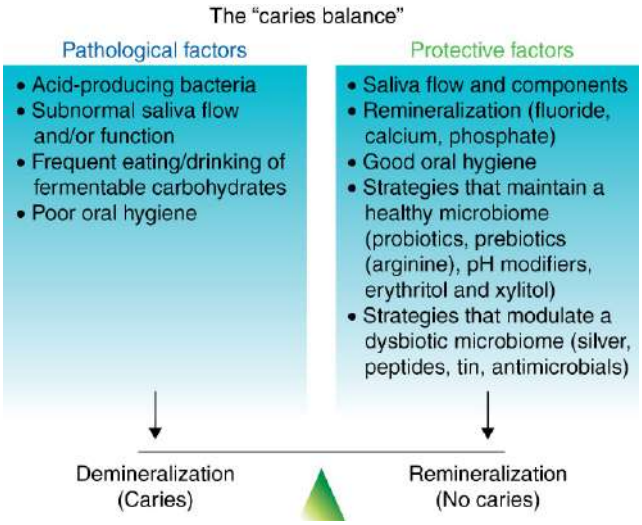



**Figure 2.4** The caries balance. The balance between demineralization and remineralization is illustrated in terms of pathologic factors (i.e. those favouring demineralization) and protective factors (i.e. those favouring remineralization).
(Modified from Featherstone JDB: Prevention and reversal of dental caries: role of low level fluoride, *Comm Dent Oral Epidemiol* 27:31–40, 1999.)

## I. Biofilm

*Dental plaque* is a term historically used to describe the soft, tenacious film accumulating on the surface of teeth. Dental plaque has been more recently referred to as a *plaque biofilm*, or simply biofilm, which is a more complete and accurate description of its composition (bio) and structure (film).[4,5]

The following are the salient attributes of dental biofilms:

- *Biofilm* is composed mostly of bacteria, their by-products, extracellular matrix and water.
- The accumulation of biofilm on teeth is a *highly organized* and *ordered sequence* of events (Figs. 2.5–2.7).
- A 'hedgehog' formation has been recently characterized[6] because of the spine of radially oriented filaments. The filaments are a mass of *Corynebacterium* filaments with *Streptococcus* at the periphery. Actinomyces are usually found at the base of the biofilm suggesting that *Corynebacterium* attaches to a pre-existing biofilm containing *Actinomyces*. In any case, it is notable that each taxon is localized in a precise and well-defined spatial zone indicating that the microbes in the oral biofilm have a precise and well-tuned interaction[6] (Fig. 2.6).
- Teeth normally have a biofilm community dominated by *Streptococcus sanguis* and *Streptococcus mitis* with significant differences in their proportion according to the various habitats within the oral cavity (Fig. 2.8 and Table 2.1).

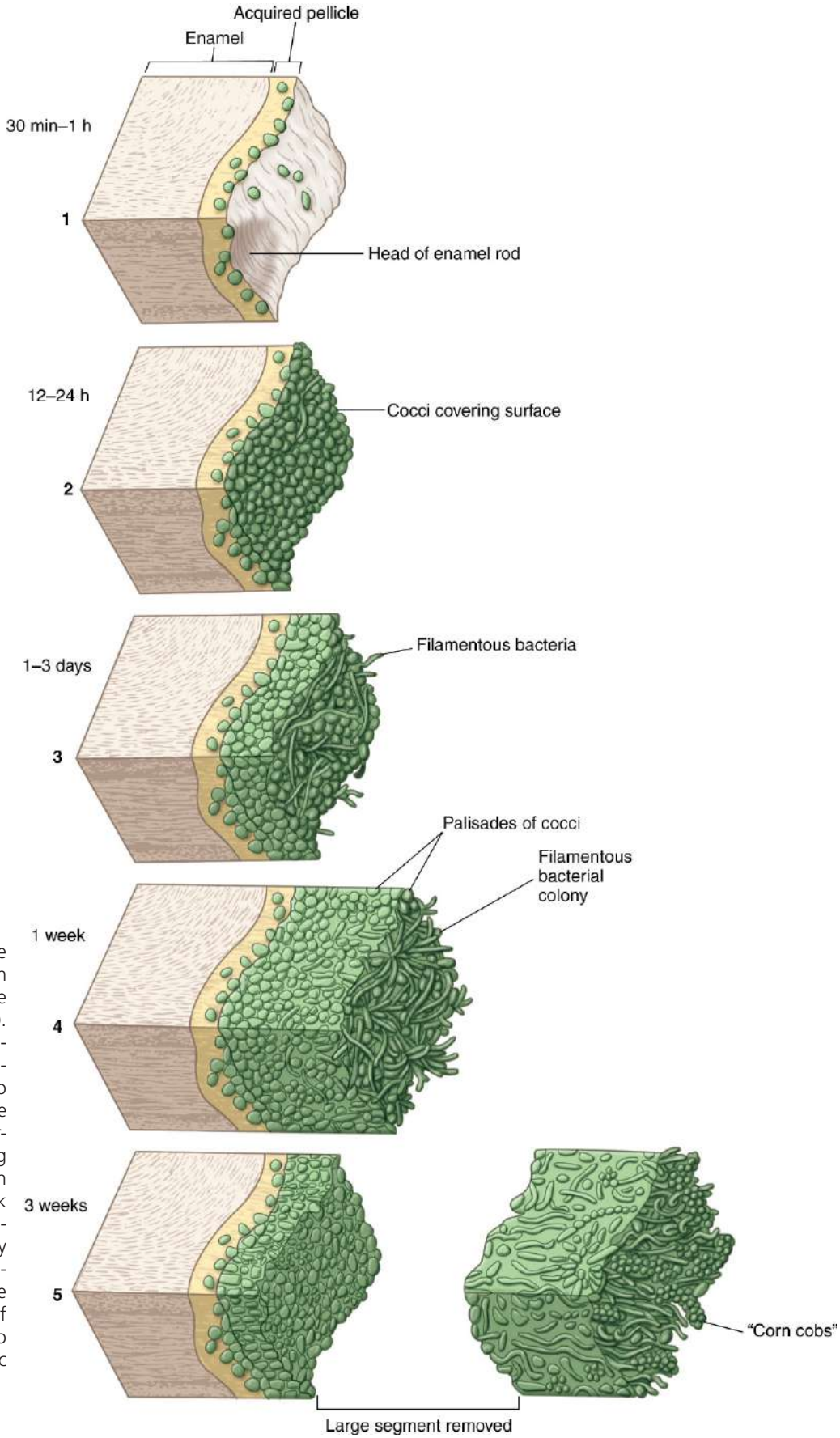



**Figure 2.5** Drawings 1–5 illustrate the various stages in colonization during plaque formation on the shaded enamel block shown in (B). The accumulated mass of bacteria on the tooth surface may become so thick that it is visible to the unaided eye. Such plaques are gelatinous and tenaciously adherent; they readily take up disclosing dyes, aiding in their visualization for oral hygiene instruction. Thick plaque biofilms (4 and 5) are capable of great metabolic activity when sufficient nutrients are available. The gelatinous nature of the plaque limits outward diffusion of metabolic products and serves to prolong the retention of organic acid metabolic by-products.

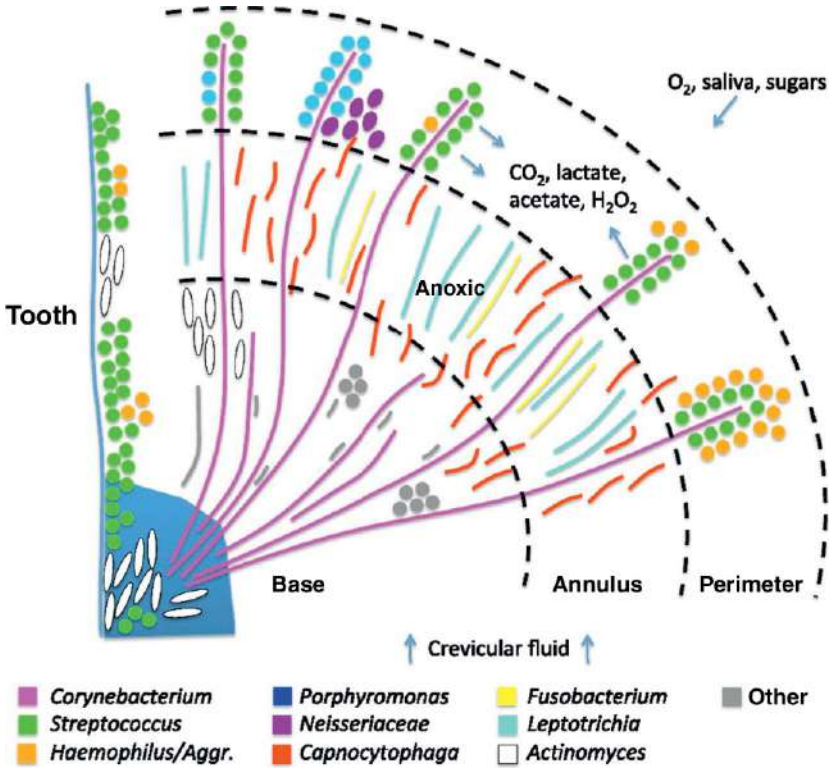



**Figure 2.6** This illustrates how different taxons inhabit specific niches on the biofilm creating microenvironments. There is a fine-tuned synergy among the cells in the oral microbial communities. The environment and the biochemical gradients drive the selection process. This can be exemplified by the role of *Streptococcus*. Where *Streptococcus* predominate, they create an environment rich in $CO_2$, lactate and acetate, containing peroxide and having low oxygen. This environment is advantageous for the growth of bacteria such as *Fusobacterium* and *Leptotrichia*.
(From Welch JL, Rossetti BJ, Riekem CW, et al.: Biogeography of a human oral microbiome at the micron scale, *Proc Natl Acad Sci USA* 9;113(6):E791–E800, 2016. doi:10.1073/pnas.1522149113.)

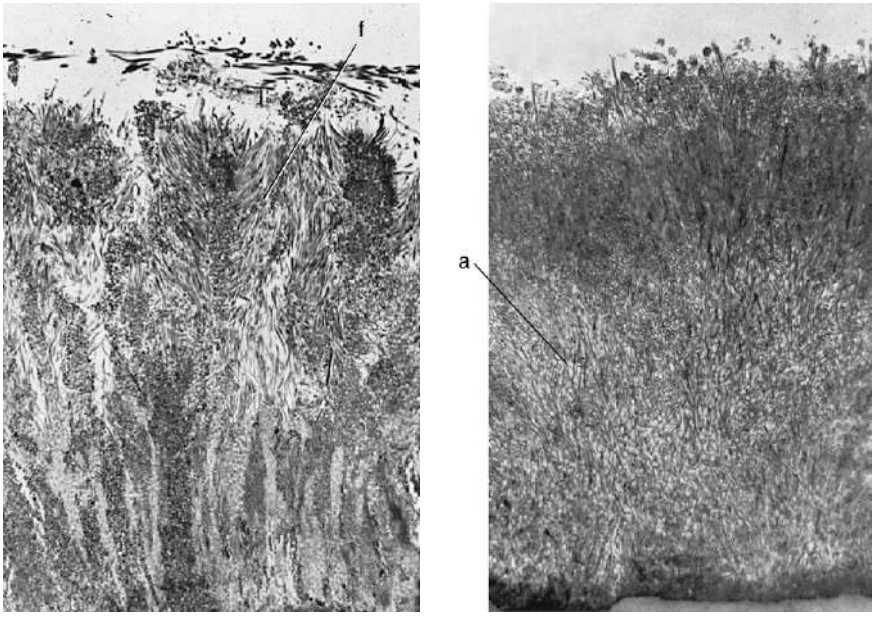



**Figure 2.7** **(A)** Plaque biofilm formation at 1 week. Filamentous bacteria (*f*) appear to be invading cocci microcolonies. Plaque near gingival sulcus has fewer coccal forms and more filamentous bacteria (860×). **(B)** At 3 weeks old, plaque biofilm is almost entirely composed of filamentous bacteria. Heavy plaque formers have spiral bacteria (*a*) associated with sub-gingival plaque (660×).
(From Listgarten MA, Mayo HE, Tremblay R: Development of dental plaque on epoxy resin crowns in man. A light and electron microscopic study, *J Periodontol* 46(1):10–26, 1975.)

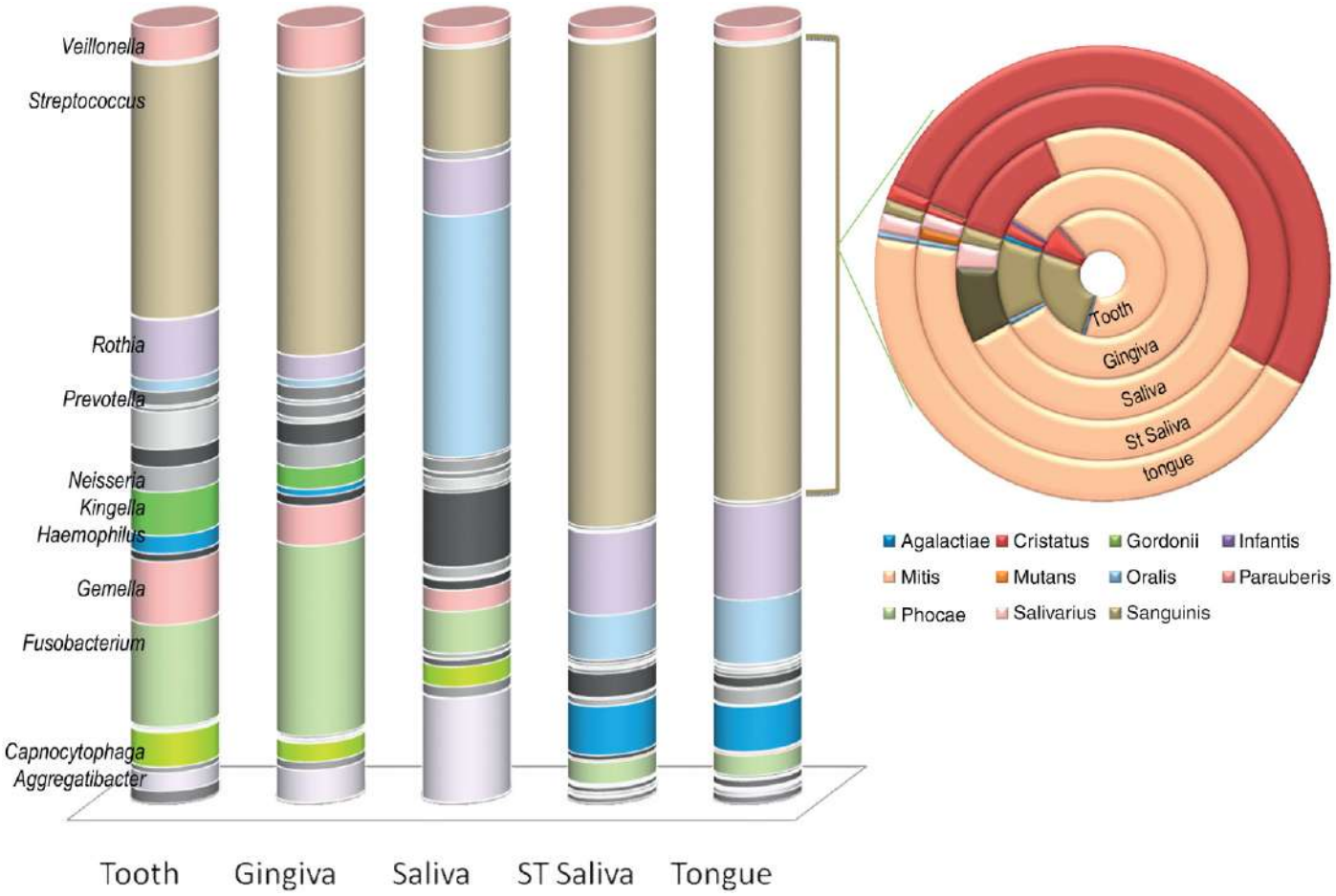



**Figure 2.8** Approximate proportional distribution of predominant cultivable flora of five oral habitats.
(From Simón-Soro Á, Tomás I, Cabrera-Rubio R, et al.: Microbial geography of the oral cavity, *J Dent Res* 92:616, 2013. doi:10.1177/0022034513488119.)

- Recent evidence indicates that there are no specific pathogens that correlate with dental caries, but rather microbial communities.[7]
- Mature plaque biofilm communities have tremendous metabolic potential and are capable of *rapid anaerobic metabolism* of any available carbohydrates.

Professional tooth cleanings are intended to control the biofilm (plaque) and prevent caries (and periodontal) disease. However, after professional removal of all organic material and bacteria from the tooth surface, a new coating of organic material begins to accumulate immediately. Within 2 h, a cell-free, organic film, the *acquired enamel pellicle* (AEP) (Figs. 2.5 and 2.6) can cover the previously denuded area completely. The pellicle is formed primarily from the selective precipitation of various components of salivary glycoproteins, enzymes and immunoglobulins. The functions of the *pellicle* are believed to be as follows:

- Protect the enamel
- Reduce friction between teeth
- Possibly to provide a matrix for remineralization[8]

## II. Tooth Habitats for Cariogenic Biofilm

The tooth surface which is covered with the AEP is the ideal surface for the attachment of many oral streptococci. If left undisturbed, biofilm rapidly builds up to sufficient depth to produce an anaerobic environment adjacent to the tooth surface. Tooth habitats favourable for harbouring pathogenic biofilm include:

I. *Pits and fissures* (Fig. 2.9)

Pits and fissures are particularly susceptible surfaces for caries lesion initiation. The pits and fissures provide excellent mechanical shelter for organisms and harbour a community dominated by *S. sanguis* and other streptococci.[9] The relative proportion of mutans streptococci

**Table 2.1** Oral habitats[a]

| Habitat | Predominant species | Environmental conditions within plaque |
|---|---|---|
| Mucosa | *S. mitis* | Aerobic |
| | *S. sanguis* | pH approximately 7 |
| | *S. salivarius* | Oxidation–reduction potential positive |
| Tongue | *S. salivarius* | Aerobic |
| | *S. mutans* | pH approximately 7 |
| | *S. sanguis* | Oxidation–reduction potential positive |
| Teeth (non-carious) | *S. sanguis* | Aerobic |
| | | pH 5.5 |
| | | Oxidation–reduction negative |
| Gingival crevice | *Fusobacterium* | Anaerobic |
| | *Spirochaeta* | pH variable |
| | *Actinomyces* | Oxidation–reduction very negative |
| | *Veillonella* | |
| Enamel caries | *S. mutans* | Anaerobic |
| | | pH <5.5 |
| | | Oxidation–reduction negative |
| Dentin caries | *S. mutans* | Anaerobic |
| | *Lactobacillus* | pH <5.5 |
| | | Oxidation–reduction negative |
| Root caries | *Actinomyces* | Anaerobic |
| | | pH <5.5 |
| | | Oxidation–reduction negative |

[a]The microenvironmental conditions in the habitats associated with host health are generally aerobic, near neutrality in pH, and positive in oxidation-reduction potential. Significantmicroenvironmental changes are associated with caries and periodontal disease. The changes are the result of the biofilm community metabolism.

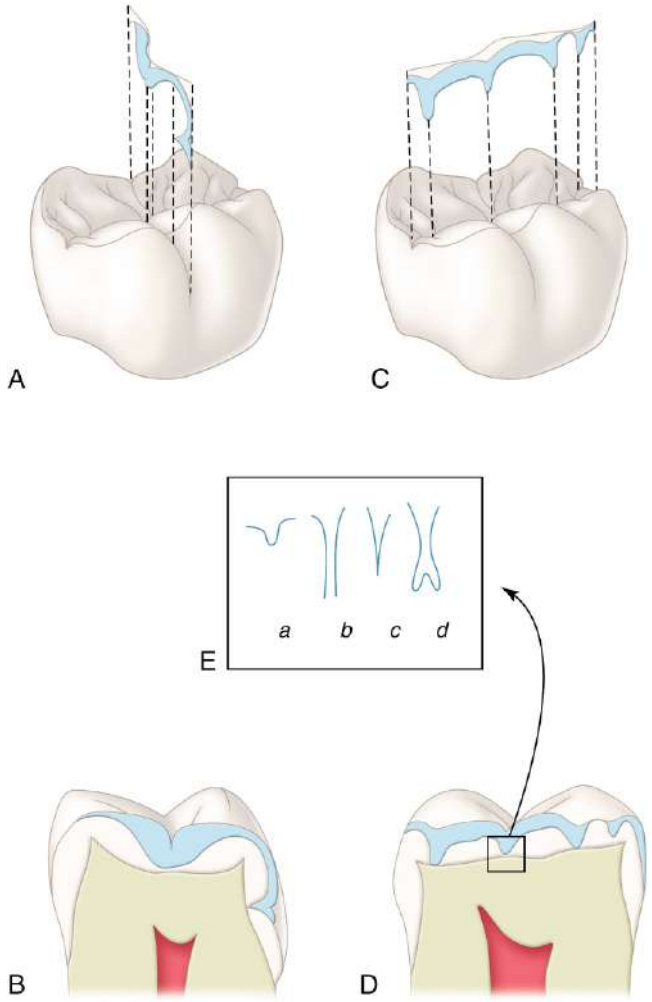



**Figure 2.9** Developmental pits, grooves and fissures on the crowns of the teeth can have complex and varied anatomy. (**A** and **B**) The facial developmental groove of the lower first molar often terminates in a pit. The depth of the groove and the pit varies. (**C** and **D**) The central groove extends from the mesial pit to the distal pit. Sometimes grooves extend over the marginal ridges. (E) The termination of pits and fissures may vary from a shallow groove (*a*) to complete penetration of the enamel (*b*); the end of the fissure may end blindly (*c*); or open into an irregular chamber (*d*). The physical nature of the pit and fissures allows for biofilm to remain stagnated. Additionally, the decreased thickness of enamel (and sometimes absence of enamel) at the bottom of the pit and fissure make these surfaces especially susceptible to dental caries.

(MS) most probably determines the cariogenic potential of the pit-and-fissure community.

In susceptible patients, sealing the pits and fissures just after tooth eruption may be the most important strategy in their resistance to dental caries on those surfaces.

II. *Smooth enamel surfaces* (Fig. 2.10)

The proximal enamel surfaces immediately gingival to the contact area are the second most susceptible areas to dental caries lesions. These areas are protected physically and are relatively free from the effects of mastication, tongue movement and salivary flow. The types and numbers of organisms composing the proximal surface biofilm community vary. Important ecologic determinants for the biofilm community on the proximal surfaces are the topography of the tooth surface, the size and shape of the gingival papillae and the oral hygiene and diet of the patient. A rough surface (caused by caries lesions, a poor-quality restoration or a structural defect) restricts adequate biofilm removal. This situation favours the occurrence and progression of caries lesions and/or periodontal disease at the site.

III. *Root surfaces*

The proximal root surface, particularly near the cementoenamel junction (CEJ), often is unaffected by the action of hygiene procedures such as flossing

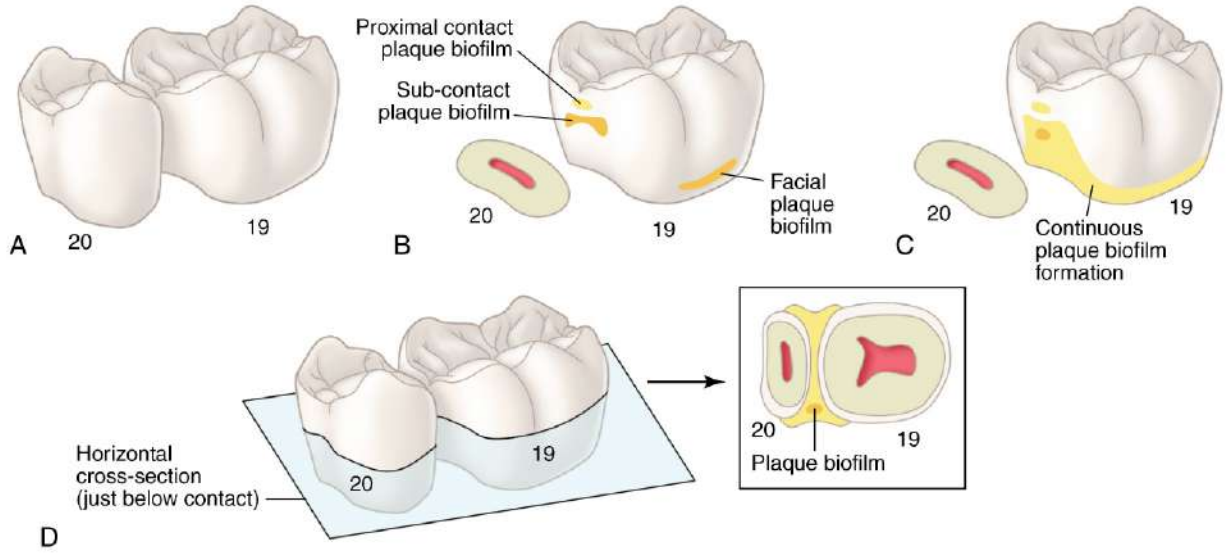



**Figure 2.10** Biofilm formation on posterior teeth and associated caries lesions. **(A)** Teeth nos. 19 and 20 in contacting relationship. **(B)** The crown of tooth no. 20 has been removed at the cervix. The proximal contact and subcontact biofilm can be seen on the mesial surface of tooth no. 19. The biofilm on the facial surfaces is illustrated. **(C)** During periods of unrestricted growth, the biofilms on the mesial and facial surfaces become part of a continuous ring of biofilm around teeth. **(D)** A horizontal cross-section through teeth nos. 19 and 20 with heavy biofilm. Inset shows the interproximal space below the contact area filled with gelatinous biofilm. This mass of interproximal biofilm concentrates the effects of biofilm metabolism on the adjacent tooth smooth surfaces. All interproximal surfaces are subject to biofilm accumulation and acid demineralization. In patients exposed to fluoridated water, most interproximal lesions become arrested at a stage before cavitation.

because it may have concave anatomic surface contours (fluting) and occasional roughness at the termination of the enamel. These conditions, when coupled with exposure to the oral environment (as a result of gingival recession), favour the formation of mature, cariogenic biofilm and proximal root-surface caries lesions.

IV. *Sub-gingival areas* (Fig. 2.11).
The facial or lingual root surfaces (particularly near the CEJ), when exposed to the oral environment (because of gingival recession), are often both neglected in hygiene procedures and usually not rubbed by the bolus of food. Consequently, these root surfaces also frequently harbour cariogenic biofilm.

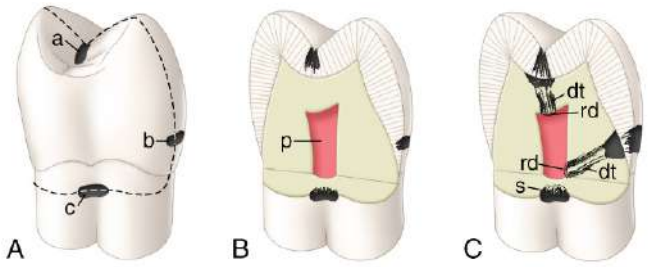



**Figure 2.11** **(A)** Caries lesions may originate at many distinct sites: pits and fissures (*a*), smooth surface of crown (*b*) and root surface (*c*). Proximal surface lesion of crown is not illustrated here because it is a special case of smooth-surface lesion. Histopathology and progress of facial (or lingual) and proximal lesions are identical. *Dotted line* indicates cut used to reveal cross-sections illustrated in **(B)** and **(C)**. Lesions illustrated here are intended to be representative of each type. No particular association between these lesions is implied. **(B)** In cross-section, the three types of lesions show different rates of progression and different morphology. Caries lesion progression and morphology follows the inclination of the enamel rods and/or dentinal tubules. Pit-and-fissure lesions have small sites of origin visible on the occlusal surface but have a wide base. Overall shape of a pit-and-fissure lesion is an inverted V. In contrast, a smooth-surface lesion is V shaped with a wide area of origin and apex of the V directed towards pulp (*p*). Root caries begins directly on dentin. Root-surface lesions can progress rapidly because dentin is less resistant to caries attack. **(C)** Although advanced caries lesions produce considerable histologic change in enamel, dentin and pulp changes in dentin can be seen even before the lesion reached the dentin. Bacterial invasion of lesion results in extensive demineralization and proteolysis of the dentin. Clinically, this necrotic dentin appears soft, wet and mushy. Deeper pulpally, dentin is demineralized and is structurally intact. This tissue appears to be dry and leathery in texture. Two types of pulp–dentin response are illustrated. Under pit-and-fissure lesions and smooth-surface lesions, odontoblasts have died, leaving empty tubules called dead tracts (*dt*). New odontoblasts have been differentiated from pulp mesenchymal cells. These new odontoblasts have produced reparative dentin (*rd*), which seals off dead tracts. Another type of pulp–dentin reaction is sclerosis (*s*)—occlusion of the tubules by peritubular dentin. This is illustrated under root-caries lesion.

**Clinical Notes**

- Root-surface caries lesions are more common in *older patients* because of niche availability and other factors sometimes associated with senescence, such as decreased salivary flow due to multiple medications and poor oral hygiene as a result of lowered manual dexterity and decreased motivation.
- Caries lesions originating on the root are alarming because they (1) have a comparatively rapid progression, (2) are often asymptomatic, (3) are closer to the pulp and (4) are more difficult to restore.

## III. Diet and Caries

Frequent ingestion of fermentable carbohydrates begins a series of changes in the local tooth environment, essentially changing the composition of the biofilm, thus favouring the growth of highly acidogenic and aciduric bacteria thereby influencing the pH of the biofilm that eventually leads to caries lesion formation (Fig. 2.12).

- High-frequency exposure of fermentable carbohydrates such as **sucrose** may be the most important factor in producing cariogenic biofilm and, ultimately, caries lesions (Fig. 2.13). Dietary sucrose has two important detrimental effects on how biofilm affects caries.
  - First, frequent ingestion of foods containing sucrose provides a change in the biofilm profile from a noncariogenic biofilm to a cariogenic biofilm.[10]
  - Second, mature biofilm exposed frequently to sucrose rapidly metabolizes it into organic acids, resulting in a profound and prolonged decline in pH.
- The frequent use of candies and lozenges during the day or night increases the risk of caries.
- Sugar-containing acidic beverages, including sport drinks, fruit juices and soft drinks, all contribute to a cariogenic diet.

Because the eventual metabolic product of cariogenic diet is *acid*, in addition to caries lesions the exposure to acidity from other sources (e.g. dried fruits, fruit drinks or other acidic foods and drinks) also may result in dental erosion. In contrast, when ingestion of fermentable carbohydrates is severely restricted or absent, biofilm growth typically does not lead to caries lesions. The dietary restriction must include all intakes that result in acidity, not just sucrose.

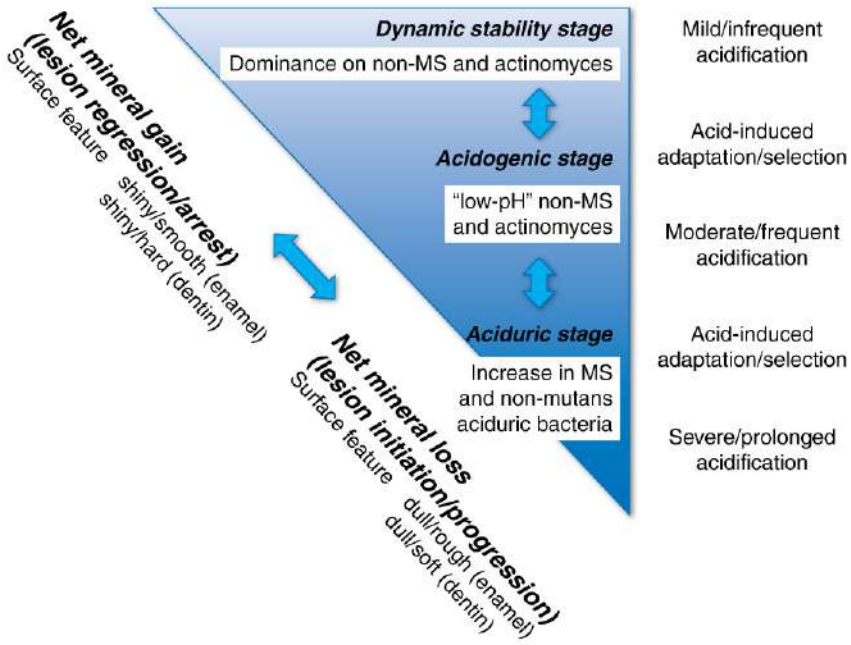



**Figure 2.12** This diagram illustrates how the diet affects the biofilm and consequences on the tooth surface. In health, the biofilm is in symbiosis. A person who has only occasional consumption of fermentable carbohydrates will have a resident microbiome dominated by non-mutans streptococci and actinomyces in a dynamic stability stage that is adapting to the relatively minor changes in pH. As the diet changes to more frequent exposures to fermentable carbohydrates, the microbiome adapts to the more acidic environment in an acidogenic stage, where bacteria that favour low pH flourish. In diets with frequent exposures to fermentable carbohydrates this constant acid-induced adaptation and selection of the microbiome continues and leads to an aciduric stage where there is an increase in MS and non-mutans aciduric bacteria, thus leading to a dysbiosis of the microbiome. At one end of the spectrum, in an environment with severe and prolonged acidification and aciduric bacteria, there is a net mineral loss at the tooth surface, enamel or dentin, causing lesion initiation or lesion progression, while at the other end of the spectrum with only mild acidification, there is a net mineral gain at the tooth surface and consequently lesion regression or arrest. Certainly this is a simplified explanation of the changes in the biofilm as the microbiome adapts to the acidic environment due to cariogenic diets, which does not consider the many other factors that are at play to either halt the process or boost the process.
(From Takahashi N, Nyvad B. The role of bacteria in the caries process: ecological perspectives, *J Dent Res* 90(3):294–303, 2011. doi:10.1177/0022034510379602.)

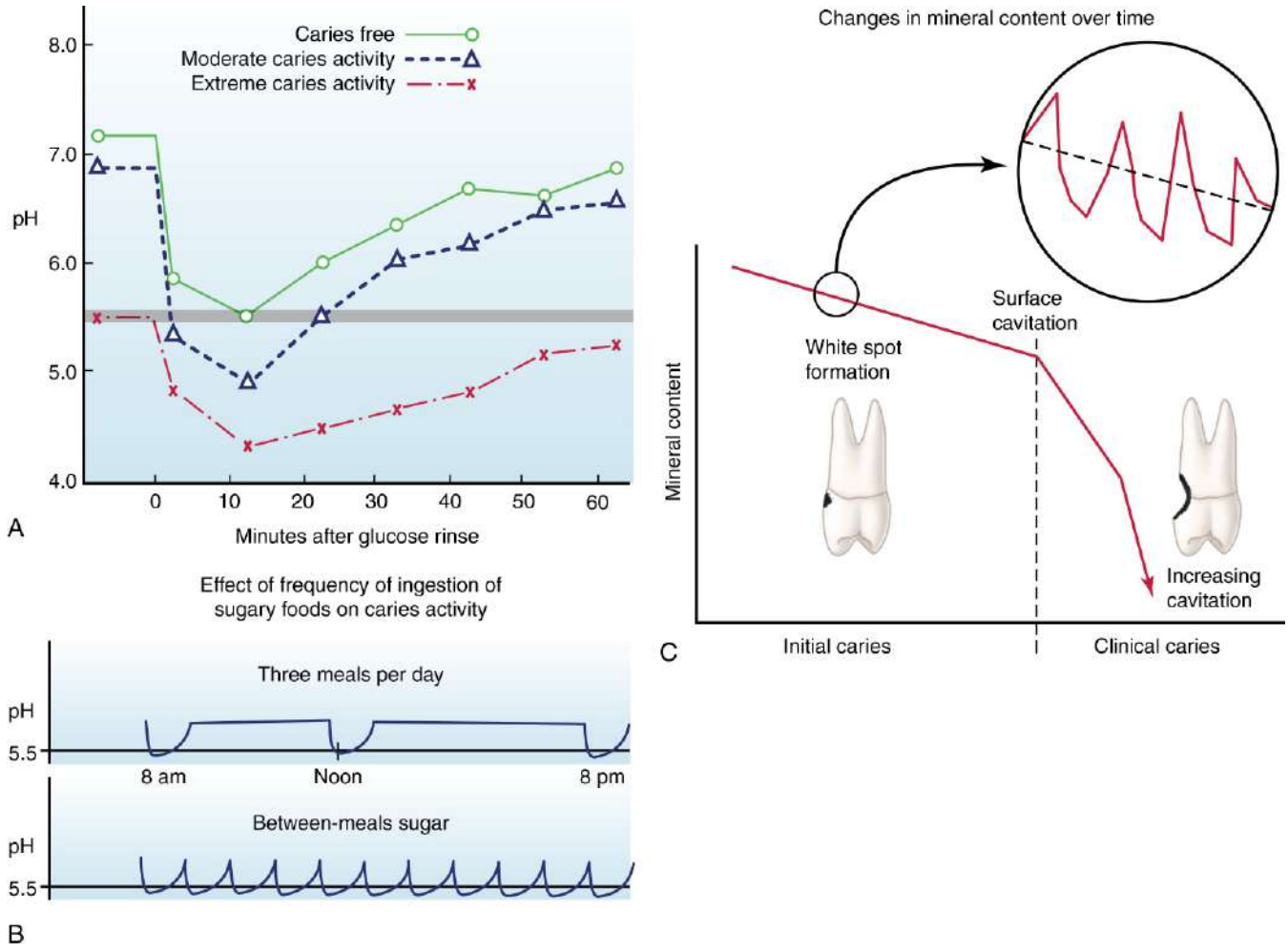



**Figure 2.13** **(A)** Mature biofilm communities have tremendous metabolic potential and are capable of rapid anaerobic metabolism of any available carbohydrates. Classic studies by Stephan show this metabolic potential by severe pH drops at the plaque–enamel interface after glucose rinse. It is generally agreed that a pH of 5.5 is the threshold for enamel demineralization. Exposure to a glucose rinse for an extreme caries activity plaque results in a sustained period of demineralization (pH 5.5). Recording from a slight caries activity biofilm shows a much shorter period of demineralization. **(B)** The frequency of sucrose exposure for cariogenic biofilm greatly influences the progress of tooth demineralization. The top line illustrates pH depression, patterned after Stephan's curves in **(A)**. Three meals per day results in three exposures of biofilm acids, each lasting approximately 1 h. The biofilm pH depression is relatively independent of the quantity of sucrose ingested. Between-meal snacks or the use of sweetened breath mints results in many more acid attacks, as illustrated at the bottom. The effect of frequent ingestion of small quantities of sucrose results in a nearly continuous acid attack on the tooth surface. (The clinical consequences of this behaviour can be seen in Fig. 2.38.) **(C)** In active caries, a progressive loss of mineral content subjacent to the cariogenic biofilm occurs. Inset illustrates that the loss is not a continuous process. Instead, alternating periods of mineral loss (demineralization) occur, with intervening periods of remineralization. The critical event for the tooth is cavitation of the surface, marked by the vertical dashed line. This event marks an acceleration in caries destruction of the tooth and irreversible loss of tooth structure. An intervention is usually required to arrest the lesion, often of the restorative nature.
(Part A, Adapted and redrawn from Stephan RM: Intra-oral hydrogen-ion concentration associated with dental caries activity, *J Dent Res* 23:257, 1944.)

**Clinical Notes**

- Dietary *sucrose* plays a leading role in the development of pathogenic biofilms and may be the *most important factor* in disruption of the normal healthy ecology of dental biofilm communities.
- Caries activity is most strongly stimulated by the frequency, rather than the quantity, of sucrose ingested.

## IV. Saliva: Nature's Anticaries Agent

Saliva is an extremely important substance for the proper digestion of foods, and it also plays a key role as a natural anticaries agent (Table 2.2). Many medications are capable of reducing salivary flow and increasing caries risk (Table 2.3).

The importance of saliva in the maintenance of the oral health is illustrated dramatically by observing changes

in oral health after therapeutic radiation to the head and neck. After radiation, salivary glands become fibrotic and produce little or no saliva, leaving the patient with an extremely dry mouth, a condition termed *xerostomia* (*xero*, dry; *stoma*, mouth). Such patients may experience near-total destruction of the teeth in just a few months after radiation treatment.[11,12] Salivary protective mechanisms that maintain the normal oral flora and tooth surface integrity include bacterial clearance, direct antibacterial activity, buffers and remineralization.[13]

### Bacterial Clearance

- Adults produce *1–1.5 L of saliva a day*, very little of which occurs during sleep. While in the mouth, saliva lubricates oral tissues and bathes teeth and the biofilm. The secretion rate of saliva may have a bearing on caries susceptibility and calculus formation.
- The flushing effect of this salivary flow is, by itself, adequate to remove virtually all microorganisms not adherent to an oral surface.
- The flushing is most effective during mastication or oral stimulation, both of which produce large volumes of saliva.

### Direct Antibacterial Activity

- Salivary glands produce an impressive array of antimicrobial products (Table 2.2). *Lysozyme*, *lactoperoxidase*, *lactoferrin* and *agglutinins* are salivary proteins that possess antibacterial activity.
- Although the antibacterial proteins in saliva play an important role in the protection of soft tissue in the oral cavity from infection by pathogens, they have little effect on caries because similar levels of antibacterial proteins can be found in caries-active and caries-free individuals.[14,15]
- It is suggested that caries susceptibility in healthy individuals is not related to saliva composition but more importantly to the *quantity of saliva*. Individuals with decreased salivary production (owing to illness, medication or irradiation) may have significantly higher caries susceptibility (Table 2.3).

### Buffer Capacity

- The volume and buffering capacity of saliva available to tooth surfaces have major roles in caries protection.[16] The buffering capacity of saliva is determined by
  - *Primary buffer*: bicarbonate ion concentration
  - *Additional buffers*: urea and sialin
- The benefit of buffering is to reduce the potential for acid formation.
- Saliva is crucial in controlling the oral flora and the mineral content of teeth, hence *salivary testing* should be done on patients with signs and symptoms of hyposalivation. Signs include lack of saliva pool on the floor of the mouth, gingivitis and mucositis (including burning mouth syndrome), cheilitis (inflammation and fissuring of the lips and/or the tongue), fissure in the soft mucosa, fungal infections in the mouth such as thrush, glossodynia (painful tongue), sialadenitis (salivary gland infection), saliva that seems thick and stringy and several caries lesions especially in uncommon areas such as the lower central incisors. Symptoms include complaints of 'cotton mouth' (xerostomia); bad breath; difficulty chewing, speaking and swallowing; a changed sense of taste and problems wearing dentures.

### Remineralization

- Saliva and biofilm fluid are supersaturated with *calcium* and *phosphate ions*. Without a means to control precipitation of these ions, the teeth literally would become encrusted with mineral deposits. Saliva contains *statherin*, a proline-rich peptide that stabilizes calcium and phosphate ions and prevents excessive deposition of these ions on teeth.[17]
- This *supersaturated state* of the saliva provides a constant opportunity for remineralizing enamel and can help protect teeth in times of cariogenic challenges.

## Clinical Characteristics of the Caries Lesion

The caries lesion is the product of disequilibrium between the demineralization and remineralization processes discussed previously. Caries can be described according to location, clinical management strategy, rate of activity, occurrence and depth of lesion. The use of correct and consistent terms when referring to caries lesions is important. Box 2.2 and Table 2.4 summarize the most common terms used in this textbook.

The characteristics of a caries lesion vary with the nature of the surface on which the lesion develops. There are three distinctly different clinical sites for caries lesion initiation:

I. Lesions originating in developmental pits and fissures of enamel.
II. Lesions originating on smooth enamel surfaces that shelter cariogenic biofilm.
III. Lesions originating on root surfaces.

Each of these areas has distinct surface morphology and environmental conditions. Consequently, each area has a distinct biofilm population. The diagnosis, treatment and prevention of these different lesion types should take into account the different aetiologic factors operating at each site.

**Table 2.2** Elements of saliva that control plaque biofilm communities

| Names | Action | Effects on plaque biofilm community |
|---|---|---|
| **Salivary enzymes** | | |
| Amylase | Cleaves—1,4 glucoside bonds | Increases availability of oligosaccharides |
| Lactoperoxidase | Catalyses hydrogen peroxide-mediated oxidation; adsorbs to hydroxyapatite in active form | Lethal to many organisms: suppresses plaque formation on tooth surfaces |
| Lysozyme | Lyses cells by degradation of cell walls, releasing peptidoglycans; binds to hydroxyapatite in active conformation | Lethal to many organisms; peptidoglycans activate complement; suppresses plaque formation on tooth surfaces |
| Lipases | Hydrolysis of triglycerides to free fatty acids and partial glycerides | Free fatty acids inhibit attachment and growth of some organisms |
| **Non-enzyme proteins** | | |
| Lactoferrin | Ties up free iron | Inhibits growth of some iron-dependent microbes |
| Secretory immunoglobulin A (IgA) (smaller amounts of IgM, IgG) | Agglutination of bacteria inhibits bacterial enzymes | Reduces numbers in saliva by precipitation; slows bacterial growth |
| Glycoproteins (mucins) | Agglutination of bacteria | Reduces numbers in saliva by precipitation |

*IgA*, Immunoglobulin A.

**Table 2.3** Medications with potential to cause hypo-salivation or dry mouth (xerostomia)

| Action/medication group | Medicaments |
|---|---|
| **Sympathomimetic** | |
| Antidepressants | Venlafaxine<br>Duloxetine<br>Reboxetine<br>Bupropion |
| **Anticholinergic** | |
| Tricyclic antidepressants | Amitriptyline<br>Clomipramine<br>Amoxapine<br>Protriptyline<br>Doxepin<br>Imipramine<br>Trimipramine<br>Nortriptyline<br>Desipramine |
| Muscarinic receptor antagonists | Oxybutynin |
| Alpha-receptor antagonists | Tamsulosin<br>Terazosin |
| Antipsychotics | Promazine<br>Triflupromazine<br>Mesoridazine<br>Thioridazine<br>Clozapine<br>Olanzapine |
| Antihistamines | Aziridine<br>Brompheniramine<br>Chlorpheniramine<br>Cyproheptadine<br>Dexchlorpheniramine<br>Hydroxyzine<br>Phenindamine<br>Cetirizine<br>Loratadine |
| **Anticholinergic, dehydration** | |
| Diuretics | Furosemide<br>Bumetanide<br>Torsemide<br>Ethacrynic acid |
| **Sympathomimetic** | |
| Antihypertensive agents | Metoprolol<br>Monoxidine<br>Rilmenidine |
| Appetite suppressants | Fenfluoramine<br>Sibutramine<br>Phentermine |
| Decongestants | Pseudoephedrine |
| Bronchodilators | Tiotropium |
| Skeletal muscle relaxants | Tizanidine |
| Antimigraine agents | Rizatriptan |
| **Synergistic mechanism** | |
| Opioids, hypnotics | Opium<br>Cannabis<br>Tramadol<br>Diazepam |
| **Unknown** | |
| H2 antagonists, proton pump inhibitors | Cimetidine<br>Ranitidine<br>Famotidine<br>Nizatidine<br>Omeprazole |
| Cytotoxic drugs | Fluorouracil |
| Anti-HIV drugs, protease inhibitors | Didanosine |

*(Adapted from the Kois Center:* Support Materials, Always Pages. *http://koiscenter.com/store/supmatlist.aspx, Accessed January 13, 2012)*

**Box 2.2 Important terminologies**

- **Caries lesion:** Tooth demineralization as a result of the caries process. Other texts may use the term *carious lesion*. Laypeople may use the term *cavity*
- **Occlusal caries:** A caries lesion on an occlusal surface
- **Proximal caries:** A caries lesion on a proximal surface
- **Coronal caries:** A caries lesion in any surface of the anatomic tooth crown
- **Primary dentin:** Sound, normal dentin that forms during tooth development; it is usually completed 3 years after tooth eruption. Histologically, primary dentin has tubules with smooth odontoblastic processes, with no intra-tubular crystals. The inter-tubular dentin has normal cross-banded collagen and normal dense apatite crystals. Clinically, primary dentin is hard, cannot be easily penetrated with a blunt explorer and can only be cut by a bur or a sharp cutting instrument.
- **Secondary dentin:** Sound, normal dentin that forms physiologically on all internal aspects of the pulp cavity throughout the life of the tooth. Histologically, secondary dentin resembles primary dentin. Clinically, secondary dentin is similar to primary dentin.
- **Tertiary dentin:** Dentin that forms in response to stimuli such as caries, attrition and operative procedures. Also known as reparative or reactive dentin. Usually appears as a localized dentin deposit on the wall of the pulp space immediately subjacent to the area of the tooth that has received the injury. Tertiary dentin is less mineralized than primary and secondary dentin and contains irregular dentinal tubules. Clinically, tertiary dentin is not as hard as primary dentin.
- **Sclerotic dentin:** Dentin that forms in response to stimuli such as aging or mild irritation (slow advancing caries). When responding to initial caries demineralization events, crystalline material precipitates in intra-tubular and inter-tubular dentin. Sclerotic dentin walls off a lesion by blocking (sealing) the dentinal tubules. This zone can be seen even before the demineralization reaches the dentin and it may not be present in rapidly advancing lesions. Clinically, sclerotic dentin is dark and harder than normal dentin.
- **Leathery dentin:** Term used to describe the clinical presentation of the transition zone between soft and firm dentin (next section). Technically, leathery dentin is part of the firm dentin zone.

## I. Pit and Fissure Caries

Pits and fissures are particularly susceptible surfaces for caries initiation (Fig. 2.9; Figs. 2.14–2.18). The type and nature of the organisms prevalent in the oral cavity determine the type of organisms colonizing pits and fissures and are instrumental in determining the outcome of the colonization. Large variations exist in the microflora found in pits and fissures, suggesting that each site can be considered a separate ecologic system. Numerous Gram-positive cocci, especially *S. sanguis*, are found in the pits and

**Table 2.4 Classification of dental caries**

I. **According to location**
   i. **Pit-and-fissure caries:** A caries lesion on a pit-and-fissure area
   ii. **Smooth-surface caries** A caries lesion on a smooth tooth surface
   iii. **Root caries** A caries lesion in the root surface

II. **According to clinical management strategy**
   i. **Initial Caries Lesion/Non-cavitated caries lesion/White Spot Lesion:** A caries lesion that has not been cavitated. In enamel caries, non-cavitated lesions are also referred to as 'white spot' lesions. (Clinically, the distinction between a cavitated and a non-cavitated caries lesion is not as simple as it may seem. Although historically any roughness detectable with a sharp explorer has been considered a cavitated lesion, more recent caries detection guidelines establish that only lesions in which a blunt probe (e.g. WHO [World Health Organization]/CPI [Community Periodontal Index]/PSR [Periodontal Screening and Recording] probe) penetrates are to be considered cavitated. This distinction has important implications on lesion management because most initial caries lesions can be arrested or remineralized without any restorative intervention
   ii. **Cavitated caries lesion:** A caries lesion that results in the breaking of the integrity of the tooth, or a *cavitation*. All cavitated lesions would require restorative intervention

III. **According to rate of activity**
   i. **Active caries lesion:** A caries lesion that is considered to be biologically active, that is, lesion in which tooth demineralization is in frank activity at the time of examination
      a. **Moderate caries lesion:** An active caries lesion that may or may not have cavitated but that has *not reached the inner one-third of dentin* is considered as a moderate caries lesion. This can be observed clinically by microcavitations in the enamel or a grey shadow
      b. **Advanced (deep) caries lesion:** A definitely cavitated active caries lesion exposing dentin and that has *reached the inner one-third of dentin* will be considered an advanced (deep) caries lesion.
   ii. **Inactive caries lesion:** A caries lesion that is considered to be biologically inactive at the time of examination, that is, in which tooth demineralization caused by caries may have happened in the past but has stopped and is currently stalled. Also referred to as *arrested caries*, meaning that the caries process has been arrested but that the clinical signs of the lesion itself are still present
   iii. **Rampant caries:** Term used to describe the presence of extensive and multiple cavitated and active caries lesions in the same person. Typically used in association with 'baby bottle caries', 'radiation therapy caries' or 'meth-mouth caries'. These terms refer to the etiology of the condition.

IV. **According to occurrence**
   i. **Primary caries:** A caries lesion not adjacent to an existing restoration or crown
   ii. **Secondary caries:** A caries lesion adjacent to an existing restoration, crown or sealant. Other term used is *caries adjacent to restorations and sealants (CARS)*. Also referred to as *recurrent caries*, implying that a primary caries lesion was restored but that the lesion reoccurred
   iii. **Residual caries:** Refers to carious tissue that was not completely excavated prior to placing a restoration. Sometimes residual caries can be difficult to differentiate from secondary caries

V. **According to depth of lesion**
   i. **Enamel caries:** A caries lesion in enamel, typically indicating that the lesion has not penetrated into dentin. (Note that many lesions detected clinically as enamel caries may very well have extended into dentin histologically.)
   ii. **Dentin caries:** A caries lesion extending into dentin

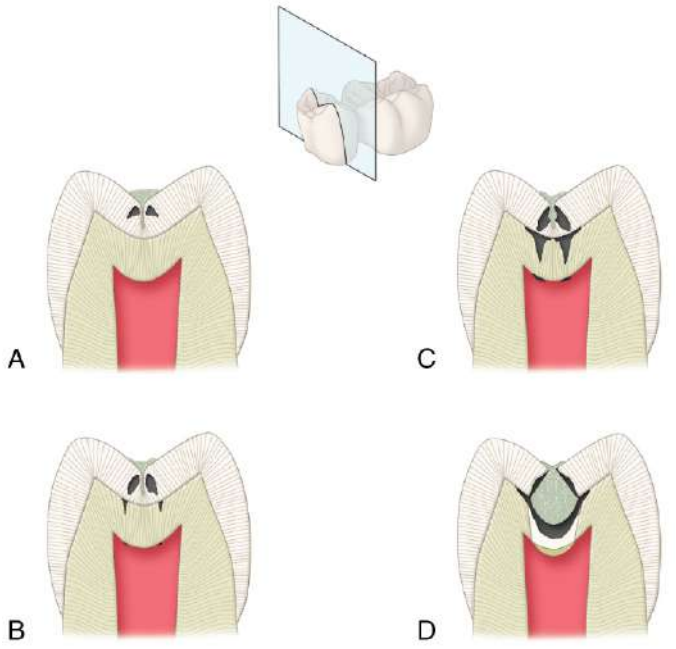



**Figure 2.14** Progression of caries in pits and fissures. (A) Initial lesions develop on the lateral walls of the fissure. Demineralization follows the direction of the enamel rods, spreading laterally as it approaches the dentinoenamel junction (*DEJ*). (B) Soon after the initial enamel lesion occurs, a reaction can be seen in the dentin and pulp. Forceful probing of the lesion at this stage can result in damage to the weakened porous enamel and accelerate the progression of the lesion. Clinical detection at this stage should be based on observation of discolouration and opacification of the enamel adjacent to the fissure. These changes can be observed by careful cleaning and drying of the fissure. (C) Initial cavitation of the opposing walls of the fissure cannot be seen on the occlusal surface. Opacification can be seen that is similar to the previous stage. Remineralization of the enamel because of trace amounts of fluoride in the saliva may make progression of pit-and-fissure lesions more difficult to detect. (D) Extensive cavitation of the dentin and undermining of the covering enamel darken the occlusal surface.

fissures of newly erupted teeth, whereas large numbers of *S. mutans* usually are found in carious pits and fissures.

The shape of the pits and fissures contributes to their high susceptibility to caries lesion. The long, narrow fissure prevents adequate biofilm removal. Considerable morphologic variation exists in these structures. Some pits and fissures end blindly, others open near the dentin, and others penetrate entirely through the enamel.

Pit-and-fissure caries lesions typically *expand* as they penetrate into the enamel. The entry site may appear much smaller than the actual lesion, making clinical detection difficult. Caries lesions of pits and fissures develop from demineralization on their walls, known as *wall lesions* (Fig. 2.14A–C). Progression of the dissolution of the walls of a pit-and-fissure lesion is similar in principle to that of the smooth-surface lesion because a wide susceptible surface area extends inward, paralleling the enamel rods. A lesion originating in a pit or fissure affects a greater area of the dentinoenamel junction (DEJ) than does a comparable smooth-surface lesion (Fig. 2.15).

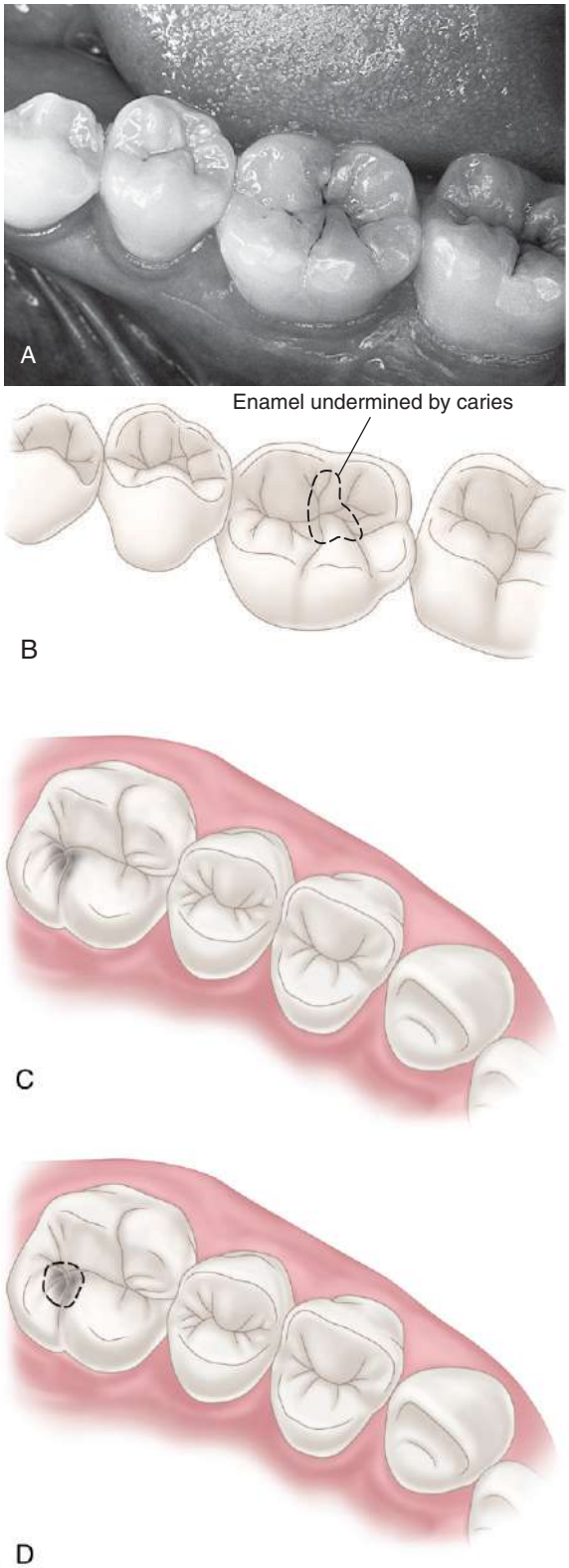



**Figure 2.15** (A) Mandibular first molar has undermined discoloured enamel due to extensive pit and fissure caries; (B) discoloured enamel is outlined by broken line in the central fossa region; (C) maxillary first molar has undermined discoloured enamel owing to extensive pit-and-fissure caries. Note the *dark shadow* due to the dentin demineralization underneath the enamel and (D) Discolored enamel is outlined by the broken line in the central fossa region.

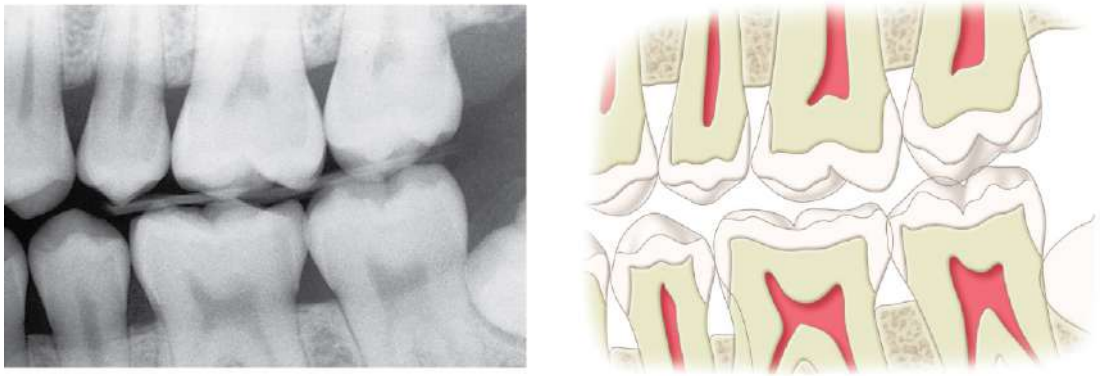


**Figure 2.16** Bitewing radiograph of normal teeth, free from caries. Note the uniform density of the enamel on the interproximal surfaces. A third molar is impacted on the distal aspect of the lower second molar.

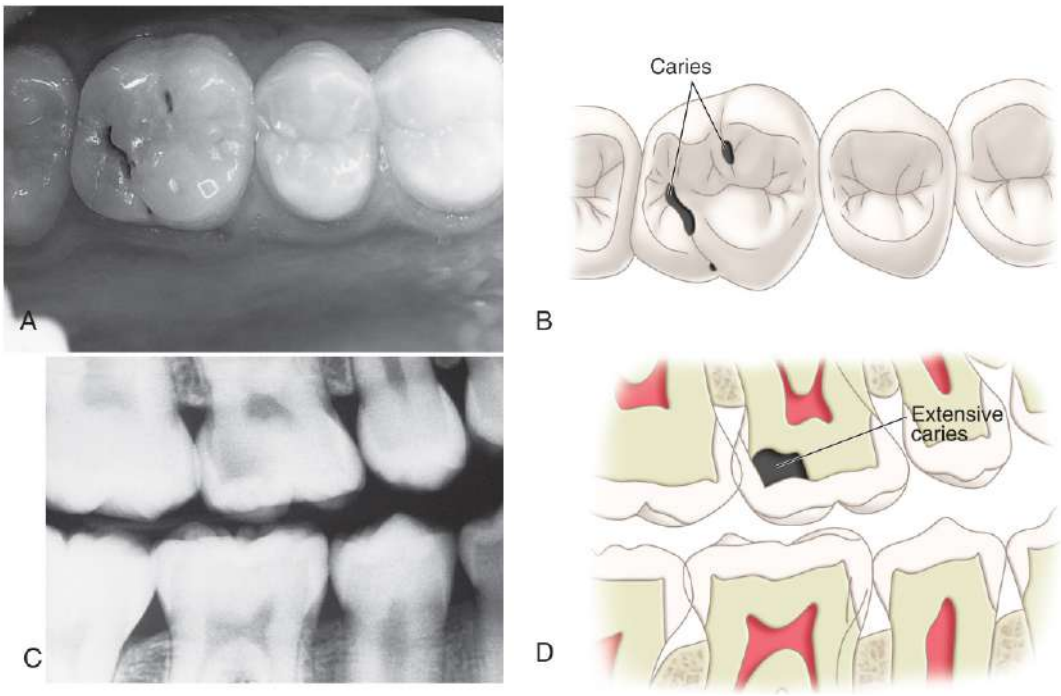



**Figure 2.17** Example of occlusal caries that is much more extensive than is apparent clinically. (A and B) Clinical example; (C and D) a bitewing radiograph further reveals an extensive area of demineralization undermining the distofacial cusp.

**Clinical Notes**

- The relative proportion of mutans streptococci (MS) most probably determines the cariogenic potential of the pit-and-fissure community.
- In cross-section, the gross appearance of a pit-and-fissure lesion is an inverted 'V' with a narrow entrance and a progressively wider area of involvement closer to the DEJ (Fig. 2.14D).
- In susceptible patients, sealing the pits and fissures just after tooth eruption may be the most important event in their resistance to caries.

## II. Smooth Surface Caries

The proximal enamel surfaces immediately *gingival to the contact area* are the second most susceptible areas to caries (Figs. 2.16–2.18; also Fig. 2.10). These areas are protected physically and are relatively free from the effects of mastication, tongue movement and salivary flow (Fig. 2.19). The types and numbers of organisms composing the proximal surface biofilm community vary.[18] Important ecologic determinants for the biofilm community on the proximal surfaces are

- Topography of the tooth surface
- Size and shape of the gingival papilla
- Oral hygiene of the patient

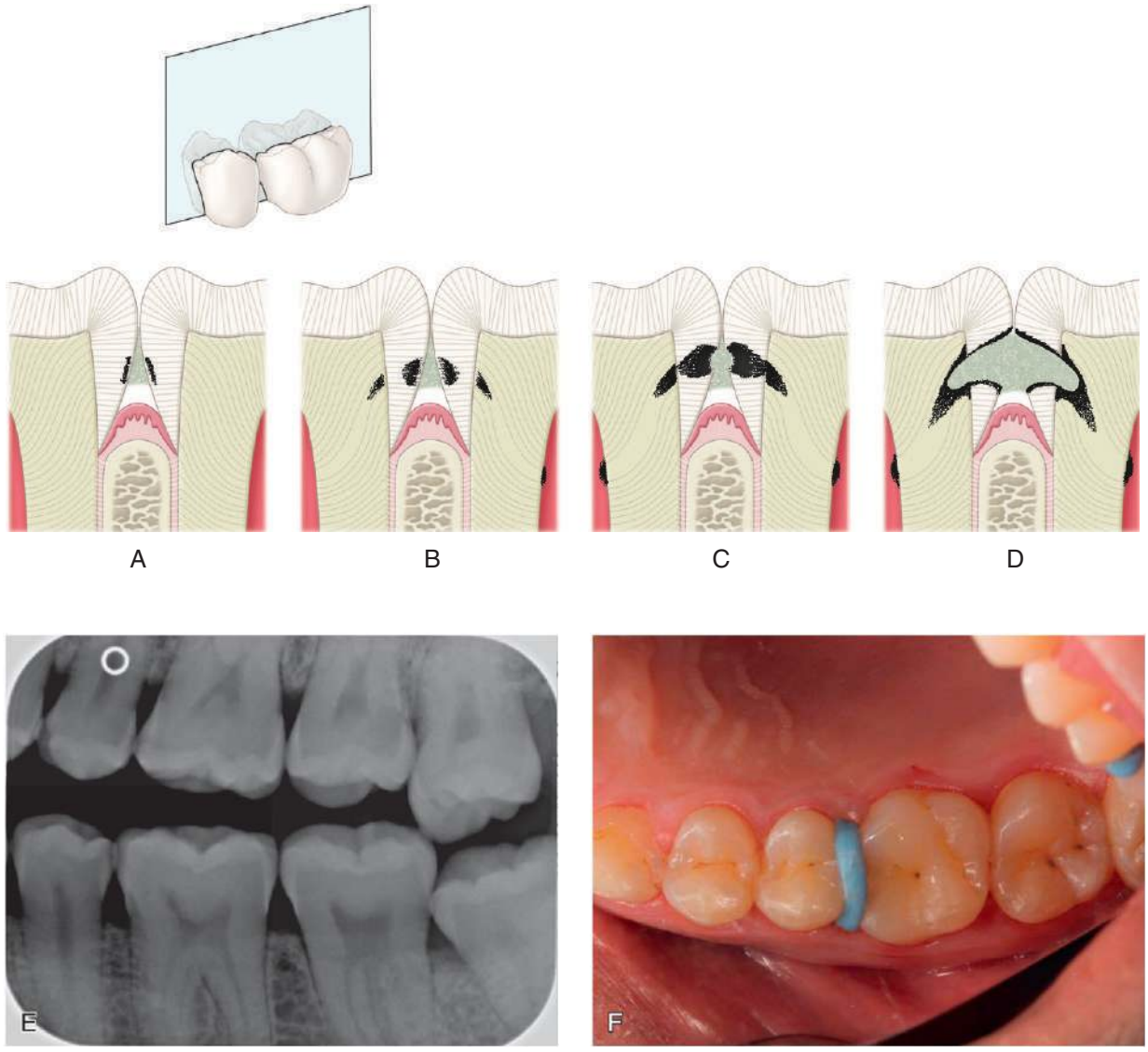



**Figure 2.18** Longitudinal sections (inset for A) showing initiation and progression of caries on interproximal surfaces. (A) Initial demineralization (indicated by the shading in the enamel) on the proximal surfaces is difficult to detect clinically or radiographically. The lesions develop right below the contact surface. Flossing of the interproximal area and thorough drying will allow for the best conditions to detect these lesions clinically. (B) When proximal caries lesions first become detectable radiographically, the enamel surface is likely still to be intact. An intact surface is essential for successful remineralization and arrest of the lesion. Demineralization of the dentin (indicated by the shading in the dentin) occurs before cavitation of the surface of the enamel. Treatment designed to promote remineralization can be effective up to this stage. (C) Cavitation of the enamel surface is a critical event in the caries process in proximal surfaces. After cavitation, lesion arrest becomes more difficult and virtually impossible in an interproximal surface contact. Restorative treatment to repair the damaged tooth surface is required. Cavitation can be diagnosed only by clinical observation. The use of a sharp explorer to detect cavitation is problematic because excessive force in application of the explorer tip during inspection of the proximal surfaces can damage weakened enamel and accelerate the caries process by iatrogenic cavitation. Separation of the teeth using orthodontic separators can be used to provide more direct visual inspection of suspect surfaces. Fibre-optic illumination although not specific for cavitation can facilitate detection; an orange shadow can be seen when lesions reach the dentin. (D) Advanced cavitated lesions require prompt restorative intervention to prevent pulpal disease, limit tooth structure loss and remove the nidus of infection of odontopathic organisms. (E) Radiographic image showing radiolucency on distal of tooth no. 13 and mesial of tooth no. 14. (F) An orthodontic separator was placed and left in place for 3 days.

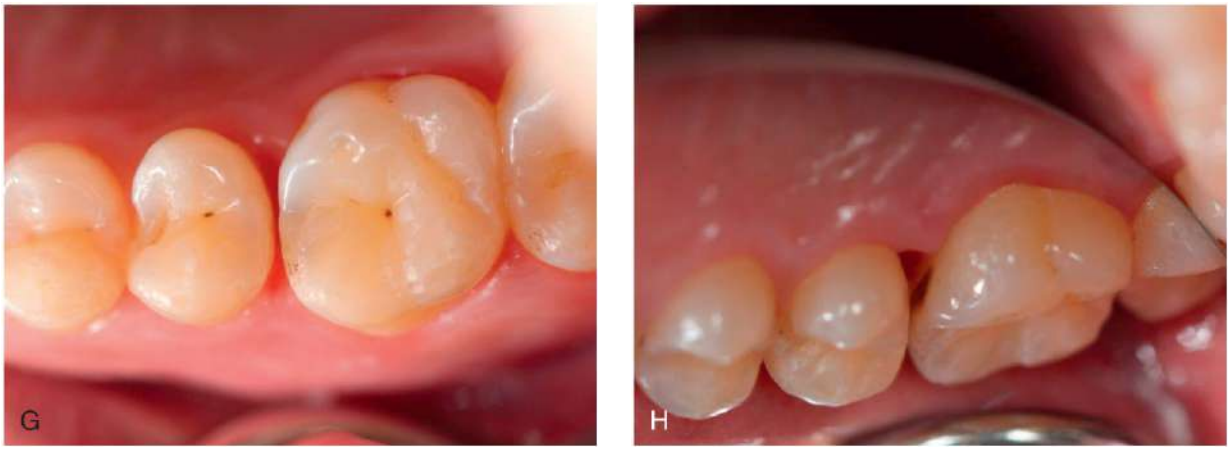



**Figure 2.18 (*cont.*)** (G) Clinical image of the teeth after separation. (H) There was no evidence of cavitation on either surface.



**Figure 2.19** Extracted tooth showing extensive caries lesion just gingival to the proximal contact area (note the slightly 'flat' contact area adjacent to marginal ridge).

Lesions starting on smooth enamel surfaces have a broad area of origin and a conical, or pointed, extension towards the DEJ.

After the demineralization reaches the DEJ, when there is enamel cavitation and bacterial contamination, the bacterial metabolic activity within the lesion causes rapid dentin demineralization both laterally and pulpally.

**Clinical Notes**

- A rough proximal surface gingival to the contact area (caused by caries, a poor-quality restoration or a structural defect) restricts adequate biofilm removal. This situation favours the occurrence of caries or periodontal disease at this site.
- A cross-section of the enamel portion of a smooth-surface lesion shows a V-shape, with a wide area of origin and the apex of the 'V' directed towards the DEJ.

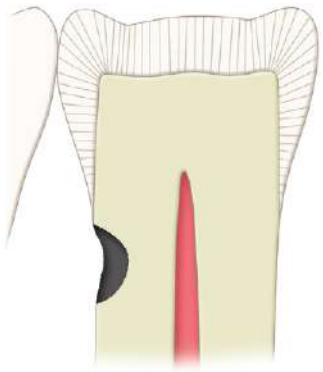


**Figure 2.20** Root surface caries.

## III. Root Caries

The root surface is rougher than enamel and readily allows cariogenic biofilm formation in the absence of good oral hygiene. The cementum covering the root surface is extremely thin and provides little resistance to caries lesion activity. In addition, the critical pH for dentin is higher than for enamel, so demineralization is likely to start even before the pH reaches the critical level for enamel (pH = 5.5). In recent years, the prevalence of root caries has increased significantly because of the increasing number of older persons who retain more teeth, experience gingival recession and usually have cariogenic biofilm on the exposed root surfaces.[19–22] (Fig. 2.20).

**Clinical Notes**

- Root caries lesions have less well-defined margins and tend to be *U-shaped* in cross-section
- Caries originating on the root is alarming because
  - It has a comparatively *rapid progression*
  - It is often *asymptomatic*
  - It is closer to the pulp
  - It is more difficult to restore

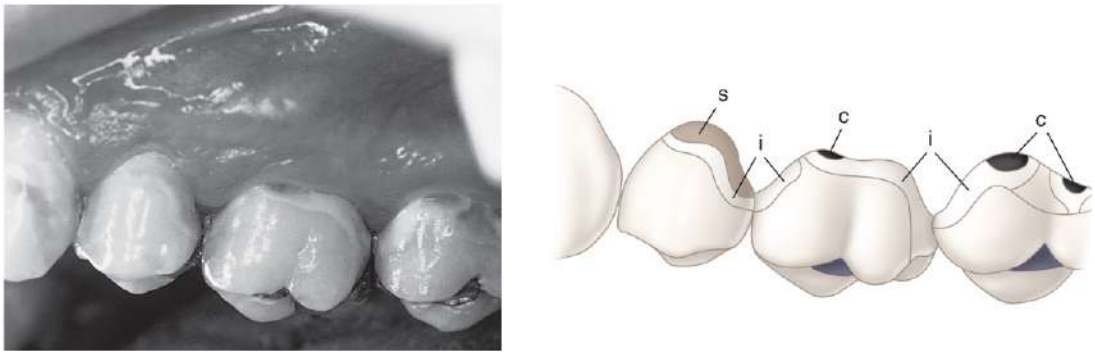



**Figure 2.21** Facial and lingual smooth-surface caries. Several levels of caries involvement can be seen, including cavitation (*c*); non-cavitated white spot lesions (*i*) and stained, roughened, partially remineralized non-cavitated lesions (*s*).

## Enamel Caries

### Non-cavitated Enamel Caries/White Spot Lesion/Initial Caries Lesion

Non-cavitated caries lesion or white spot lesion (WSL) or incipient caries lesion is the *first evidence* of caries activity in enamel and appear as *chalky white opaque areas on enamel when air-dried and seems to disappear when the tooth is rehydrated or made wet* (Fig. 2.21).

- This lesion may be characterized as *reversible*, and the enamel surface is fairly hard, intact and smooth to the touch.
- These areas of enamel lose their translucency because of the *extensive subsurface porosity* caused by demineralization.
- These lesions usually are observed on the *facial* and *lingual* surfaces of teeth. They can also occur in the proximal surfaces but are difficult to detect.
- Care must be exercised in distinguishing white spots of initial caries lesions from developmental white spot hypocalcifications or other developmental defects of enamel. Initial (white spot) caries lesions partially or totally disappear visually when the enamel is hydrated (wet), whereas hypocalcified enamel is affected less by drying and wetting (Table 2.5). Hypocalcified enamel does not represent a clinical problem except for its potential esthetically objectionable appearance.
- Non-cavitated enamel lesions sometimes can be seen on radiographs as a *faint radiolucency* that is limited to the superficial enamel.
- Initial lesions can also occur on the proximal smooth surfaces, although their detection by visual examination is more challenging. Tooth separation with orthodontic separators can facilitate visual examination of proximal tooth surfaces.
- When a proximal lesion is clearly visible radiographically, the lesion may have advanced significantly, and histologic alteration of the underlying dentin probably already has occurred, whether the lesion is cavitated or not.
- It has also been shown experimentally and clinically that non-cavitated caries of enamel *can remineralize* if immediate corrective measures alter the oral environment, including plaque removal and fluoride therapy.[23,24] The remineralization mechanism of WSL is summarized in Box 2.3, while Tables 2.5 and 2.6 list the characteristics of enamel at various stages of demineralization.

**Clinical Notes**

- During clinical examination of teeth it is important to dry or dessicate the tooth surfaces to diagnose white spot lesions.
- Non-cavitated (white spot) caries partially or totally *disappears visually* when the enamel is hydrated (wet), whereas hypocalcified enamel is affected less by drying and wetting (Table 2.6).
- Hypocalcified enamel does not represent a clinical problem except for its esthetically objectionable appearance.
- Injudicious use of an explorer tip can cause actual cavitation in a previously non-cavitated area, requiring, in most cases, restorative intervention.

**Box 2.3 Remineralization mechanism of a noncavitated caries lesion or incipient caries or white spot lesion (WSL)**

The *supersaturation* of saliva with calcium and phosphate ions serves as the driving force for the remineralization process

↓

Non-cavitated enamel lesions retain most of the original crystalline framework of the enamel rods, and the etched crystallites serve as nucleating agents for remineralization

↓

Calcium and phosphate ions from saliva can penetrate the enamel surface and precipitate on the highly reactive crystalline surfaces in the enamel lesion

↓

The presence of trace amounts of *fluoride ions* during this remineralization process greatly enhances the precipitation of calcium and phosphate, resulting in the remineralized enamel becoming more resistant to subsequent caries attack because of the incorporation of more acid-resistant *fluorapatite*

**Table 2.5** Clinical characteristics of normal and altered enamel

| | Hydrated | Desiccated | Surface texture | Surface hardness |
|---|---|---|---|---|
| Normal Enamel | Translucent | Translucent | Smooth | Hard |
| Hypocalcified Enamel | Opaque | Opaque | Smooth | Hard |
| Active Initial Caries | Translucent (early lesions) | Opaque | Rough | Softened |
| | Opaque (more established initial lesions) | Opaque | Rough | Softened |
| Arrested Initial Caries | Shiny and/or dark | Shiny and/or dark | Smooth | Hard |
| Active Moderate Caries | Opaque | Opaque | Rough | Softened |
| Arrested Moderate Caries | Shiny and/or dark | Shiny and/or dark | Smooth | Hard |
| Active Advanced Caries | Opaque | Opaque | Rough | Softened |
| Arrested Advanced Caries | Shiny and/or dark | Shiny and/or dark | Smooth | Leathery or hard |

**Table 2.6** Clinical significance of enamel lesions

| | Plaque biofilm | Enamel structure | Non-restorative, therapeutic treatment (e.g. remineralization, antimicrobial, pH control) | Restorative treatment |
|---|---|---|---|---|
| Normal enamel | Normal | Normal | Not indicated | Not indicated |
| Hypocalcified enamel | Normal | Abnormal but not weakened | Not indicated | Only for esthetics |
| Non-cavitated caries | Cariogenic | Porous, weakened | Yes | Not indicated |
| Active caries | Cariogenic | Cavitated, very weak | Yes | Yes |
| Inactive caries | Normal | Remineralized, strong | Not indicated | Only for esthetics |

## Zones of Enamel Caries

The four zones in a sectioned enamel caries (Fig. 2.22 and 2.23) lesion when examined in quinoline by transmitted light are as follows:

### Zone 1: Surface zone

- The surface zone is relatively unaffected by the caries attack.
- The intact surface over incipient caries is a phenomenon of the caries demineralization process rather than any special characteristics of the superficial enamel. Nevertheless, the importance of the intact surface cannot be overemphasized because it serves as a barrier to bacterial invasion.
- It has been hypothesized that *hypermineralization* and *increased fluoride content* of the superficial enamel are responsible for the relative immunity of the enamel surface.
- It has a lower pore volume than the body of the lesion (<5%) and a radiopacity comparable to unaffected adjacent enamel.

### Zone 2: Body of lesion

- The body of the lesion is the *largest portion* of the incipient lesion while in a demineralizing phase.
- It has the *largest pore volume*, varying from 5% at the periphery to 25% at the centre.
- The striae of Retzius are well marked in the body of the lesion, indicating preferential mineral dissolution along these areas of relatively higher porosity. The first penetration of caries enters the enamel surface via the striae of Retzius.
- Bacteria may be present in this zone if the pore size is large enough to permit their entry.

### Zone 3: Dark zone

- The next deepest zone is known as the dark zone because it does not transmit polarized light.
- This light blockage is caused by the presence of many tiny pores too small to absorb quinoline.
- These smaller air-filled or vapour-filled pores make the region opaque.
- The total pore volume is 2%–4%.
- Caries is an episodic disease with alternating phases of demineralization and remineralization. Experimental remineralization has shown increases in the size of the dark zone at the expense of the body of the lesion.
- The size of the dark zone is probably an indication of the amount of remineralization that has recently occurred.

### Zone 4: Translucent zone

- The translucent zone is the deepest zone and represents the advancing front of the enamel lesion.
- The name refers to its *structureless appearance* when perfused with quinoline solution and examined with polarized light.

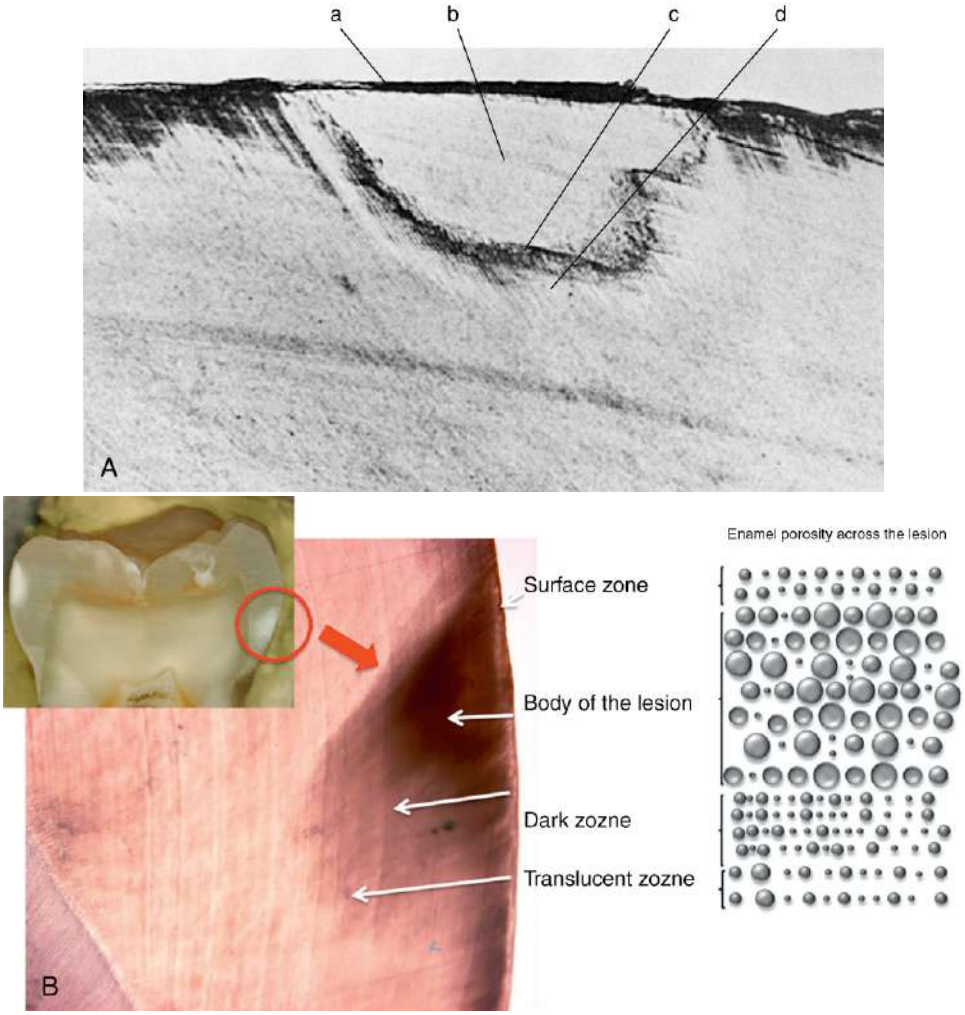



**Figure 2.22** (A) Cross-section of small caries lesion in enamel examined by transmitted light (100×). Surface (*a*) appears to be intact. Body of lesion (*b*) shows enhancement of striae of Retzius. Dark zone (*c*) surrounds body of lesion, whereas translucent zone (*d*) is evident over entire advancing front of lesion. (B) The porosity of the lesion changes across the different zones. Thus, a sharp explorer can easily penetrate the outer enamel and cause a small cavitation on an active lesion.
(From Silverstone LM, et al. editors: *Dental caries*, London and Basingstoke, 1981, Macmillan, Ltd.)

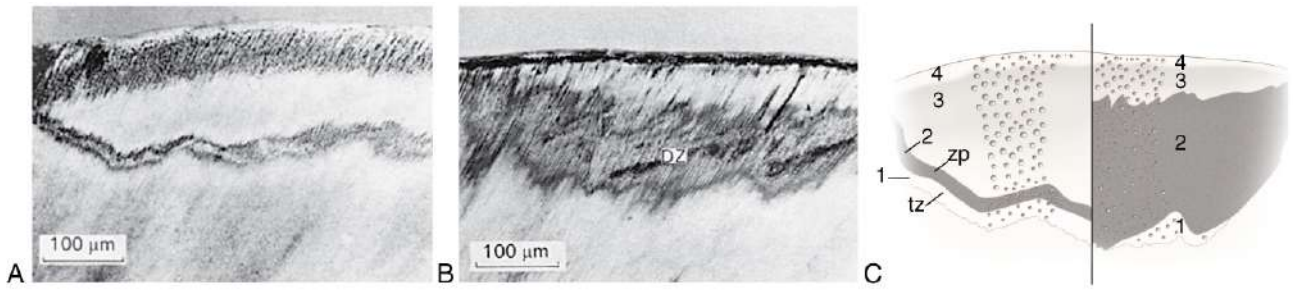



**Figure 2.23** (A) Cross-section of small caries lesion in enamel examined in quinoline with polarized light (100×). Advancing front of lesion appears as a dark band below body of lesion. (B) Same section after exposure to artificial calcifying solution examined in quinoline and polarized light. Dark zone (DZ) covers a much greater area after remineralization has occurred (100×). (C) Schematic diagram of (A) and (B). Left side indicates small extent of zones 1 and 2 before remineralization. Small circles indicate relative sizes of pores in each zone. Right side indicates increase in zone 2, the dark zone, after remineralization. This micropore system must have been created where previously the pores were much larger.
(From Silverstone LM, et al, editors: Dental caries, London and Basingstoke, 1981, Macmillan, Ltd. Reproduced with permission of Palgrave Macmillan. C was redrawn.)

- In this zone, the pores or voids form along the enamel prism (rod) boundaries, presumably because of the ease of hydrogen ion penetration during the carious process. When these boundary area voids are filled with quinoline solution, which has the same refractive index as enamel, the features of the area disappear.
- The pore volume of the translucent zone of enamel caries is 1%, 10 times greater than normal enamel.

## Caries Lesion Progression

- The progression and morphology of the caries lesion vary depending on the site of origin and the conditions in the mouth (Figs. 2.11,2.14 and 2.18).
- Peak rates for the incidence of new lesions occur 3 years after the eruption of the tooth.
- Occlusal pit-and-fissure lesions develop in *less time* than smooth-surface caries lesions.
- Radiation-induced *hyposalivation* (dry mouth) can lead to development of caries lesions in 3 months from the onset of the radiation.
- Caries lesion progression in healthy individuals is usually slow compared with the progression in compromised persons.

**Clinical Notes**

- Poor oral hygiene and frequent exposures to sucrose-containing or acidic food can produce non-cavitated initial ('white spot') lesions (first clinical evidence of demineralization) in *3 weeks*.
- The time for progression from a non-cavitated caries lesion to a cavitated caries lesion on smooth surfaces is estimated to be *18 months (±6 months)*.[18]

## Cavitated Enamel Caries

- Cavitated enamel lesions can be initially detected as *subtle breakdown* of the enamel surface. More advanced cavitated enamel lesions are more obviously detected as *gross enamel breakdown* (Fig. 2.24).
- These lesions are very sensitive to probing and can be easily enlarged by using sharp explorers and excessive probing force.
- In cavitated caries, the enamel surface is broken (not intact), and usually the lesion has advanced into dentin. Although some cavitated enamel lesions can be arrested and may not progress to larger lesions, *most cavitated caries lesions require restorative treatment*.
- The cavitation of the tooth surface produces a synergistic acceleration of the growth of the cariogenic biofilm community and the expansion of the demineralization with ensuing expanded cavitation. This sheltered, highly acidic and anaerobic environment provides an ideal niche for cariogenic bacteria. This situation results in a rapid and progressive destruction of the tooth structure.



**Figure 2.24** Advance cavitated active caries lesions on a 35-year-old patient with high caries risk.
(Courtesy: Dr Ayesha Swarn, DDS, Operative Dentistry Graduate Program at UNC School of Dentistry.)

# Dentin Caries Lesions

## Progression of Caries in Dentin

- Progression of dental caries in dentin is different from progression in enamel because of the structural differences of dentin. Caries advances more *rapidly* in dentin than in enamel because
  - Dentin provides much less resistance to acid attack owing to less mineralized content.
  - Dentin possesses microscopic tubules that provide a pathway for the ingress of bacteria and egress of minerals (Fig. 2.25).
- Increasing demineralization of the body of the enamel lesion results in the weakening and eventual collapse of the surface enamel. The resulting *cavitation* provides an even more protective and retentive habitat for the cariogenic biofilm, accelerating the progression of the lesion (Fig. 2.26).
- When enamel demineralization advances to the DEJ, rapid lateral expansion of the caries lesion along the DEJ may occur if there is bacterial contamination. The *lateral spread* of caries at the DEJ has been attributed to
  - DEJ's lower mineral content compared to primary dentin.[25]
  - More recently, it has been shown that a high concentration of MMPs is found at the DEJ. MMPs are implied in dentin caries lesion progression, although this may be a host defence mechanism and not necessarily linked to an increase in caries susceptibility.[26]

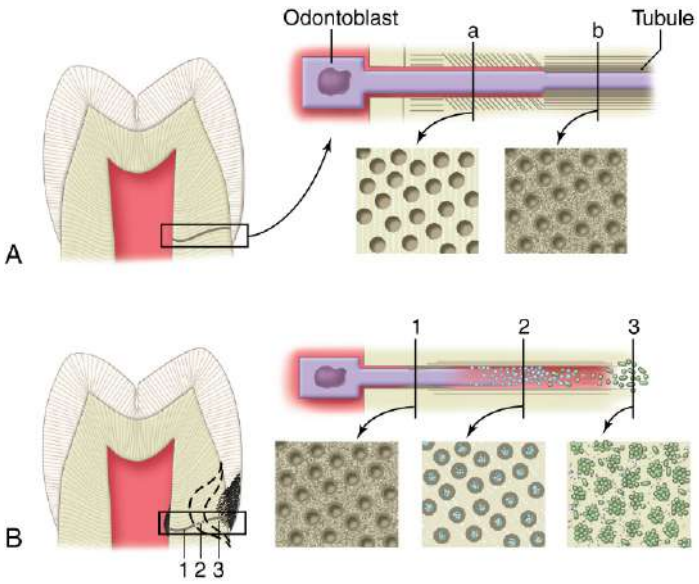



**Figure 2.25** Normal and carious dentin. (A) Normal dentin has characteristic tubules that follow a wavy path from the external surface of dentin, below enamel or cementum, to the inner surface of dentin in the pulp tissue of the pulp chamber or pulp canal. Dentin is formed from the external surface and grows inward. As dentin grows, odontoblasts become increasingly compressed in the shrinking pulp chamber, and the number of associated tubules becomes more concentrated per unit area. The more recently formed dentin near the pulp (*a*) has large tubules with little or no peritubular dentin and calcified inter-tubular dentin filled with collagen fibres. Older dentin, closer to the external surface (*b*), is characterized by smaller, more widely separated tubules and a greater mineral content in inter-tubular dentin. The older dentin tubules are lined by a uniform layer of mineral termed *peritubular dentin*. These changes occur gradually from the inner surface to the external surface of the dentin. Horizontal lines indicate predentin; diagonal lines indicate increasing density of minerals; darker horizontal lines indicate densely mineralized dentin and increased thickness of peritubular dentin. The transition in mineral content is gradual, Carious dentin undergoes several changes. The most superficial infected zone of carious dentin (3) is characterized by bacteria filling the tubules and granular material in the inter-tubular space. The granular material contains very little mineral and lacks characteristic cross-banding of collagen. As bacteria invade dentinal tubules, if carbohydrates are available, they can produce enough lactic acid to remove peritubular dentin. This doubles or triples the outer diameter of the tubules in infected dentin zone. Pulpal to (below) the infected dentin is a zone where the dentin appears transparent in mounted whole specimens. This zone (2) is the advancing demineralization front, affected carious dentin and is characterized by loss of mineral in the inter-tubular and peritubular dentin. Many crystals can be detected in the lumen of the tubules in this zone. The crystals in the tubule lumen render the refractive index of the lumen similar to that of the inter-tubular dentin, making the zone transparent. Normal dentin (1) is found pulpal to (below) transparent dentin.

  - As a result of increased bacterial metabolic activity within the dentinal lesion.[27–29]
- Dentinal caries is *V-shaped in cross-section* with a wide base at the DEJ and the apex directed pulpally (Fig. 2.18).




**Figure 2.26** Cavitated caries in both anterior (A) and posterior (B) teeth.

## Clinical Symptoms

- Episodes of short-duration *sensitivity or mild discomfort* may be felt occasionally during earlier stages of dentin caries. The pain is caused by stimulation of pulp tissue by the movement of fluid through the dentinal tubules that have been opened to the oral environment by cavitation.
- When bacterial invasion of the dentin is close to the pulp, toxins and possibly a few bacteria enter the pulp, resulting in inflammation of the pulpal tissues and, thus, *pulpal pain*.

## Dentinal Reaction to Caries

The three levels of dentinal reaction to caries that can be recognized are as follows:

I. Reaction to a long-term, low-level acid demineralization associated with a slowly advancing lesion: *sclerotic dentin*
II. Reaction to a moderate-intensity attack: *reparative dentin*

III. Reaction to severe, rapidly advancing caries characterized by very high acid levels: *pulpal necrosis*

## I. Sclerotic dentin: *Reaction to a long-term, low-level acid demineralization associated with a slowly advancing lesion*

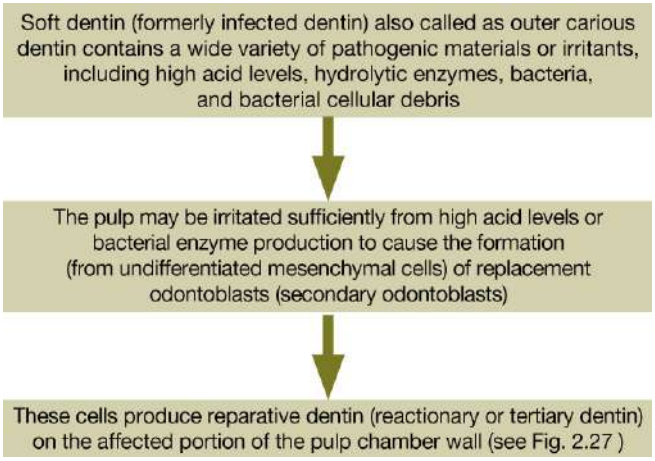
- Initial stages of caries lesions or mild caries activity produce long-term, low-level acid demineralization of dentin. In slowly advancing caries lesions, a vital pulp can repair demineralized dentin by remineralization of the inter-tubular dentin and by apposition of peritubular dentin.
- Dentin responds to the stimulus of its first caries demineralization episode by *deposition of crystalline material from the inter-tubular dentin* in the lumen of the tubules in the advanced demineralization front (formally called affected dentin) (Figs. 2.25 and 2.27).
- The refractive index of the dentin changes and the *inter-tubular dentin with more mineral content than normal dentin is termed sclerotic dentin*.
- The apparent function of sclerotic dentin is to wall off a lesion by blocking (sealing) the tubules. The permeability of sclerotic dentin is greatly reduced compared with normal dentin because of the decrease in the tubule lumen diameter.[30]
- Sclerotic dentin is usually *shiny and darker in colour* but feels hard to the explorer tip and may be seen under an old restoration. By contrast, normal, freshly cut dentin lacks a shiny, reflective surface and allows some penetration from a sharp explorer tip.
- Hypermineralized areas may be seen on radiographs as zones of *increased radiopacity* (often S-shaped following the course of the tubules) ahead of the advancing infected portion of the lesion.
- When these affected tubules become *completely occluded* by the mineral precipitate, they appear clear when a histologic section of the tooth is evaluated. This portion of dentin has been termed *translucent dentin* (Zones of dentinal caries lesions section) and is the result of mineral loss in the inter-tubular dentin and precipitation of this mineral in the tubule lumen.

**Clinical Notes**

- Sclerotic dentin can be formed only if the tooth pulp is vital.
- Translucent dentin is softer than normal dentin (Fig. 2.27)[31] and is called *firm dentin* (formerly affected dentin), in contrast to sound dentin that is 'hard' dentin.

## II. Reparative dentin: *Reaction to a moderate-intensity attack*

- The next level of dentinal response is to moderate-intensity irritants by forming *reparative dentin*. The mechanism of reparative dentin formation is explained in Flowchart 2.1.




**Flowchart 2.1** Mechanism of reparative dentin formation.

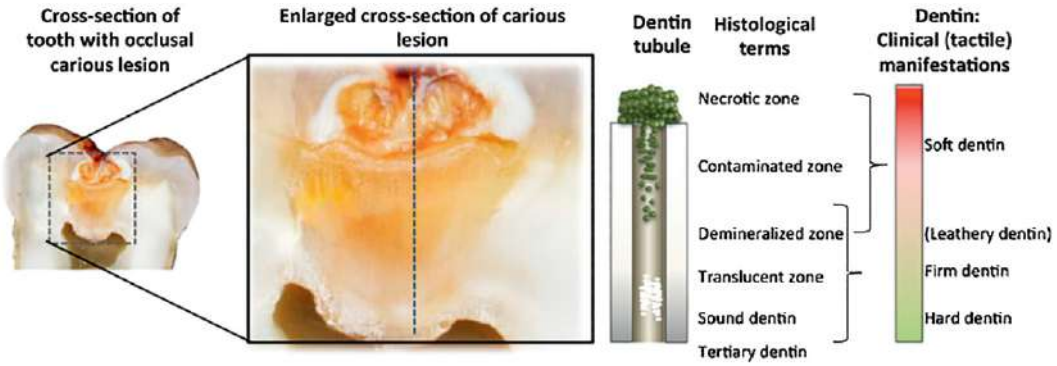



**Figure 2.27** A cross-section of a caries lesion in dentin and the histologic and clinical manifestations. The most superficial outer layer of dentin is heavily contaminated, necrotic, with irreversible demineralization and denatured collagen fibres. Clinically this is very soft, wet dentin. This soft, contaminated layer gets progressively harder as the dentin approaches the pulp. The still demineralized dentin (formerly affected dentin) can feel leathery to the touch or firm, but not as hard as sound dentin or tertiary dentin (closest to the pulp, dentin formed in reaction to the caries process). During caries excavation, the goal is to remove only the soft, outer carious dentin (infected dentin), while the inner carious dentin (affected dentin) is remineralizable and can be maintained.
(Adapted from Ogawa K, Yamashita Y, Ichijo T, et al.: The ultrastructure and hardness of the transparent layer of human carious dentin, *J Dent Res* 62(1):7–10, 1983; Courtesy: Dr. T. Fusayama. Copyright Ishiyaku EuroAmerica, Inc., Tokyo, 1993.)

- Reparative dentin is different from the normal dentinal apposition that occurs throughout the life of the tooth by primary (original) odontoblasts. The structure of reparative dentin varies from well-organized tubular dentin (less often) to very *irregular atubular dentin* (more often), depending on the severity of the stimulus.
- Reparative dentin is an *effective barrier* allowing limited diffusion of material through the tubules and is an important step in the repair of dentin. Severe stimuli also can result in the formation within the pulp chamber of unattached dentin, termed *pulp stones*, in addition to reparative dentin.

**Clinical Notes**

- The success of dentinal reparative responses, either by sclerotic dentin or by reparative dentin, depends on the severity of the caries challenge and the ability of the pulp to respond.

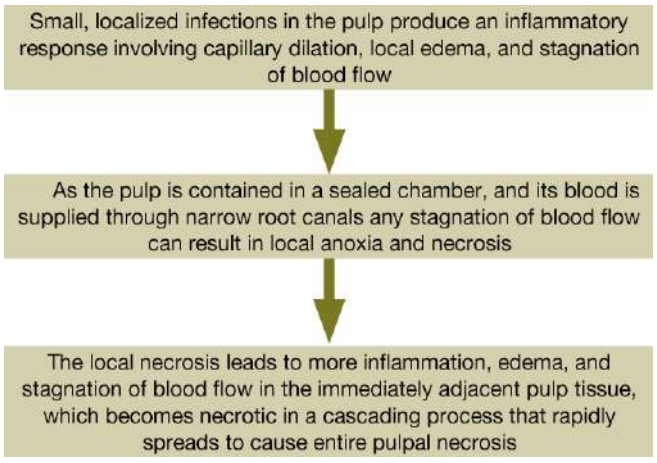
### III. Pulpal Necrosis: *Reaction to severe, rapidly advancing caries characterized by very high acid levels*

The third level of dentinal response is to severe irritation. Acute, rapidly advancing caries with high levels of acid production overpowers dentinal defences and results in *infection*, *abscess* and *death of the pulp*.

The mechanism of pulpal necrosis is explained in Flowchart 2.2.

**Clinical Notes**

- Recently erupted teeth with large pulp chambers and short, wide canals with large apical foramina have a much more favourable prognosis for surviving pulpal inflammation than fully formed teeth with small pulp chambers and small apical foramina.




**Flowchart 2.2** Mechanism of pulpal necrosis.

## Zones of Dentin Caries Lesions

Caries advancement in dentin proceeds through three changes:

I. Demineralization of dentin by weak acids and exposing the organic matrix of dentin.
II. The exposed organic matrix of dentin, particularly collagen, is denatured and degraded.
III. Loss of matrix structural integrity followed by invasion of bacteria.

Three different zones have been described in moderate and advanced (deep) caries lesions (Figs. 2.27 and 2.28). These zones are most clearly distinguished in slowly advancing lesions. In rapidly progressing caries, the difference between the zones becomes less distinct:

### Zone 1: Hard Dentin

- Hard dentin represents the deepest zone of a caries lesion—assuming the lesion has not yet reached the pulp and may include *tertiary dentin*, *sclerotic dentin* and *normal (or sound) dentin*.
- Clinically, this dentin is hard, cannot be easily penetrated with a blunt explorer and can only be removed by a bur or a sharp cutting instrument. (See definitions in Box 2.2 for histologic characteristics of hard, tertiary and sclerotic dentin.)

### Zone 2: Firm dentin (formerly affected dentin)

- Also called *inner carious dentin*, firm (affected) dentin is primarily characterized by demineralization of inter-tubular

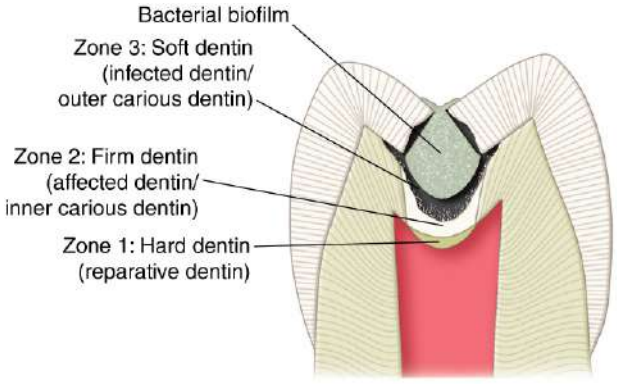



**Figure 2.28** Cross-section of occlusal caries. The occlusal enamel appears intact, with a small opening in the occlusal fissure. Enamel is darkened where it is undermined by demineralization. The surface of enamel may not be clinically cavitated, but usually some discolouration is observed at the entrance of the fissure. The lesion is filled with a bacterial plug containing high numbers of MS and lactobacilli. When there is a break in the enamel, the dentin underneath the enamel will be highly contaminated below the plug. Deeper dentin is not as heavily contaminated but is extensively demineralized. Reparative dentin is being formed below the lesion.

dentin and of initial formation of intra-tubular fine crystals at the advancing front of the caries lesion.
- As the tubule lumen becomes filled with minerals it will give a *transparent* appearance in a section observed in a light microscope.
- Histologically, this zone may be referred to as *demineralized*. Because of the caries demineralization process, firm dentin is softer than hard, normal dentin. Although organic acids attack the mineral and organic contents of dentin, the collagen cross-linking remains intact in this zone and can serve as a template for remineralization of inter-tubular dentin. Therefore, provided that the pulp remains vital, firm (affected) dentin is remineralizable.
- Clinically, firm dentin is resistant to hand excavation and can only be removed by exerting pressure. The transition between soft and firm dentin can have a leathery texture, particularly in slowly advancing lesions, and has been called *leathery dentin*. Clinically, leathery dentin does not deform upon pressure from an instrument but can be excavated with hand instruments such as spoons and curette without much pressure.

### Zone 3: Soft dentin (formerly infected dentin)

- Also called *outer carious dentin*, soft (infected) dentin is primarily characterized by bacterial contamination. This is the carious dentin closer to the tooth surface, characterized by the presence of bacteria, low mineral content and irreversibly denatured collagen.
- Histologically, this zone may be referred to as *necrotic* and *contaminated*.
- Soft dentin typically does not self-repair.
- Clinically, soft dentin lacks structure and can be easily excavated with hand and rotary instrumentation.
- This zone cannot be remineralized, and its removal is essential to sound, successful restorative procedures and the prevention of spreading the infection.[32,33]

#### Clinical Considerations

- In restorative procedures, it is convenient to term dentin as either *soft dentin* (formerly *infected dentin*) which requires removal, or *firm dentin* (formerly *affected dentin*), which does not require removal. When sclerotic dentin is encountered, it represents the ideal final excavation depth because it is a natural barrier that blocks the penetration of toxins and acids.
- To clinically distinguish soft dentin from firm dentin, the operator traditionally observes the *degree of discolouration* (extrinsic staining) and tests the area for *hardness* by the feel of an explorer tine or a slowly revolving bur. Some difficulties occur with this approach because
  - The discolouration may be slight and gradually changeable in acute (rapid) caries.
  - The hardness (softness) felt by the hand through an instrument may be an inexact guide.
- To differentiate between remineralizable (firm dentin) and nonremineralizable dentin (soft dentin); staining the carious dentin was proposed by Fusayama. Commonly employed *caries detection solutions* include 1% acid red 52 (acid rhodamine B or food red 106) in propylene glycol.[34] However, current evidence shows that *dye staining is not a good indicator of bacterial invasion of the dentin, and its use can lead to over-preparation of a cavity and it should therefore be avoided.*

## Inactive Caries Lesion/Arrested Caries

- Inactive caries lesions are remineralized (arrested) lesions that may be observed clinically as either intact, smooth white lesions or discoloured, usually brown or black, spots (Fig. 2.29).
- The change in colour is presumably caused by trapped organic debris and metallic ions within the enamel.
- These discoloured, remineralized, arrested caries lesion areas are more resistant to subsequent caries activity than the adjacent unaffected enamel.
- They should not be restored unless they are esthetically objectionable.

**Clinical Notes**

- The slow rate of caries allows time for extrinsic pigmentation and an inactive/arrested enamel lesion is usually *brown-to-black in colour* (Fig. 2.29).
- An arrested, dentinal lesion typically is *open* (allowing debridement from tooth brushing), dark and hard, and this dentin is also termed as *eburnated dentin*.

## Rampant Caries

Rampant (or acute) caries is a rapidly developing caries process usually involving several teeth and is a sign of gross dietary inadequacy, a complete absence of oral hygiene practice, systemic illness or a combination of these pathologic factors (Fig. 2.30A–C).

Rampant caries that is present primarily on proximal surfaces may point more to diet as the main driving factor, whereas rampant caries in the cervical and interproximal areas may point to diet and hygiene as the driving factors. The presence of rampant caries indicates the need for comprehensive patient evaluation.

## Root Caries

Root caries is a pervasive problem in a high percentage of older patients.[35–37] Approximately 38% of patients between the ages of 55 and 64 years have root caries, and 47% of those between 65 and 74 years have experienced

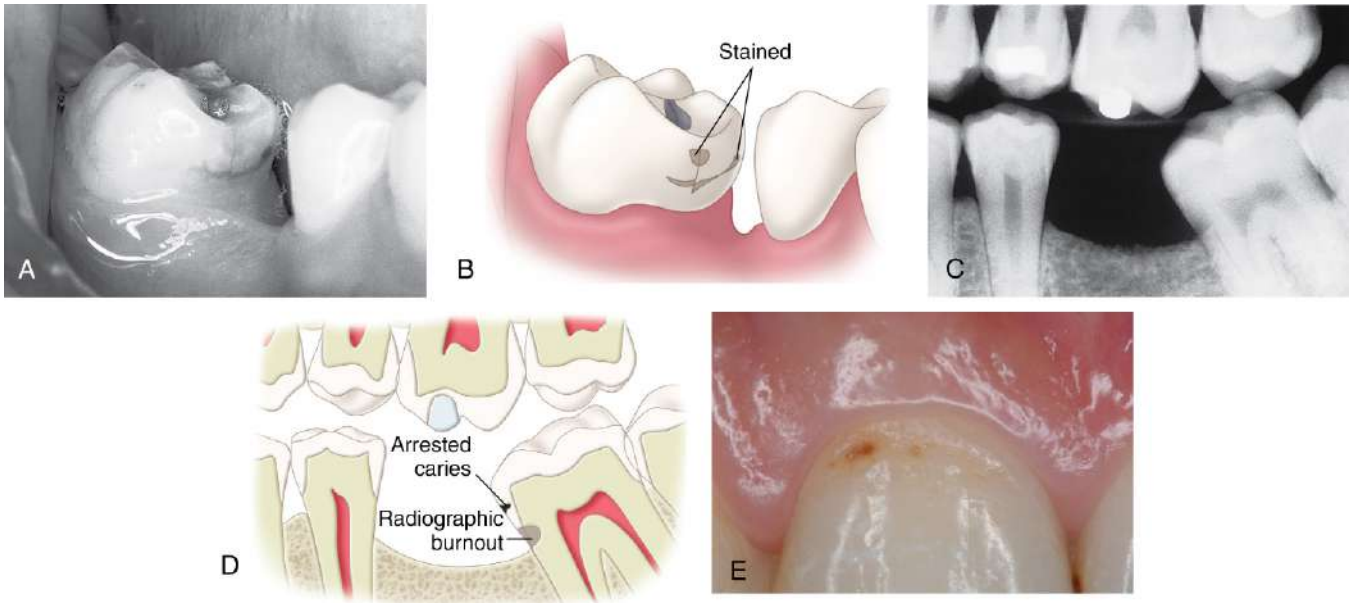



**Figure 2.29** (A and B) Example of arrested caries on the mesial surface of a mandibular second molar. The area below the proximal contact ((A and B) mirror view of tooth 37) is partly opaque and stained. Clinically the surface is hard and intact, yet the area is more radiolucent than the enamel above or below the stain. (C and D) In a different clinical case, caries diagnosis based only on the radiograph would lead to a false-positive diagnosis (i.e. caries present when it is not). The radiolucency is caused by the broad area of subsurface demineralization that extends from the facial to the lingual line angles. The X-ray beam was directed parallel to the long axis of demineralization and consequently produced a sharply demarcated zone of radiolucency in the enamel. This example illustrates the shortcomings of radiographic diagnosis. Were there not visual access to the mesial surface of the second molar, it would be easy to diagnose active caries incorrectly and consequently restore the tooth. (E) Cavitated inactive (arrested) enamel caries lesion on the cervical one-third of a central incisor of a 27-year-old patient with low caries risk. This lesion, if not esthetically offensive, does not require a restoration and should be monitored.




**Figure 2.30** (A) Rampant caries in a preschool child. (B) Rampant caries in a school-age child. (C) Rampant caries in an adult. (A, From Dean JA, Avery DR, McDonald RE: *McDonald and Avery's dentistry for the child and adolescent*, ed 9, St Louis, 2011, Mosby.)

root caries.[38] The incidence of root caries in more older adults (over 75 years) is even higher.[39]

## Etiology

I. One of the primary aetiologic factors for these patients is their use of prescription drugs for a wide variety of systemic medical problems.
II. The *critical pH of dentin* (pH at which dentin begins to demineralize) is between 6.2 and 6.7, whereas that of enamel is about 5.5.[40] As a result, root dentin will demineralize in very weak acids, and root caries progresses at about *twice the rate* of coronal caries.

## Risk Factors

Risk factors for root caries include the following:

I. Gingival recession
II. Poor oral hygiene
III. Cariogenic diet
IV. Presence of multiple restorations or multiple missing teeth
V. Existing caries
VI. Xerogenic medications
VII. Compromised salivary flow rates

## Clinical Protocol

The clinical protocol is based upon four primary strategies for the prevention of root caries.

- The *first* strategy is to try to *improve salivary flow rates* and increase the buffering capacity.
- The *second* strategy is to *modulate the cariogenic biofilm* in the oral cavity.
- The *third* strategy is to *reduce the quantity and number of exposures* of ingested refined carbohydrates.
- The *fourth strategy* is to attempt to *remineralize non-cavitated lesions* and prevent new lesions from developing.

In addition to following the aforementioned protocol, three additional considerations are important:

I. *Use of powered toothbrushes and irrigation devices*. It is critical that patients susceptible to root caries practice meticulous oral hygiene. However, many of these patients have physical and visual deficiencies, and this makes it difficult for them to adequately cleanse the mouth. For these patients, a powered toothbrush may be advantageous (Fig. 2.31).[41] Additionally, daily use of a water irrigation device (Waterpik, Water Pik Inc., Fort Collins, CO) may be beneficial (Fig. 2.32). Although the device will not remove biofilm, studies have shown that daily use will change the composition of the biofilm in a beneficial way. However, since the toothbrush essentially provides the means of fluoride delivery, use of a water irrigation device alone is not indicated unless used with a fluoridated rinse (e.g. ACT) (Fig. 2.33).




**Figure 2.31** Powered toothbrush.




**Figure 2.32** Water irrigation device (Waterpik, Waterpik Inc.).

ADA American Dental Association®
America's leading advocate for oral health

## Caries Risk Assessment Form (Age >6)

| | | | | |
|---|---|---|---|---|
| Patient name: | | | | |
| Birth date: | | | Date: | |
| Age: | | | Initials: | |
| | | Low risk | Moderate risk | High risk |
| Contributing conditions | | Check or circle the conditions that apply | | |
| I. | **Fluoride exposure** (through drinking water, supplements, professional applications, toothpaste) | ☐Yes | ☐No | |
| II. | **Sugary foods or drinks** (including juice, carbonated or non-carbonated soft drinks, energy drinks, medicinal syrups) | Primarily at mealtimes ☐ | | Frequent or prolonged between meal exposures/day ☐ |
| III. | **Caries experience of mother, caregiver and/or other siblings** (for patients ages 6-14) | No carious lesions in last 24 months ☐ | Carious lesions in last 7-23 months ☐ | Carious lesions in last 6 months ☐ |
| IV. | **Dental home:** established patient of record, receiving regular dental care in a dental office | ☐Yes | ☐No | |
| General health conditions | | Check or circle the conditions that apply | | |
| I. | **Special health care needs** (developmental, physical, medical or mental disabilities that prevent or limit performance of adequate oral health care by themselves or caregivers) | ☐No | Yes (over age 14) ☐ | Yes (ages 6-14) ☐ |
| II. | **Chemo/Radiation therapy** | ☐No | | ☐Yes |
| III. | **Eating disorders** | ☐No | ☐Yes | |
| IV. | **Medications that reduce salivary flow** | ☐No | ☐Yes | |
| V. | **Drug/Alcohol abuse** | ☐No | ☐Yes | |
| Clinical conditions | | Check or circle the conditions that apply | | |
| I. | **Cavitated or non-cavitated** (incipient) **carious lesions or restorations** (visually or radiographically evident) | No new carious lesions or restorations in last 36 months ☐ | 1 or 2 new carious lesions or restorations in last 36 months ☐ | 3 or more carious lesions or restorations in last 36 months ☐ |
| II. | **Teeth missing due to caries in past 36 months** | ☐No | | ☐Yes |
| III. | **Visible plaque** | ☐No | ☐Yes | |
| IV. | **Unusual tooth morphology** that compromises oral hygiene | ☐No | ☐Yes | |
| V. | **Interproximal restorations - 1 or more** | ☐No | ☐Yes | |
| VI. | **Exposed root surfaces** present | ☐No | ☐Yes | |
| VII. | **Restorations with overhangs** and/or **open margins; open contacts** with food impaction | ☐No | ☐Yes | |
| VIII. | **Dental/Orthodontic appliances** (fixed or removable) | ☐No | ☐Yes | |
| IX. | **Severe dry mouth (xerostomia)** | ☐No | | ☐Yes |

**Overall assessment of dental caries risk:** ☐ Low ☐ Moderate ☐ High

Patient instructions:

A

© American Dental Association, 2009, 2011. All rights reserved.

**Figure 2.33** A, Example of a caries risk assessment form recommended by the American Dental Association. B, Example of a caries risk assessment form used by the University of North Carolina, Department of Operative Dentistry. C, Another caries assessment form used by the University of North Carolina, Department of Operative Dentistry. This form is very useful for patient communication and compliance. (A, Copyright 2009, 2011 the American Dental Association.)

## Caries Risk Initial Assessment

Name_____

Date_____

| Risk | Low risk | At risk | High risk |
|---|---|---|---|
| Risk rating score | ≤ 3 | ≥4 and ≤8 | ≥9 |

**Patient interview assessment**

| Dental history | | | Risk rating |
|---|---|---|---|
| 1. Had nonemergency dental care in last year | Yes | No | –1 |
| 2. Brushes teeth at least twice daily | Yes | No | –3 |
| 3. Uses fluoridated toothpaste or product daily | Yes | No | –3 |
| 4. New caries lesions within the last 3 years | Yes | No | 8 |
| 5. Patient has teeth sensitive to hot, cold, sweets | Yes | No | 3 |
| 6. Patient avoids brushing any part of mouth | Yes | No | 3 |

Supplemental notes for dental history

| Dietary assessment | | | |
|---|---|---|---|
| 1. Water supply currently fluoridated | Yes | No | –2 |
| 2. Frequent snacking with sugary foods, acidic foods, fermentable carb foods | Yes | No | 8 |
| 3. Sugary drinks including soft drinks, juice, sports drinks, medicinal syrups | Yes | No | 8 |
| 4. Tobacco use of any kind | Yes | No | 3 |
| 5. Excessive alcohol or recreational drug use | Yes | No | 8 |
| 6. Eating disorders | Yes | No | 5 |

Supplemental notes for dietary assessment

| Xerostomia assessment | | | |
|---|---|---|---|
| 1. Patient is aware of dry mouth or reduced saliva | Yes | No | 10 |
| 2. Medications taken that reduce salivary flow | Yes | No | 8 |
| 3. Medical conditions affecting salivary flow/content | Yes | No | 8 |
| 4. Saliva flow or content visibly abnormal | Yes | No | 10 |

Xerostomia assessments with scores of 8-10 indicate baseline salivary testing is required

| Patient clinical assessment<br>Clinical oral findings | | | |
|---|---|---|---|
| 1. Readily visible biofilm/plaque | Yes | No | 5 |
| 2. Visible cavitated lesions | Yes | No | 10 |
| 3. Interproximal enamel lesions or radiolucencies | Yes | No | 10 |
| 4. Visible white spots | Yes | No | 5 |
| 5. Visible brown spots or non cavitated caries lesions | Yes | No | 3 |
| 6. Deep pits or grooves | Yes | No | 5 |
| 7. Radiographic cavitated lesions | Yes | No | 10 |
| 8. Restorations with overhangs and/or margin concerns or open contacts | Yes | No | 3 |
| 9. Prosthesis ortho, fixed or removable | Yes | No | 3 |

Clinical assessments with scores of 10 indicate that baseline bacterial testing is required

Risk rating total =

| 10. Clinician's impression of patient's risk | Low | At risk | High |
|---|---|---|---|

This is clinician's impression of patient's risk if different than would be indicated by risk factors marked. Describe in box below.

**Patient compliance assessment**

**Patient's attitude and general assessment of patient's ability to comply for each of the following categories:**

| | | |
|---|---|---|
| Oral hygiene compliance patient limitations | Yes | No |
| Dietary recommendation patient limitations | Yes | No |
| Therapeutic homecare products limitations | Yes | No |
| Special needs health care (physical or mental compliance issues) | Yes | No |

This is clinician's assessment of any perceived limitations for the patient to comply with oral hygiene, dietary or using home care products. Could be lifestyle, physical or economic reasons. Describe in box below.

**Patient clinical in office tests indicated from risk score**

| | | | | |
|---|---|---|---|---|
| Cariscreen meter | Completed | At risk | Low risk | Reading _____ |
| GC America Saliva Check | Completed | At risk | Low risk | |
| GC America Strep Mutans | Completed | At risk | Low risk | |
| GC America Plaque Indicator | Completed | At risk | Low risk | |
| UNC Biologic Testing Lab | Completed | At risk | Low risk | |

Notes from oral findings, patient's attitude, office tests, special circumstances that would influence caries risk or management

B

Figure 2.33 *cont.*

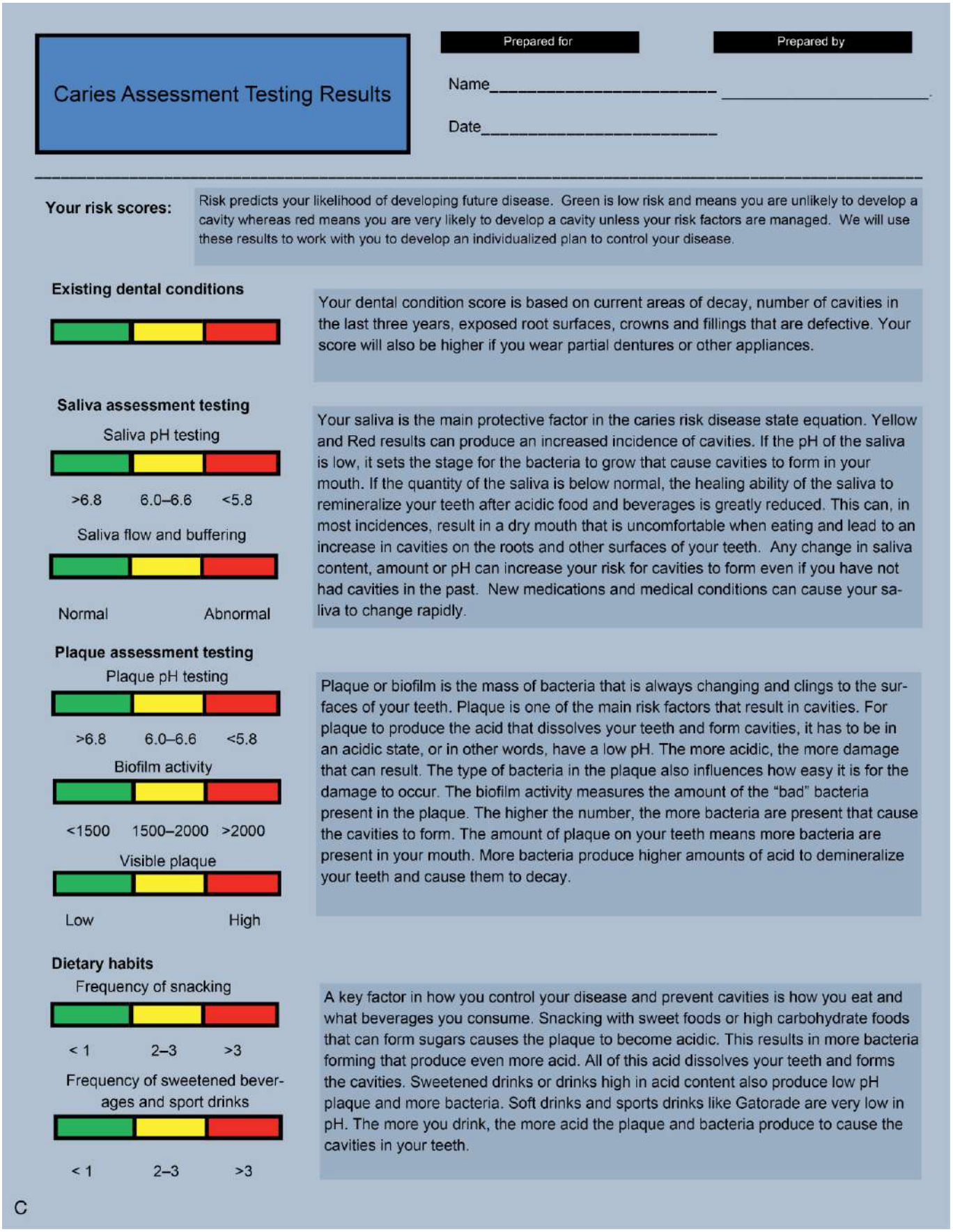



Figure 2.33 *cont.*

II. *Restore all root caries lesions with a fluoride-releasing material*. Resin-modified glass ionomer materials are preferred for definitive restorations primarily because they bond effectively to both enamel and dentin and they act as reservoirs for fluoride which can be re-released into the oral cavity.[42,43] They are effective as anticaries materials *only* if patients reload the material a *minimum of 3 times a day by brushing with fluoride-containing toothpaste* or by using other fluoride-containing products.[44] Alternatively, these same root caries lesions can be arrested by application of silver diamine fluoride (SDF).

III. Educating patients of the necessity for three exposures to fluoride per day and for reloading the fluoride-releasing materials can assist in motivating them to improved levels of compliance.[44]

In summary, many older patients are experiencing an epidemic of root caries, primarily as a result of the hyposalivatory effects of medications prescribed for systemic illnesses. Many root caries lesions occur in locations that make them difficult if not impossible to restore. The dental profession has a strong track record of prevention, and it is clear that with root caries, prevention is much better than restoration.

**Clinical Notes**

It has been estimated that 63% of the 200 most commonly prescribed medications have *dry mouth* as an adverse effect. It is the subsequent reduction in salivary flow rates and concomitant diminished buffering capacity resulting from use of these medications that is primarily responsible for the increase in root caries in older patients.

## Caries Risk Assessment and Management

Dental caries is a multifactorial medical disease process, and the caries lesions are the expression of that disease process involving the patient as a whole. It is critical to remember that clinicians treat the entire patient and not just individual teeth and caries lesions. Equally important in the management of caries as a disease entity is the ability to individualize caries treatment or interventions for each patient. To do this, the clinician must formulate a caries risk assessment profile that is based on the patient's current risk factors and risk indicators.

## Surgical versus Medical Model of Caries Management

Historically, dentistry has used a *surgical model* for the management of dental caries, which mainly involved the biomechanical removal of caries lesions and the restoration of the resultant *tooth preparation* to form and function with a restorative material. Management of caries disease by a surgical model consisted of waiting until cavitations were detected and treating the cavitations with restorations.

Eventually, it became apparent that dealing only with the end result of the disease and not addressing its etiology for each individual patient was not successful in controlling the caries disease process. Since surgical management alone was not successful, an evidence-based system has been developed using caries management strategies. These assessments look at each patient's unique set of pathologic factors and protective factors. The *medical model* looks at *individualized caries risk assessments* and uses this information to design treatment plans according to the risk assessment findings (Table 2.7).

**Table 2.7 Caries management based on the medical model**

| Primary etiology | Cariogenic biofilm resultant from a cariogenic diet |
|---|---|
| Symptoms | Demineralization lesions in teeth |
| Treatment, therapeutic | Improvement of host resistance by (1) biofilm modification, (2) elevating biofilm pH and (3) enhancing remineralization |
| Treatment, symptomatic | Remineralization/arrest of lesions, restoration of cavitated lesions |
| Posttreatment assessment, therapeutic | Reevaluation of aetiologic conditions and primary and secondary risk factors; and continuous management based on findings |
| Posttreatment assessment, symptomatic | Examination of teeth for new lesions, assessment of lesion activity, assessment of lesion progression |

## Caries Management by Risk Assessment (CAMBRA)

### Definitions

- A *risk factor* is defined as an environmental, behavioural or biologic factor that directly increases the probability that a disease will occur and, the absence or removal of which reduces the possibility of disease.[45,46]
- *Risk indicators* may refer to existing signs of the disease process, or signs that the disease process has occurred, *but are not part of the disease causal chain*. For example, existing caries lesions would be risk indicators, as they indicate a risk status, but are not per se part of the causal chain so they are not risk factors.
- *Caries management by risk assessment* (CAMBRA) represents a medical model of caries management that provides an individualized evaluation of a patient's pathologic factors and protective factors and assesses the patient's risk for developing future disease.

Risk factors are part of the causal chain of the disease process, or they expose the host to the causal chain; but

once the disease occurs, removal of a risk factor may not always result in the disease process being halted. Caries is a complex multifactorial disease with no single risk factor, which is highly predictive of future caries. However, caries risk assessment is necessary to inform what (if any) interventions are needed to lower the patient's caries risk, which is the ultimate goal of caries management in a *medical model*.[47,48] It is very important to spend the time with the patient to uncover all relevant risk factors currently present. Understanding and controlling risk factors and protective factors can be very important in the prevention of new caries lesions and to slow down or arrest the progression of existing caries lesions. Some risk factors and protective factors can be adjusted and *modified* by either the patient or the clinician, such as sucrose intake and fluoride exposure; other risk factors are *not modifiable*, such as such as hyposalivation as the result of a needed medication or an existing condition such as Sjögren syndrome. The caries balance model is based on minimizing pathologic factors while maximizing protective factors to attain a balance that favours no disease occurring, or health.

**CAMBRA** is evolving into the standard of care in caries management and represents a management philosophy that manages the caries disease process using a medical model. This process provides an individualized evaluation of a patient's pathologic factors and protective factors and assesses the patient's risk for developing future disease. The risk assessment is then used to develop an *individualized* evidence-based caries management plan that involves aspects of non-surgical therapeutics and dental surgical interventions. Both risk assessment and patient-centred interventions are based on the concept of caries balance as discussed earlier in this chapter (Fig. 2.1–2.4). The caries balance model is based on minimizing pathologic factors while maximizing protective factors to attain a balance that favours no disease occurring, or health. With the use of CAMBRA for patient management, mounting evidence suggests that early damage to teeth may be reversed and that the incidence of disease manifestations can be significantly reduced or prevented.

- CAMBRA is a concept that has gained interest in the United States and is evolving into the standard of care in caries management.[49–51]
- Additionally, the *International Caries Classification and Management System*[52–54] has endeavoured to merge caries detection and assessment with an individualized management plan.
- Another caries risk assessment method that has been validated for children is the *Cariogram*.[55,56]

**Clinical Notes**

- Caries risk assessments are specific for adults and adolescents older than 6 years and for children under 6 years of age.
- One of the most widely used systems is **CAMBRA** (**ca**ries **m**anagement **b**y **r**isk **a**ssessment).[47,49]

The ideal caries risk assessment model should be inexpensive, easy to use and have high degrees of accuracy in predictive value. The clinician must gather all appropriate data from both the interview with the patient and the clinical examination for caries detection to formulate an individualized caries risk assessment. This would involve the following three attributes:

I. Identification of risk factors
II. Identification of risk indicators
III. Assessment of caries lesion activity

## I. Identification of Risk Factors

The term *risk factor* is associated with the variables studied that have value for prediction purposes, which means that the risk factor is present *before* the disease occurs.

Risk factors can exist for a patient without the disease being expressed at the current point in time. The health history (Table 2.8) and clinical examination findings (Table 2.9) associated with increased caries risk are to be noted.

**Table 2.8 Health history factors associated with increased caries risk**

| History factor | Risk-increasing observations |
|---|---|
| Age | Childhood, adolescence, senescence |
| Fluoride exposure | No fluoride in public water supply<br>No fluoride toothpaste |
| Smoking | Risk increases with amount smoked |
| Alcohol | Risk increases with amount consumed |
| General health | Chronic illness and debilitation decreases ability to give self-care |
| Medication | Medications that reduce salivary flow |

**Table 2.9 Clinical examination findings associated with increased caries risk**

| Clinical examination | Risk-increasing findings |
|---|---|
| General appearance | Appearance: sick, obese or malnourished |
| Mental or physical disability | Inability or unwillingness to comply with dietary and oral hygiene instruction |
| Mucosal membranes | Dry, red, glossy mucosa, suggesting decreased salivary flow |
| Active caries lesions | Cavitation and softening of enamel and dentin; circumferential chalky opacity at gingival margins |
| Plaque biofilm | High plaque scores |
| Gingiva | Puffy, swollen and inflamed gingiva; bleeds easily |
| Existing restorations | Large numbers, indicating past high caries rate; poor quality, indicating increased habitat for cariogenic organisms |

Factors that have been identified as contributing to increased caries risk in patients ***above the age of 6 years*** include

*Dietary habits*. Quantity and frequency of sucrose intake[57]
*Caries*. Past caries experience is the *best predictor* of future caries activity[58]
*Fluoride*. Decreased fluoride exposure
*Habits*. Increased smoking, alcohol consumption
*General health*. Use of medications that cause xerostomia
*Age*. Children and older adults have increased risks
*Lower economic status*
*Lower educational attainment*

In addition to all the above risk factors for adults and adolescents, age-specific risk factors and indicators that should be considered for ***children under 6 years*** include

- Presence of active caries in the primary caregiver in the past year
- Feeding on demand for the past 1 year of age
- Bedtime bottle or sippy cup with anything other than water
- Non-supervised brushing
- Severe enamel hypoplasia

## II. Identification of Risk Indicators

- Risk indicators are *existing signs* of the disease process.
- They are clinical observations and detection modalities used to identify risk-level status.
- Risk indicators that classify a patient as 'at risk' or 'high risk' include
  - Visible cavitations in pits and fissures or in proximal surfaces of teeth
  - Brown spots, active white spots or cavitated lesions on free smooth surfaces
  - Restorations placed in the past 3 years

## III. Assessment of Caries Lesion Activity

One of the roles of caries risk assessments in caries management is to assist in determining the current caries lesion activity.

The diagnosis of caries lesions should include whether they are

I. **Active caries lesion**: progressing lesion
II. **Inactive caries lesion**: non-progressing lesion

**Clinical Notes**

With a positive shift in protective factors, change in oral hygiene and reduction of negative risk factors, it is possible for caries lesions to change from an active lesion to an inactive lesion. These inactive lesions may remineralize and not require operative intervention.

Caries risk assessment is performed by doing a combination of the following analyses as shown in Box 2.4.

**Box 2.4 Caries risk analysis**

**Primary analysis**
I. Dietary analysis
II. Dental clinical analysis (dental exam)
III. Salivary analysis
**Supplemental analysis**
I. Bacterial biofilm analysis

### *Primary Analysis*

#### *Dietary Analysis*

The following aspects are to be elicited while taking the patient history:

- *Quantity of sugar intake* in the form of fermentable carbohydrates and increased frequency of intake are conditions that increase risk for caries.
- The use of *candies* and *lozenges* frequently during the day or night increase the risk.
- *Acidic beverages*, including sport drinks, fruit juices and soft drinks, all contribute to increasing risk by providing energy to the acidogenic and aciduric bacteria and by influencing the pH of the biofilm to support cariogenic bacteria.
- *Frequency of snacking* and the frequency of consuming these foods and beverages all support an increase in biologic caries risk factors by modifying the biofilm to support a lower pH environment.

### *Dental clinical analysis (dental exam)*

The dental examination determines *risk indicators* more than risk factors. This is also important as many of the indicators are directly related to the current caries activity.

The *risk indicators* to be examined include

- Visible cavitated caries lesions
- White spots on teeth
- Brown spots on teeth

The risk *factors* to be examined include

- Visible plaque or biofilm
- Exposed root surfaces
- Deep pits or grooves
- Fixed, removable prosthesis or orthodontic appliances used
- Poor quality existing restoration with open contacts, open margins or overhangs

### *Salivary analysis*

The following tests are performed as part of salivary analysis:

I. Salivary flow rate
II. Salivary buffering capacity
III. Salivary pH

Patients with good saliva flow and adequate buffering capacity can still have caries. However, when patients have dry mouth symptoms and a salivary flow analysis leads to

a diagnosis of hyposalivation, the decreased salivary flow is a predictive risk factor for root caries in older patients with recession and for increased caries in general in other populations.

### *Supplemental Analysis*

### *Bacterial biofilm analysis*

I. *S. mutans or lactobacilli count in saliva or plaque*. Use of supplemental tests to analyse the bacterial component of the biofilm has little value to determine the patient's risk level.
II. *Adenosine triphosphate (ATP) activity of the biofilm bacteria*. The measurement of ATP activity of the biofilm bacteria as a surrogate measure of caries activity has also not been identified to be a strong predictor of risk.

**Clinical Notes**

- Dietary analysis and dental clinical analysis form the foundational basis of caries risk assessment.
- The caries risk predictive value for salivary tests is not fully demonstrated by strong evidence in all circumstances.
- Bacterial biofilm analysis may be useful for patient communication by providing insight into the specifics of the biofilm (bacterial types and resultant environment). However, the predictive value (for caries risk) of the evidence of the tests requires further validation by controlled research studies.

## Classification of Patients According to Degree of Risk

### Low Caries Risk

If a clinician finds no detectable or active caries lesions and minimum or no identifiable risk factors, that patient would be assigned a *low caries risk*. In this situation, in the current state of the patient's health, the protective factors for not developing caries lesions outweigh the risk factors that could lead to new caries lesions.

### At Risk

These are patients who might not be having any carious lesions right now but are *at risk* to develop caries in the near future if an imbalance occurs between the protective factors and risk factors present in the patient.

### High Risk

These are patients in whom the caries balance is tilted towards demineralization. These patients will in all probability develop caries in the near future. The strongest predictor for caries risk for patients in the *at-risk* and *high-risk* categories are the *number of caries lesions* being detected for the patient over the last 2–3 years along with past history of caries lesions in the patient's lifetime.[4]

The caries prevention or management program should be recommended on the basis of the level of caries risk (Table 2.10 and Box 2.5).

**Table 2.10 Suggested risk-based interventions for adults[a]**

| Caries risk category | Office-based interventions | Home-based interventions |
|---|---|---|
| High | 3-Month re-care examination and oral prophylaxis<br>Fluoride varnish at each re-care visit<br>Individualized oral hygiene instructions and use of specialized cleaning aids (e.g. powered toothbrush, Waterpik)<br>Dietary counselling<br>Bitewing radiographs every 6–12 months[b] | Brush with prescription fluoride dentifrice (e.g. 1/1%/5000 ppm NaF)<br>Use sugar substitutes (e.g. xylitol, erythritol)<br>Apply calcium-phosphate compounds (e.g. MI Paste)<br>On selected cases use agents to modulate a dysbiotic microbiome (e.g. xylitol gum or lozenge, chlorhexidine rinse). Use agents to encourage a healthy microbiome (probiotics, prebiotics, like arginine)<br>If xerostomic, increase salivary function (e.g. xylitol gum, rinses, oral moisturizers) |
| Moderate | 4–6-Month re-care examination and oral prophylaxis<br>Fluoride varnish at each recall<br>Reinforce proper oral hygiene<br>Dietary counselling | Brush with fluoride dentifrice (e.g. 1450 ppm fluoride)<br>OTC fluoride rinse (e.g. 0.05% NaF) |
| Low | 9–12-Month re-care examination and oral prophylaxis<br>Reinforce good oral hygiene | Brush with fluoride dentifrice |

*NaF*, Sodium fluoride; *OTC*, over the counter; *ppm*, parts per million.
[a]These are general guidelines and should be customized based on the specific patient's needs and on weight of individual risk factors uncovered with a caries risk assessment instrument.
[b]Data from US Department of Health and Human Services, Public Health Service, Food and Drug Administration; and American Dental Association, Council on Dental Benefit Programs, Council on Scientific Affairs: *The selection of patients for dental radiographic examinations*. Rev. ed. 2004. Available at www.ada.org/prof/resources/topics/radiography.asp. Accessed January 20, 2012.
*(Modified from Shugars DA, Bader JD:* MetLife quality resource guide: risk-based management of dental caries in adults*, ed 3, Bridgewater, NJ, 2009–2012, Metropolitan Life Insurance Co., p 6.)*

Box 2.5 **Sample preventive protocol for a high-risk patient with cavitated caries lesions**

I. A comprehensive oral and radiographic evaluation is conducted charting caries lesions, periodontal pocket probing, existing restorations and oral cancer exam; the medical and dental histories are reviewed. In this hypothetical patient, multiple caries lesions, poor oral hygiene and generalized marginal gingivitis are noted.

II. A *caries risk assessment* is completed with emphasis on discovering the risk factors that are contributing to the causative aspect of the caries problem and discovering the risk factors for the patient that are predictors of future caries risk. A discussion of these risk factors with the patient is essential for understanding of the caries disease process and the patient's role and the provider's role in controlling the disease.

III. Diagnosis and treatment planning procedures are completed and discussed with the patient.

IV. Nonoperative and operative treatments will be completed in three phases: (i) a control phase, (ii) a definitive treatment phase and (iii) a maintenance reassessment phase.

V. *Control phase* (2–4 weeks):
   i. Oral hygiene procedures are explained and reviewed at each patient visit. The frequency of the visits in this initial phase is determined by severity of the disease. This could be weekly or more frequent, depending on the provider's evaluation. When attempting behaviour modification, repetition is essential. Review of patient's current home-care techniques and frequency of home care are reviewed along with lifestyle issues that might impede compliance. Use of a *powered toothbrush* and an *oral irrigator* for possibly improving patient compliance and technique are discussed. Evaluation of the *patient's motivation* along with the person's mental and physical capacity to comply with recommendations for home care must be noted and considered by the provider. Specific recommendations are listed for the patient to use at home, and the patient agrees that this is practical for his or her life situation.
   ii. Prescription *fluoride toothpaste* is prescribed (5000 parts per million [ppm]) and the patient is instructed to brush with it 3 times per day and to use according to given instructions (do not rinse after use, only expectorate excess). Any products used for home care should be carefully reviewed for the pH of the product. Products with pH lower than 6 should be carefully considered whether their use would contribute to the pH shift of the biofilm to pathologic for caries lesion formation.
   iii. A *diet analysis* is completed, analysed and reviewed with the patient. Cariogenic foods and beverages are identified and alternatives suggested. Also acidic foods and beverages that are contributing to the pH shift to a lower oral pH environment are identified and discussed. Options for foods that have impact to raise the oral pH are suggested, such as foods rich in arginine. Again the patient's motivation and abilities to fit the necessary diet modifications into his or her lifestyle must be evaluated and discussed.
   iv. A *microbiologic survey* may be completed for patient motivation. This can be accomplished using either the adenosine triphosphate (ATP) chairside device or formal saliva samples to identify specific mutans streptococci (MS) and lactobacilli counts.
   v. A *saliva analysis* is conducted to determine stimulated flow rate, salivary pH and buffering capacity, and viscosity. Treatment protocols for hyposalivation are recommended for patients with an analysis that indicates deficiencies in the above areas. Patients with low salivary pH would need to use baking soda rinses during the day and use xylitol gum or other recommended products to raise pH levels, increase flow and increase buffering capacity.
   vi. Caries control is completed. This may involve use of *silver diamine fluoride (SDF)* for cavitated lesions or partial caries removal and interim restoration of all cavitated caries lesions (usually during one appointment and restoring with glass ionomer) or a combination of both. This is critical to reduce the bacterial load and prevent further progression on the lesion.
   vii. A prophylaxis is performed and *5% sodium fluoride (NaF) varnish* is applied to all teeth.
   viii. If SDF is used, the lesion should be reassessed 1–2 weeks after first application to determine if it has arrested. If the lesion has not arrested a new application of SDF is advised. Lesions that will not be restored need to have a new SDF application every 6 months.
   ix. The *microbiologic testing*, if done, may be repeated 2–4 weeks after initiation of treatment. A reduction in counts can be a great motivational tool for patients to continue with their modifications in behaviour (diet or oral hygiene). Diet analysis is followed up and reviewed again. Successes in diet modifications are positively reinforced. Shortcomings in diet modifications are discussed and options explored with the patient to rectify diet issues, where needed. Home-care regimens are reviewed again with patients and refined. It is important to listen to the patient and work to incorporate diet and home care into the patient's lifestyle and abilities to mentally and physically comply with the recommendations.

VI. *Definitive treatment phase*:
   i. The *glass ionomer provisional restorations are replaced* (usually by quadrants) with definitive restorations.
   ii. *Oral hygiene procedures* are reinforced at each visit. Flossing and brushing 3 times per day with prescription toothpaste is recommended.
   iii. Patients with root caries may benefit from *chewing xylitol chewing gum* with at least 1 g of xylitol per piece three to 6 times per day, preferably after meals and snacks.
   iv. Some patients with low salivary flow or low salivary buffering capacity may benefit from applying casein phosphopeptide–amorphous calcium phosphates (*CPP–ACP*) to teeth after brushing and flossing prior to retiring to bed.
   v. If reduced salivary flow rates are considered to be a major aetiologic factor, the patient should be instructed to *chew sugar-free mints several times a day* or use other recommended products for xerostomia treatment. Prescribing *pilocarpine* or other salivary stimulant should be considered.
   vi. When all definitive restorations are completed, the patient then enters the maintenance reassessment phase.

VII. *Maintenance reassessment phase*:

(*Continued*)

**Box 2.5 Sample preventive protocol for a high-risk patient with cavitated caries lesions (*cont.*)**

i. The patient should be *recalled every 3 months*. Oral hygiene and home-care procedures are reviewed and evaluated. Recommendations for improvement and modifications to home care are evaluated and discussed.
ii. Prophylaxis followed by *fluoride varnish* application is accomplished.
iii. Caries risk assessment is completed again; changes are noted in risk factors that have been controlled and those risk factors still listed as causative and predictive factors.
iv. *Diet analysis* and recommendations from previous visits are reviewed and evaluated.
v. Patient continues use of prescription *5000-ppm toothpaste*, *CPP–ACP paste* and *xylitol chewing gum* as advised. Any other recommendations to changes or additions to the product protocols are reviewed, discussed and implemented.
vi. Every 6 months, salivary evaluations are repeated. Microbiologic evaluations may also be repeated to keep patient motivation.
vii. *Bitewing radiographs* are taken on an annual basis or more frequently if new lesions continue to be detected.
viii. It is critical for the patient to understand that caries is a disease that is only controlled and not 'cured'. The protocol that is determined to be currently successful may have to be periodically reviewed, updated and changed. More importantly, the patient will be much like a patient with diabetes, requiring lifetime medication and therapy, diet control and lifestyle management for disease stability and will need to be dedicated to a lifetime of careful management of caries risk factors to keep the disease controlled.

## Caries Management and Protocols or Strategies for Prevention

The primary goals of a caries prevention program are

I. to modulate the biofilm into a nonpathogenic biofilm and
II. to create an environment conducive to remineralization.

Prevention should start with a consideration to maintain the patient with a noncariogenic biofilm. Although the general health of the patient, fluoride exposure history and function of the immune system and salivary glands have a significant impact on the patient's caries risk, the patient may have little control over these factors. The patient usually is capable of controlling other factors such as diet, oral hygiene, use of agents to modulate the biofilm and dental care (which may include use of sealants and restorations). This section presents a variety of factors that may have an impact on caries prevention.

A caries prevention and management program is a complex process involving multiple interrelated factors (Tables 2.11 and 2.12).

## Clinical Considerations in Caries Prevention

### I. General Health

The effectiveness of a patient's immune system depends on overall health status.

Patients undergoing radiation or chemotherapy treatment have significantly decreased immunocompetence and are at risk for increased caries.

Medically compromised patients should be examined for changes in the following: plaque index, salivary analysis, oral mucosa, gingiva and teeth.

Early signs of increased risk include increased plaque biofilm; puffy, bleeding gingiva; dry mouth with red, glossy mucosa and demineralization of teeth.

Acute and chronic systemic illnesses necessitate medications, which cause hyposalivation (Table 2.3) and is responsible for the dramatic increase in plaque biofilm.

### II. Diet

Dietary sucrose has two important detrimental effects on how plaque biofilm affects caries:

I. Frequent ingestion of foods containing sucrose provides a stronger potential for *colonization* by *mutans streptococci*, enhancing the caries potential of the biofilm.
II. Mature plaque biofilm exposed frequently to sucrose rapidly metabolizes it into organic acids, resulting in a profound and *prolonged decline in plaque pH*.

Rampant caries that is present primarily on *interproximal surfaces* may point more to diet as the main causative factor, whereas rampant caries in the cervical and interproximal areas may point to diet and hygiene as the causative factors.

For high-risk patients, a formal *diet analysis* should routinely be undertaken to identify cariogenic foods and beverages that are frequently ingested. This analysis may take place by asking the patient to recall everything that was ingested in the past 24 h (the '24-h diet interview'). Alternatively, a more detailed diet analysis may be facilitated by requesting the patient to record a diet diary, which consists usually of a 5- to 7-consecutive-day period with 2 days surveyed being weekend days as patients' diets frequently change considerably on weekends. A form should

**Table 2.11** Methods of caries treatment by the medical model

| Method and indications | Rationale | Techniques or material |
|---|---|---|
| **A. Limit cariogenic substrate** | | |
| **Indications**<br>Frequent sucrose exposure<br>Poor-quality diet | Reduce number, duration and intensity of acid attacks<br>Reduce selection pressure for MS | Diet diary<br>Eliminate sucrose from between-meal snacks<br>Substantially reduce or eliminate sucrose from meals |
| **B. Modify microflora** | | |
| **Indications**<br>High MS counts<br>High lactobacilli counts | Modify a dysbiotic biofilm and encourage a healthy biofilm<br>Reduce overall biofilm burden to improve gingival health | Bactericidal mouthrinse (chlorhexidine)<br>Topical fluoride treatments; probiotics and prebiotics |
| **C. Disorganize plaque biofilm** | | |
| **Indications**<br>High-plaque biofilm scores<br>Puffy red gingiva<br>High bleeding point score | Prevents plaque biofilm succession<br>Decreases plaque biofilm mass<br>Promotes buffering | Brushing<br>Flossing<br>Other oral hygiene aids as necessary (e.g. electric toothbrush) |
| **D. Modify tooth surface** | | |
| **Indications**<br>Non-cavitated lesions<br>Surface roughening | Increase resistance to demineralization<br>Decrease plaque biofilm retention | Systemic fluorides<br>Topical fluorides<br>Smooth surface |
| **E. Stimulate saliva flow** | | |
| **Indications**<br>Dry mouth with little saliva<br>Red mucosa<br>Medication that reduces salivary flow | Increases clearance of substrate and acids<br>Promotes buffering | Eat noncariogenic foods that require lots of chewing<br>Sugarless chewing gum<br>Medications to stimulate salivary flow<br>Use dry mouth topical agents, oral lubricants etc. |
| **F. Seal susceptible surfaces** | | |
| **Indications**<br>Moderate and high caries risk individuals<br>Teeth with susceptible anatomy (deep grooves)<br>Initial non-cavitated enamel lesions in high-risk patients (smooth-surface sealants) | Prevents colonization (infection) of pit-and-fissure system with cariogenic plaque biofilm<br>Inhibits progression of smooth-surface lesion | Use pit-and-fissure and smooth-surface resin sealants |
| **G. Restore active cavitated surfaces** | | |
| **Indications**<br>Cavitated lesions<br>Defective restorations | Eliminate nidus of cariogenic biofilm<br>Deny habitat for cariogenic biofilm reinfection | Apply SDF to caries lesions as indicated Restore all cavitated lesions<br>Correct all defects (e.g. marginal crevices, proximal overhangs) |

*MS*, Mutans streptococci.

**Table 2.12** Treatment strategies

| Examination findings | Non-restorative, therapeutic treatment | Restorative treatment | Follow-up[a] |
|---|---|---|---|
| Normal (no lesions) | None | None | 1-Year clinical examination |
| Hypocalcified enamel (developmental white spot) | None for nonhereditary lesions; hereditary lesions (dentinogenesis imperfecta) may require special management | Treatment is elective; esthetics (restore defects) | 1-Year clinical examination |
| Non-cavitated enamel lesions only; bitewing radiographs indicated (demineralized white spot) | Techniques A–E in Table 2.11, as indicated | Seal defective pits and fissures as indicated | 3 months; evaluate: oral flora, MS counts, progression of white spots, presence of cavitations |
| Possible cavitated lesions (active caries) and other non-cavitated lesions present; bitewing radiographs indicated | Techniques A–E in Table 2.11, as indicated | Techniques F and G (restorations, sealants) in Table 2.11 as indicated | 3 months; evaluate: oral flora, MS counts, progression of white spots, presence of new cavitations, pulpal response |
| Inactive caries; no active (new cavitations) or non-cavitated lesions | None | Treatment is elective; esthetics (restore defects) | 1-Year clinical examination |

*MS*, Mutans streptococci.
[a]These are only generalized follow-up times. Particular circumstances may dictate shorter or longer follow-up intervals, depending on presence of primary and secondary modifying risk factors (Fig. 2.1).

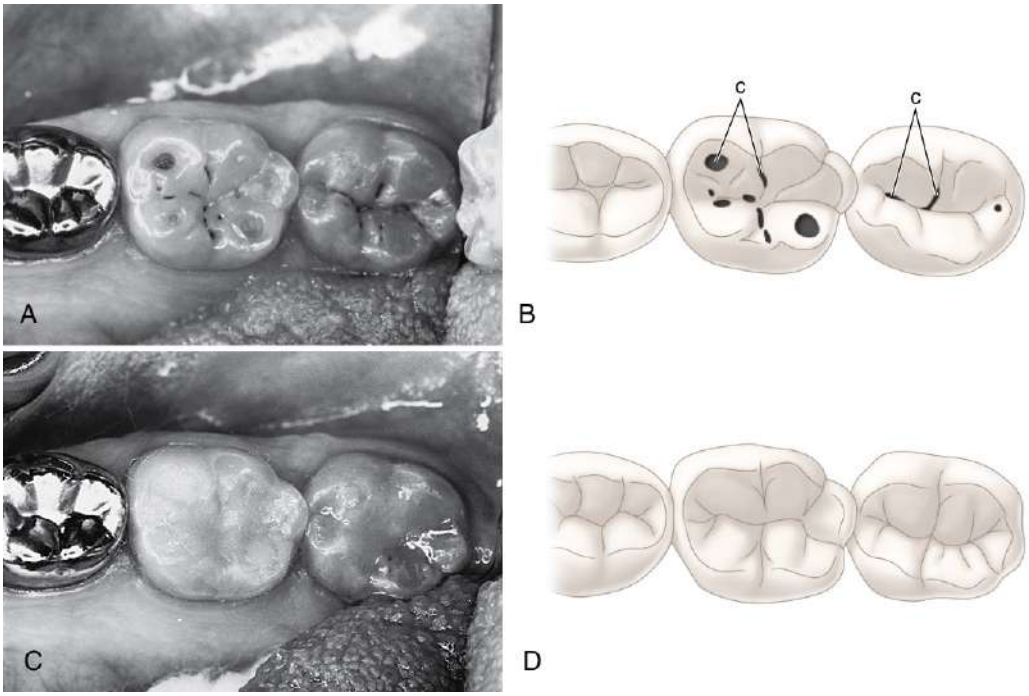



**Figure 2.34** (A and B) Photograph of the occlusal surfaces of the teeth and (C and D) after cessation of oral hygiene procedures, caries (c) rapidly developed in the exposed dentin and fissures on the occlusal surfaces. Caries was treated conservatively by excavation of softened dentin and restoration of the excavations and fissures with composite.

be provided that divides each day into six segments (breakfast, morning, lunch, afternoon, dinner and evening) and the patient should be instructed to write down everything ingested, including medications and the amount. The diary is then analysed by the dentist, and a discussion is held with the patient to suggest appropriate alternatives.[59]

**Clinical Notes**

Caries activity is most strongly stimulated by the frequency, rather than the quantity, of sucrose ingested.

## III. Oral Hygiene

Biofilm-free tooth surfaces do not decay (Fig. 2.34). Löe established supragingival plaque as the aetiologic agent of gingivitis.[60] Effective plaque control by oral hygiene measures using a fluoridated toothpaste results in resolution of the gingival inflammation and remineralization of any initially demineralized enamel surface.

The oral flora on the teeth of patients with good plaque control has a high percentage of *S. sanguis* or *S. mitis* and is much less cariogenic than older, mature plaque communities, which have a significantly higher percentage of *S. mutans*.

Krasse showed that a combination of *oral hygiene* and *diet counselling* is effective in children.[61] In this classic study, children in two schools were monitored for *Lactobacillus* levels. The children in one school were given feedback about the results of the studies and proper preventive oral hygiene and dietary instruction. After 18 months, the children in the school receiving preventive counselling had an average of 3.3 new restorations, whereas the control school children, who received no counselling, averaged 8.2 new restorations. This study is an excellent demonstration that *good oral hygiene* and *dietary improvements* can be effective when using microbiologic testing as a motivational tool.

Other adjuncts to regular brushing and flossing is the regular use of electric toothbrushes and oral irrigation devices.[62,63]

**Clinical Notes**

Daily removal of biofilm by tooth brushing with a fluoridated toothpaste and dental flossing is the best patient-based measure for preventing caries and periodontal disease.

## IV. Fluorides

The highest level of evidence for caries prevention and reduction supports the exposure of teeth to fluoride. Fluoride in trace amounts increases the resistance of tooth structure to demineralization and is particularly important for caries prevention[64] (Fig. 2.35).

When available to humans, fluoride produces spectacular decreases in caries incidence. The availability of fluoride for caries risk reduction has primarily been achieved through the fluoridation of community water systems. Fluoride exposure may occur by means of diet, toothpastes, mouth rinses and professional topical applications.

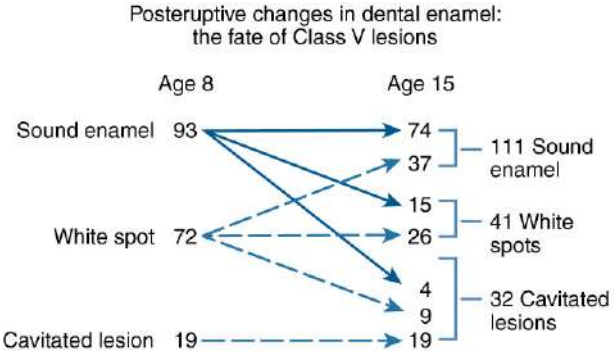



**Figure 2.35** White spot lesions of enamel may remineralize, remain unchanged or progress to cavitated lesions. In this study, done in a community with a fluoridated public water supply, only 9 of 72 non-cavitated lesions became cavitated. More than half of non-cavitated lesions (37 of 72) regressed to become indistinguishable from normal enamel.
(Redrawn from Backer DO: Posteruptive changes in dental enamel, *J Dent Res* 45:503–511, 1966.)

The optimal fluoride level for public water systems is *0.7 ppm of fluoride*.[65] At 0.1 ppm (parts per million) and less, the preventive effect is lost, while excessive fluoride exposure (≥10 ppm) results in fluorosis, which initially causes enamel to become white but may eventually cause a brownish discolouration, a condition termed *mottled enamel*.

## Mechanism of Action

Fluoride exerts its anticaries effect by three different mechanisms:

I. First, the presence of fluoride ion greatly enhances the precipitation into tooth structure of *fluorapatite* from calcium and phosphate ions present in saliva. This insoluble precipitate replaces the soluble salts containing manganese and carbonate that were lost because of bacteria-mediated demineralization. This exchange process results in the enamel becoming more acid resistant (Fig. 2.36).
II. Second, initial caries lesions are remineralized by the same process.
III. Third, fluoride has antimicrobial activity. In low concentrations, fluoride ion inhibits the enzymatic production of glucosyltransferase. Glucosyltransferase promotes glucose to form extracellular polysaccharides, which increases bacterial adhesion. Intracellular polysaccharide formation also is inhibited, preventing storage of carbohydrates by limiting microbial metabolism between the host's meals. In high concentrations (12,000 ppm) used in topical fluoride treatments, fluoride ion is directly toxic to some oral microorganisms, including MS. Suppression of growth of MS after a single topical fluoride treatment may last several weeks.[66] It is possible to lengthen this suppression greatly by a change in dietary habits (especially eliminating sucrose) and by the patient's conscientious application of a good oral hygiene program.

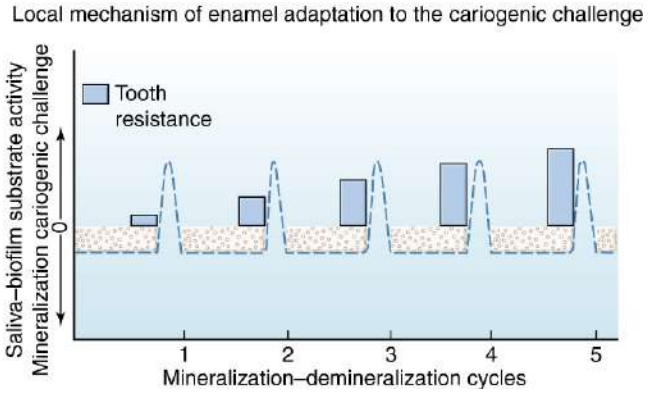



**Figure 2.36** Diagrammatic representation of enamel adaptation reaction. Enamel interacts with its fluid environment in periods of undersaturation and supersaturation presented here as periodic cycles. Undersaturation periods dissolve most soluble mineral at the site of cariogenic attack, whereas periods of supersaturation deposit most insoluble minerals if their ionic components are present in immediate fluid environment. As a result, under favourable conditions of remineralization, each cycle could lead towards higher enamel resistance to a subsequent challenge.
(Redrawn from Koulouirides T: In Menaker L, editor: *The biologic basis of dental caries*, New York, 1980, Harper & Row.)

## Fluoride Treatment Protocols

All of the methods for fluoride exposure (Table 2.13) are effective to some degree. The clinician's goal is to choose the most effective combination for each patient. This choice must be based on the patient's age, caries experience, general health and oral hygiene.

### *Public water supply fluoridation*

Ideal concentration for public water supply fluoridation is 0.7 ppm.

Children with developing permanent teeth benefit the most by this method. In regions without adequate fluoride in the water supply, dietary supplementation of fluoride is indicated for children and sometimes for adults. The amount of fluoride supplement must be determined individually. This is of particular importance in rural areas with individual wells because the fluoride content of well water can vary greatly within short distances.

### *Professional application of fluoride*

#### *Sodium fluoride varnish (5%)*

Fluoride varnishes are professionally applied and provide a high uptake of the fluoride ion into enamel and are widely accepted as the *vehicle of choice and the most cost-effective means for fluoride delivery to young adults and older adults alike*. These varnishes are effective bactericidal and caries-preventive agents.[67–75]

*Indications:*

- For high-risk patients of caries, fluoride varnish should be applied every 3 months.
- For moderate-risk patients, application every 6 months is indicated.

**Table 2.13** Fluoride treatment modalities[a]

| Route | Method of delivery | Concentration (ppm) | Caries reduction (%) |
|---|---|---|---|
| Systemic topical | Public water supply | 0.7 | 50–60 |
| | ***Self-application*** | | |
| | Low-dose/high-frequency rinses (0.05% sodium fluoride daily) | 225 | 30–40 |
| | High-potency–low-frequency rinses (0.2% sodium fluoride weekly) | 900 | 30–40 after 2 years |
| | Fluoridated dentifrices (daily) | 1000–1450 | 20 |
| | Prescription-strength fluoridated dentifrices (daily) | 4950 | 32 |
| | ***Professional application*** | | |
| | Sodium fluoride varnish (5%) | 22,500 | 30 |
| | Acidulated phosphate fluoride gel (1.23%) annually or semiannually | 12,300 | 40–50 |
| | Sodium fluoride solution (2%) | 20,000 | 40–50 |
| | Stannous fluoride solution (8%) | 80,000 | 40–50 |
| | SDF (38%) | 44,800 | ~96.1 for caries arrest<br>~70.3 for caries prevention |

*ppm*, Parts per million; *SDF*, silver diamine fluoride.
[a]Caries reduction estimates for topically administered fluorides indicate their effectiveness when used individually. When they are combined with systemic fluoride treatment, they can provide some additional caries protection.

- Fluoride varnish is not indicated for low-risk patients.

*Mechanism of action*: The fluoride varnish deposits large amounts of fluoride on an enamel surface, especially on a demineralized enamel surface. Calcium fluoride precipitates on the surface, and often fluorapatite is formed. The high concentration of surface fluoride also may provide a reservoir for fluoride, which promotes remineralization.

*Method of application*:

- Only tooth brushing, rather than prophylaxis, is necessary before application.
- Because the fluoride varnish sets when contacting moisture, thorough isolation of the area is not required.
- When applying fluoride varnish, the clinician dries off saliva from teeth and applies a thin layer of fluoride varnish directly onto teeth.

The main disadvantage of fluoride varnish is that a temporary change in tooth colour may occur. Patients should avoid eating for several hours and avoid brushing until the next morning after the varnish has been applied.

**Clinical Notes**

Current evidence indicates that fluoride varnishes with the concentration of 5% sodium fluoride are the most efficacious of all topically applied fluoride products.[69,76]

### *Acidulated phosphate fluoride gel (1.23%)*

Topical application of fluoride should be done periodically for children and adults who are at high risk for caries development.

Flossing followed by tooth brushing is recommended before the application of topical fluorides.

Acidulated phosphate fluoride gel (APF gel) is effective, but the potential risk of swallowing excessive amounts of fluoride exists, particularly in young children. APF gel is available in thixotropic gels and has a long shelf life.

### *Stannous fluoride gel (8% F)*

Although the tin ion in stannous fluoride may be responsible for staining the teeth, it may be beneficial in arresting root caries.

It has a bitter, metallic taste; may burn the mucosa and has a short shelf life.

### *Silver diamine fluoride (SDF 38%)*

SDF 38% is a topical solution used as a caries-arresting and anti-hypersensitivity agent.[77,78]

*Mechanism of action*. Both silver and fluoride play active roles in their mechanisms of arresting caries development and treatment of tooth hypersensitivity.[79–81] Silver has an antibacterial action that slows demineralization and enhances remineralization.[82]

*Indications:*

- SDF, due to its ease of use, has been recommended to *arrest large cavitated lesions*, in groups of patients that either *do not have access to traditional restorative care* or to whom delivery of the standard treatment is challenging.
- Some researchers have studied the use of SDF on *root caries* and have found it to be effective at preventing root caries lesions.[83]

*Method of application*: Application of SDF usually only requires removal of the biofilm and application of the product with a microbrush for 3 min, then either rinsing the area with water or covering the lesion with fluoride varnish.

Gingival tissues and lips should be covered with petroleum jelly to avoid staining. The main drawback of SDF

is that with the precipitation of silver, the carious dentin becomes stained black.

This approach may give patients who would otherwise be unable to receive treatment a low-cost alternative to arrest caries lesions and preserve their dentition (Fig. 2.37). However, the anticaries effect of SDF may be reduced over time. It has been reported that up to 50% of arrested lesions were 'reactivated' by 24 months after one application of SDF.[84] The authors reported that this was likely due to the biofilm-retaining nature of these cavitated lesions. Thus, there is a rationale for restoring these arrested lesions from both the caries management approach and for function and esthetics.

## Self-administered fluoride

### *Fluoride mouth rinses*

Self-administered fluoride rinses have an additive effect (about 20% reduction) when used in conjunction with topical or systemic fluoride treatment. Fluoride rinses are indicated in high-risk patients and patients exhibiting a recent increase in caries activity. Two varieties of fluoride rinses have similar effectiveness:

I. *High dose (0.2% F)—low-frequency rinses*. They are best used in supervised weekly rinsing programs based in public schools.
II. *Low dose (0.05% F)—high-frequency rinses*. They are best used by individual patients at home. A high-risk or caries-active patient should be advised to use the rinse daily. The optimal application time is in the evening. The rinse should be forced between teeth many times and then expectorated, not swallowed. Eating and drinking should be avoided after the rinse.

### *Fluoridated dentifrices*

Routine use of over-the-counter fluoride containing dentifrice 3 times per day is recommended for all patients. Two types of fluoride dentifrices are available:

I. *1000–1450 ppm fluoridated dentifrices*. This is recommended for low-risk patients and they generally contain 0.32% sodium fluoride (1450 ppm).
II. *5000 ppm fluoridated dentifrices*. For moderate-risk and high-risk patients *6 years or older*, prescription dentifrices containing higher concentrations of fluoride are recommended. These products typically contain 1.1% sodium fluoride (5000 ppm) and can be safely used up to 3 times per day in this age group.[85]

For most benefit, patients should be instructed to not rinse after brushing and avoid eating or drinking for 30 min after use.




**Figure 2.37** (A) Young high caries risk patient. Tooth no. 17 with extensive caries lesion. Tooth was asymptomatic and roots were still developing. (B) Unsupported enamel was removed, soft caries was left on the pulpal floor to avoid exposing the pulp. (C) SDF was applied and left for 3 min. Darkened dentin can be observed. Tooth was restored with amalgam and remained asymptomatic. (D) Tooth with PFM crown and secondary root caries on distal surface with difficult access for restoration. (E) SDF applied and darkened dentin can be observed.
(Part C, Courtesy: Dr. Nguyen Ngoc, resident Graduate Operative Program at UNC; Part E, Courtesy: Dr. Epure, resident Graduate Operative Dentistry Program at UNC.)

## V. Sealants (Pit-and-Fissure Sealants)

### Concept

Pits and fissures typically result from an incomplete coalescence of enamel and are particularly prone to caries.

Although fluoride treatments are most effective in preventing smooth-surface caries, they are less effective in preventing pit-and-fissure caries. These areas can be sealed with a low-viscosity fluid resin after acid etching. Based on the available scientific literature, pit-and-fissure sealants are appropriate for prevention of caries in susceptible teeth and, within limits, for arresting initial caries lesions.[86–91] There is mounting evidence that sealing initial lesions is an effective means to arrest dental caries.[92]

### Mechanism of Action

I. First, sealants mechanically fill pits and fissures with a resin-based polymer. They *eliminate the geometry* that harbours bacteria and to prevent nutrients reaching bacteria in the base of the pit or fissure. The principal feature of a sealant required for success is *adequate retention*. The sealant blocks bacterial accumulation occurring in otherwise non self-cleansing locations (Fig. 2.38). Complete penetration of the sealant is not absolutely critical. It is possible to occlude only the neck region of a fissure and produce clinically acceptable results (Fig. 2.39).

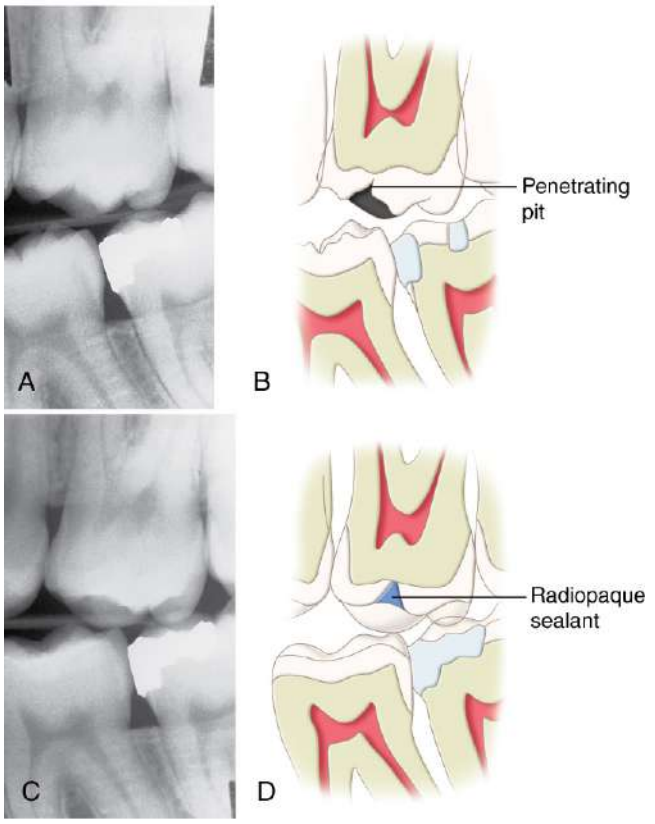



**Figure 2.38** (A and B) Radiograph of a maxillary first molar with a deep central fossa pit that appears to penetrate to the dentin and (C and D) the central pit was sealed with a highly filled, radiopaque sealant. The sealant is readily visible on the radiograph.

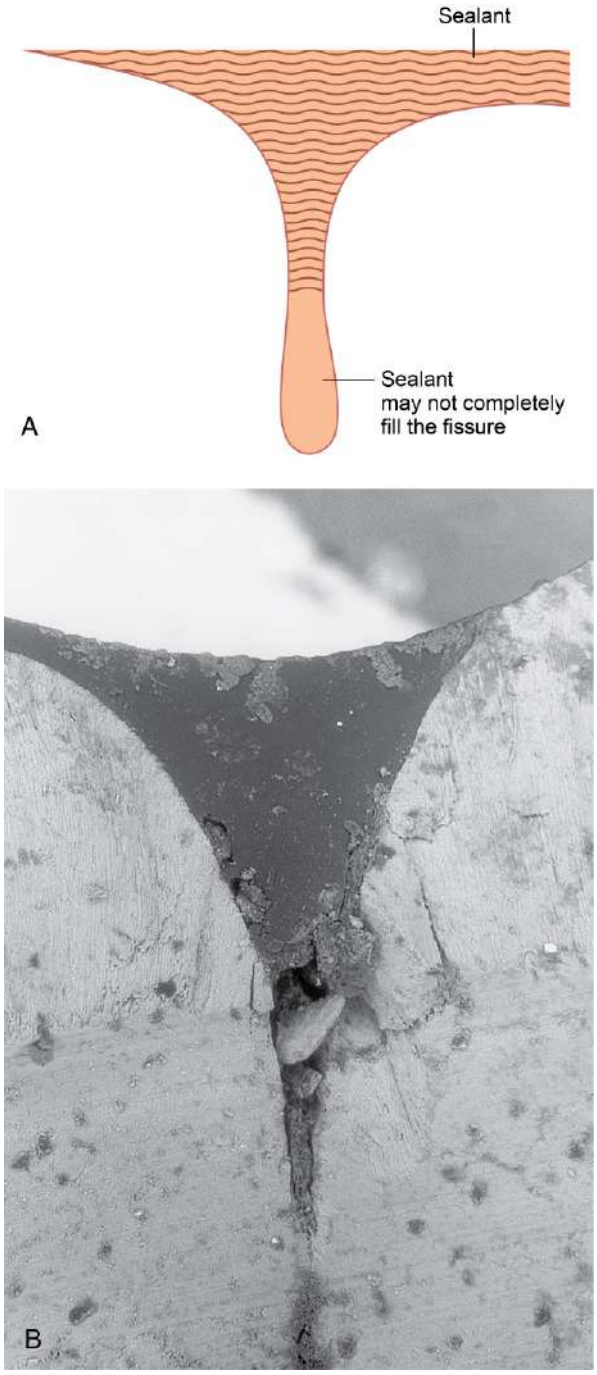



**Figure 2.39** Fissure sealants. (A) Schematic view of idealized fissure after sealing and after loss of excess sealant and (B) scanning electron microscopy cross-section of sealed fissure. The sealant does not penetrate into the entire fissure. Excess sealant on occlusal surface has been mostly worn away to the boundary of the self-cleansing zone.
(Part B, Courtesy: S. Mitchell, School of Dentistry, University of North Carolina, Chapel Hill, NC.)

**Table 2.14 Indications for use of sealants**

| Criteria | Seal | Do not seal |
|---|---|---|
| Tooth age | Recently erupted teeth | Teeth that have remained caries-free for ≥4 years; staining is usually present in pits and fissures |
| Tooth type | Molars | Premolars except when patient is caries-active |
| Occlusal morphology grooves | Deep, retentive, narrow pits and wide fissures | Well-coalesced fossae and easily cleaned grooves |
| Recent caries activity caries-free | Teeth showing signs of softening or opacity in pit or fissure, Occlusal or smooth surface proximal cavitated lesions on tooth to be sealed , Proximal cavitated lesion on tooth to be sealed; lesions on other teeth. | Teeth that have remained caries-free for ≥4 years |
| Availability of other deficient; preventive measures recommended | Patient's water supply is fluoride deficient and patient is not cooperating in caries-preventive program (restoration of pits and fissures is preferred) | Patient receiving appropriate systemic or topical fluoride therapy or both |

II. Second, sealants render the surface of the tooth, where the pits and fissures are located, easier to clean by toothbrushing and mastication.

## Indications

- Sealants are indicated, regardless of the patient's age, for either preventive or therapeutic uses, depending on the patient's caries risk, tooth morphology or presence of incipient enamel caries.
- Sealants should be used primarily for the prevention of caries rather than for the treatment of existing caries lesions.
- Only the following sites currently are recommended for treatment with pit-and-fissure sealants[21]:
  - Caries-free pits and fissures
  - Incipient lesions in enamel not extending to the DEJ
- In children, sealants are most effective when applied to the pits and fissures of permanent posterior teeth immediately on eruption of the clinical crowns, provided proper isolation can be achieved.
- Adults also can benefit from the use of sealants if the individual experiences an increase in caries susceptibility because of a change in diet, lack of adequate saliva or a particular medical condition (Table 2.14).

### Clinical considerations before placing a sealant

- In assessing the occlusal surfaces of posterior teeth as potential candidates for a sealant procedure, the primary decision is based on *whether a cavitated lesion exists*. This decision is based primarily on radiographic and clinical examinations, although other adjunctive technologies for occlusal caries detection are available. Explorers must be used judiciously in the detection of caries, as a sharp explorer tine may cause a cavitation. The clinical examination should be primarily focused on visual assessments of a clean, dry tooth surface preferably under adequate light and magnification.
- A *bitewing radiograph* should be obtained and evaluated before sealant placement to ensure that no dentinal or proximal caries lesion is evident.
- When no cavitated caries lesion is diagnosed, the treatment decision is either to pursue no treatment or to place a pit-and-fissure sealant, particularly if the surface is at high risk for future caries. If a small caries lesion is detected, and the adjacent grooves and pits, although sound at the present time, are at risk for caries in the future, a preventive resin restoration (Chapter 19) or conservative composite restoration of the active cavitated lesion is recommended.
- The *patient's caries risk* also should be a factor when considering treatment options.

**Clinical Notes**

- Explorers must be used judiciously in the detection of caries, as a sharp explorer tine may cause a cavitation.

## Advantages

They provide an *alternative* to tooth preparation and restoration techniques for elimination of caries prone pits and fissures on occlusal surfaces.

## Materials Employed

Most currently used sealant materials are as follows:

I. Light-activated urethane dimethacrylate resin.
II. Light-activated bisphenol A–glycidyl methacrylate resin.

## Clinical Technique

The following discussion deals with a fissure present on a mandibular first permanent molar (Fig. 2.40A).

**Step 1**. The tooth is isolated with a rubber dam (or another effective isolation method such as cotton rolls or isolite). If proper isolation cannot be obtained, the bond of the sealant material to the tooth surface will be




**Figure 2.40** Steps in application of pit-and-fissure sealant. (A) After isolation and thorough cleaning of the occlusal surface to be sealed; (B) after acid-etching, rinsing and drying and (C) with sealant applied.

compromised, resulting in either loss of the sealant or caries under the inadequately bonded sealant.

**Step 2**. The area is cleaned with a slurry of pumice on a bristle brush and the tooth is rinsed thoroughly, while the explorer tip is used carefully to remove residual pumice or additional debris.

**Step 3**. The tooth surface is dried, and etched with *37% phosphoric acid gel for 15–30 s*. Properly acid-etched enamel surface has a lightly frosted (matte) appearance (Fig. 2.40B). Any brown stains that originally may have been in the pits and fissures still may be present and should be allowed to remain. *Fluoride-rich, acid-resistant enamel* may need to be etched for a longer time.

**Step 4**. The sealant material is then applied and gently teased into place, to avoid entrapping air and it should overfill slightly all pits and fissures, not extending onto unetched surfaces (Fig. 2.40C). If too much sealant is applied, excess may be removed with a dry microbrush prior to light activation.

**Step 5**. The sealant is light-activated for 15 s.

**Step 6**. After light activation and removal of the rubber dam, if used, the occlusion is evaluated by using articulating paper. If necessary, a round carbide finishing bur or white stone is used to remove any excess sealant. The surface usually does not require further polishing.

**Clinical Notes**

Based on the available scientific literature, pit-and-fissure sealants are appropriate for prevention of caries in susceptible teeth and, within limits, for arresting initial caries lesions (Fig. 2.41).

### *Resin infiltrants*

Another strategy recently introduced is the use of extremely low-viscosity resin sealants for the infiltration of white-spot caries lesions on smooth surfaces[93] (Fig. 2.42). In situ studies demonstrate the ability of resin sealants, also called *infiltrants*, to prevent further demineralization under cariogenic conditions.[94]

Resin infiltrants have reportedly been used in free (i.e. facial and lingual) as well as in interproximal enamel surfaces. However, similar to the sealant technique used in pit-and-fissure areas, the technique requires careful attention to detail. A practice-based randomized clinical research trial has indicated that infiltration was more effective at preventing lesion progression than instructions on oral hygiene, flossing and fluoride use.[95]

## VI. Saliva

Saliva is nature's first line of defence against dental caries. Saliva works by the following mechanisms.

- Diluting the acid produced in plaque biofilm
- Washing the acid away (swallowing)
- Buffering the produced acid (biocarbonate + phosphate)
- Assisting in remineralization (calcium + phosphate)

Normal aging does not result in reduced salivary flow rates. However, many older patients have compromised salivary flow rates as a result of medications taken for systemic conditions.[96]

One important strategy for the prevention of caries in such patients is to increase the salivary flow rate and, by this, the concomitant buffering capacity. For these patients, gathering initial baseline data on salivary flow rates is critical. Commercial kits are available for the assessment of salivary flow. These kits provide data on simulated flow rates, the pH of saliva and the level of buffering capacity. If saliva samples are sent for microbiologic testing, specific MS and lactobacilli counts may be determined. Specific strategies for improving flow rates and reducing bacterial counts may then be initiated. Subsequent retesting is necessary to identify the relative efficacy of the strategies.

When attempting to improve salivary flow rates, a consultation with the patient's physician may be in order. It may be possible that a less xerogenic drug be prescribed or the dose of a current drug reduced. Changing the time of taking the medication(s) may be useful. Prescribing salivary stimulants such as *pilocarpine* or *cevimeline* may be very beneficial in patients with functioning salivary glands but who have medication-induced hyposalivation.

Other strategies to improve salivary flow rates include increased water intake, use of sugar-free candies/mints several times a day and/or use of xylitol chewing gum.

**Clinical Notes**

- When normal salivary flow rates are compromised, patients are usually at high risk for developing caries.

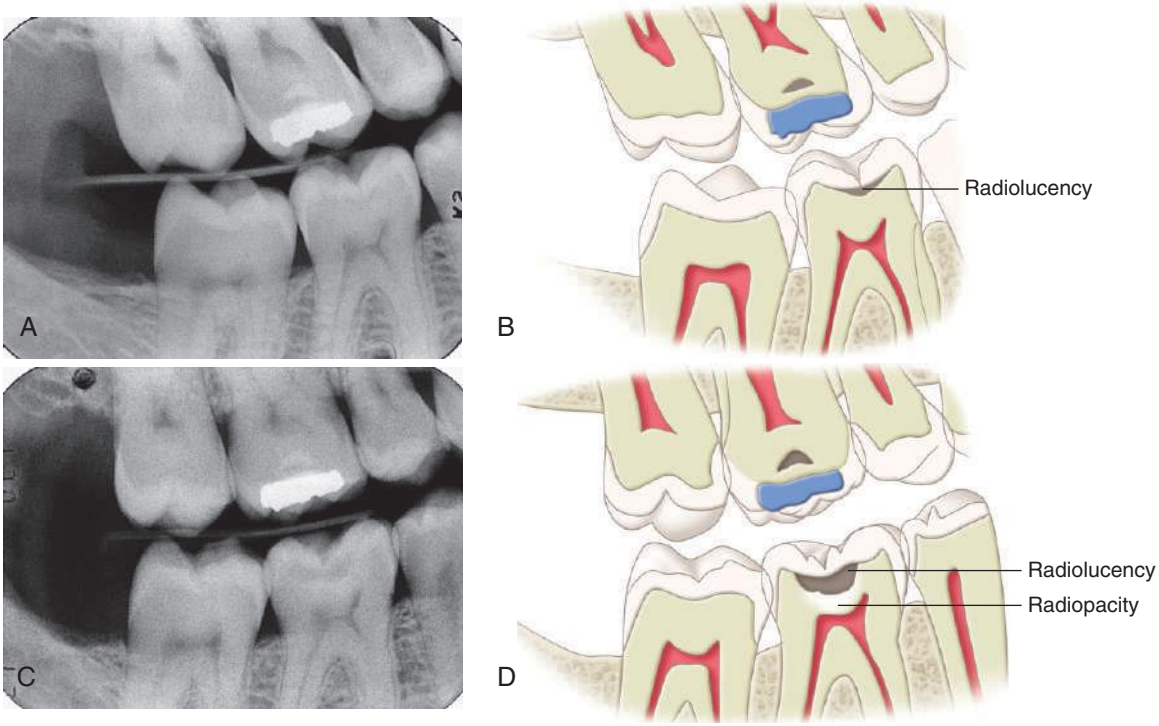



**Figure 2.41** (A–D) Progression of pit-and-fissure caries. (A) The mandibular right first molar (tooth no. 30) was sealed. Note radiolucent areas under the occlusal enamel in (A) and (B). The sealant failed, and caries progressed slowly during the next 5 years; the only symptom was occasional biting-force pain. This underscores the need to monitor the sealant for failures. (C and D) Note the extensive radiolucency under the enamel and an area of increased radiopacity below the lesion, suggesting sclerosis.




**Figure 2.41 (*cont.*)** (E–L) Conversely, a well-sealed surface can prevent lesion progression. (E and F) This maxillary first molar (tooth no. 14) with a shadow and a radiolucency indicative of dental caries was sealed as part of a clinical study. (G–J) The tooth was followed clinically and by bitewing radiographs over the next 8 years. (K and L) No progression of the lesion was observed.
(Courtesy: Dr. Azam Bakhshandeh, Department of Odontology, Faculty of Health and Medical Sciences, University of Copenhagen, Department of Cariology, Endodontics, and Pediatric Dentistry, Copenhagen, Denmark.)

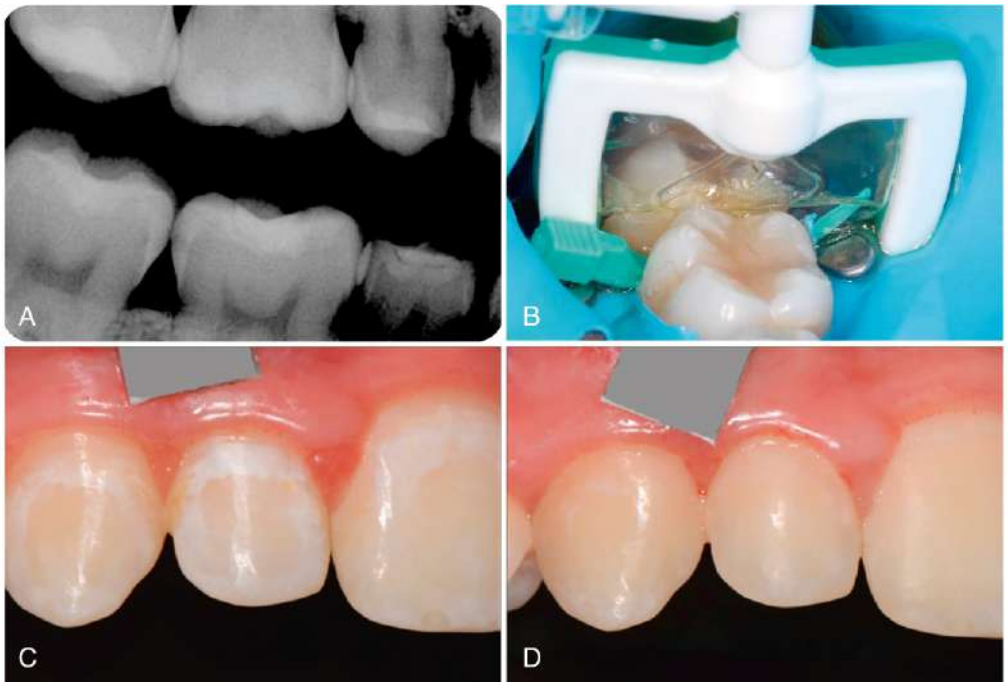



**Figure 2.42** (A) Radiographic image of first mandibular molar with a proximal lesion on distal surface. (B) Tooth was infiltrated according to manufacturer's instructions (Icon, DMG America). (C and D) Clinical images of smooth-surface lesions on anterior teeth before and after infiltration, with Icon.
(A, Courtesy Dr. Andrea Cortes, Operative Dentistry Specialization Program, El Bosque University. B, Courtesy Dr. Olga Lucia Zarta (chair) and Dr. Ainimsay Benitez (resident), Operative Dentistry Specialization Program, Dental School, El Bosque University, Bogotá, Colombia. C and D, Reprinted with permission from Meyer-Lueckel H, Paris, *Zahnmedizin up2date* 267–290, 2017.)

## VII. Chemical Agents

A variety of antimicrobial agents are available to help prevent caries (Table 2.15). These agents have been shown to reduce MS levels but this alone has not been shown to modify caries outcomes.[97] Two differing strategies have been suggested for reducing bacterial counts.

### Chlorhexidine

The traditional approach is the use of chlorhexidine (CHX) mouthwash, varnish or both, along with prescription

**Table 2.15** Antimicrobial agents

| | Mechanism of action | Spectrum of antibacterial activity | Persistence in mouth | Adverse effects |
|---|---|---|---|---|
| **Antibiotics** | | | | |
| Vancomycin | Blocks cell wall synthesis | Narrow | Short | Increases Gram-negative flora |
| Kanamycin | Blocks protein synthesis | Broad | Short | Can increase caries activity |
| Actinobolin | Blocks protein synthesis | Streptococci | Long | Unknown |
| **Bis Biguanides** | | | | |
| Alexidine | Antiseptic; prevents bacterial adherence | Broad | Long | Bitter taste; stains teeth and tongue brown; mucosal irritation |
| Chlorhexidine | Antiseptic; prevents bacterial adherence | Broad | Long | Bitter taste; stains teeth and tongue brown; mucosal irritation |
| **Halogens** | | | | |
| Iodine | Bactericidal | Broad | Short | Metallic taste |
| Fluoride | 1–10 parts per million (ppm) reduces acid production; 250 ppm bacteriostatic; 1000 ppm bactericidal | Broad | Long | Increases enamel resistance to caries attack; fluorosis in developing teeth with chronic high doses |

fluoride toothpaste. When using this approach, it may be prudent to use toothpaste free from sodium laurel sulphate (SLS), which causes the foaming action in dentifrices. Although data are equivocal, evidence demonstrates that SLS reduces the ability of CHX to reduce biofilm formation.[98]

*Mechanism of action*. CHX gluconate (0.12%) solution is effective in its ability to chelate cations and, as a result, disrupts cell adhesion, cell membrane function and the ability of MS to uptake glucose and produce glucans and its metabolism resulting in a decrease in MS counts.

CX was first as a rinse and was initially used for periodontal therapy. It was prescribed as a 0.12% rinse to high-risk patients for short-term use. It is also available as a varnish, and the most effective mode of varnish use is as a professionally applied material.[99] Emilson concluded that CHX varnishes provide effective reduction in MS, although recent evidence is not as conclusive in favour of a CHX rinse.[100,101]

**Clinical Notes**

Despite some favourable evidence, because there is only a weak correlation with a reduction in caries, CHX currently has very limited application as a regularly used caries management antimicrobial agent.

## Xylitol

Xylitol is a natural five-carbon sugar obtained from birch trees. It is usually recommended that patients chew two pieces of xylitol gum containing a total of 1 g xylitol 3–6 min after eating or snacking.

*Mechanism of action*.

I. MS cannot ferment (metabolize) xylitol, so no acid is produced. Over time, xylitol reduces the number of MS in the biofilm.
II. Chewing any sugar-free gum after meals reduces the acidogenicity of the biofilm because chewing stimulates salivary flow, which improves the buffering of the pH drop that occurs after eating.[42] Reductions in caries rates are greater however when xylitol is used as the sugar substitute.[43]

Although there is some suggestion that xylitol may reduce caries rate,[102] enhance remineralization and help arrest dentinal caries,[38,103,104] studies in adults have not been able to confirm its role on caries control[105] except for *root caries*.[106] The evidence supporting the use of xylitol products for the purpose of caries reduction in both adults and children is very weak.[107]

A myriad of other chemical agents have been suggested to selectively kill Gram-positive cariogenic coaggregates of bacteria. Among these are *propolis*, *nutraceutical phenols* from licorice root, *arginine* and bactericidal products such as *sodium hypochlorite 0.2%* in the form of an oral rinse.[108–110] However, despite the effect of these strategies on decreasing cariogenic bacteria counts, they have not been shown to actually decrease caries incidence in the absence of fluoride.




**Figure 2.43** CPP–ACP remineralizing compound (GC Tooth Mousse, GC Asia).

Caries is a biofilm disease modulated by diet; therefore, changes to bacterial counts will have a short-term impact on the caries process. In the absence of other changes, for example, a significant reduction in fermentable carbohydrate intake, the biofilm microbiome will continue to adapt to the acidogenic and aciduric environment caused by a highly cariogenic diet; and these antimicrobial strategies will have little impact on caries outcomes.

## VIII. Calcium and Phosphate Compounds

Amorphous calcium-phosphate (ACP) products have become commercially available and reportedly have the potential to remineralize tooth structure.[111] Remineralization products that use casein phosphopeptide (CPP) as a vehicle to deliver and maintain a supersaturation state of ACP at or near the tooth surface have recently been introduced. Some of these products contain other caries-preventive agents such as fluoride and xylitol (e.g. MI Paste Plus, GC America, Alsip, IL; Fig. 2.43). Gum, lozenges and topically applied solutions containing CPP–ACP have also been reported to remineralize white spots.[112,113]

The mechanism of action of the ACP–CPP compounds is shown in Box 2.6.

**Box 2.6 Mechanism of remineralization action of ACP–CPP compound**

Casein phosphopeptide (CPP) is a milk-derived protein that binds to the tooth's biofilm and is used to stabilize Amorphous Calcium Phosphate (ACP)

↓

ACP is a reactive and soluble calcium phosphate compound that releases calcium and phosphate ions to convert to apatite and remineralize the enamel when it comes in contact with saliva[46]

↓

Remineralization products use CPP as a vehicle and maintains a supersaturation state of ACP at or near the tooth surface

Mounting evidence indicates that CPP–ACP complexes, when used regularly, are effective in enamel remineralization;[114–117] however, most of these studies have not looked at its effect independent of fluoride.

**Clinical Notes**

The CPP–ACP products must not be prescribed to patients with allergies to dairy (milk) proteins.

## IX. Probiotics

One novel approach for reducing dental caries that has emerged in recent years is that of probiotics. The fundamental concept is to inoculate the oral cavity with bacteria that will compete with cariogenic bacteria and eventually replace them. Obviously, the probiotic bacteria must not produce significant adverse effects.

A number of commercial products have been introduced and have been demonstrated to be safe in short-term studies. However, their relative level of efficacy remains unknown. It has been speculated that for the probiotic microorganisms to gain dominance, existing pathogens must first be eliminated.

**Clinical Notes**

The concept of probiotics holds significant promise but considerably more research is required.

## X. Restorations

The status of a patient's existing restorations may have an important bearing on the outcome of preventive measures and caries treatment. Old restorations that are rough and plaque retentive should be smoothed and polished or replaced. Restoration defects such as overhangs, open proximal contacts and defective contours contribute to plaque formation and retention. These defects should be corrected, usually by replacement of the defective restoration.

Detection of caries around restorations and sealants (CARS) ('secondary caries') may be difficult around old restorations. Discolouration of the enamel adjacent to a restoration suggests the potential for CARS, but is not a definitive diagnosis. This condition appears as a localized opalescent area next to the restoration margins. (*Exception*: A bluish colour of facial or lingual enamel that directly overlies an old, otherwise acceptable amalgam restoration does not indicate replacement unless there is a need for improvement of esthetics. Such a discolouration may be caused by the amalgam itself.)

Because metallic restorations are radiopaque, the radiolucency of CARS may be masked. The restoration of cavitated active caries lesions is preventive only because it removes large numbers of cariogenic organisms and some of the sites in which they may be protected, and it restores adequate tooth contours and surface smoothness that may facilitate oral hygiene.

Once caries has produced cavitation of the tooth surface, unless the cariogenic biofilm can be removed from a cavitated caries lesion or the lesion has been treated with SDF, preventive measures will be inadequate to prevent further progression of the lesion. Excision of the lesion (tooth preparation) and proper tooth restoration will not only stop the progression of active, cavitated lesions but also restore tooth function and esthetics.

**Clinical Notes**

- Detection of secondary caries (CARS) can be difficult around old restorations.

## XI. Immunization

For many years, investigators have been trying to develop an effective anticaries vaccine. Several prototypes have been tested in animals, but at this time, neither the safety nor efficacy of such vaccines has been demonstrated in humans.[118,119] Even if an anticaries vaccine was developed, the following concerns may affect its widespread use.

- The *potential adverse effects* of a vaccine must be identified.
- Its *cost* must be compared with that of public water fluoridation, which is inexpensive and already effective at reducing caries.
- The limitations imposed by *governmental regulatory agencies* may affect the widespread use of an anticaries vaccine.

# Clinical Considerations for Caries Removal

For a complete description of current caries classification criteria based on lesion severity and activity, please see Chapter 3. Under normal operative clinical circumstances, in a vital permanent tooth with no symptoms or signs of pulpal pathology, the extent of the dentin caries removal (excavation) is dictated primarily by lesion severity and depth:

## I. Caries removal to firm dentin

Moderate lesions (lesions not reaching the inner one-third of dentin and with no anticipated risk of pulp exposure) should be excavated to a caries-free DEJ and firm dentin.

In slowly advancing lesions, it is expedient to remove the soft (infected) dentin until the readily identifiable hard (sclerotic) dentin is reached.

## II. Selective caries removal (SCR) to soft dentin

Advanced (deep) lesions (lesions reaching the inner one-third of dentin and with anticipated risk of pulp exposure) should be excavated to a caries-free DEJ and soft dentin, following a ***SCR protocol***.[120,121]

*Indications*

- The SCR protocol is a professionally recognized and accepted tooth-level caries control treatment.[121–123]
- In a tooth with a deep advanced caries lesion, no history of spontaneous pain, normal responses to thermal stimuli and a vital pulp, a deliberate, selective caries lesion removal (as noted) may be indicated. This procedure is supported by a large body of evidence.[124–131]

*Concept*

- Removal of dentin up to hard dentin in deep, advanced caries lesions does not ensure a 'sterile' dentin as bacteria have been found to be present in all dentinal layers in deep caries lesions. Nevertheless, even when bacteria are present, compounding evidence indicates that when a good seal is present the lesion will arrest.[132–135]
- SCR allows a restoration to be placed while avoiding pulpal exposure.[136]
- Avoiding a pulpal exposure has a great impact on the lifetime prognosis of the tooth and long-term treatment costs.[137] Although the residual dentin thickness cannot be accurately assessed clinically, its preservation is a significant factor in avoiding pulpal distress.[27,138]

*Clinical protocol*

- SCR consists of complete caries removal peripherally to a sound, caries-free DEJ; axially and pulpally, caries is removed to within approximately 1 mm of the pulp within soft dentin.
- A glass ionomer (e.g. Fuji IX, GC America, Alsip, IL) temporary restoration or a definitive restoration is then placed.
- Growing evidence suggests that temporization followed by re-entry does not contribute to improved clinical outcomes;[124,139] therefore, current research supports the placement of a definitive restoration.[125–129,131]

Therefore, it is not necessary to remove all of the dentin that has been compromised by the caries process. Although caries detection solutions such as 1% acid red 52 (acid rhodamine B or food red 106) in propylene glycol[34] have been developed to help stain the infected layer, these dyes bind and stain the demineralized dentin matrix and do not stain bacteria exclusively.

**Clinical Notes**

- Complete removal of all stained tooth structure in the preparation therefore ultimately leads to significantly larger preparations than the traditional visual-tactile method of evaluating for caries removal, so their use is no longer recommended.[140–142]

The following are additional clinical considerations for caries lesion removal:

I. Regardless of which protocol is used for caries removal (caries removal to firm dentin or SCR to soft dentin), the tooth may be restored with a final restoration (Figs. 2.44 and 2.45). However, the patient should be clearly informed that in the treatment of advanced (deep) dentin caries some leathery and soft dentin may remain under the restoration. This remaining caries-affected dentin has many implications, including higher risk for endodontic complications[143] (not because 'caries was left under the restoration' but because deep caries was present to begin with) and a radiographic presentation that may suggest secondary or residual caries.

II. Although there is overwhelming scientific evidence in support of the SCR protocol, this is still a controversial clinical procedure in certain countries[126] with potential liability implications as well as implications for clinical board examinations, most of which still require

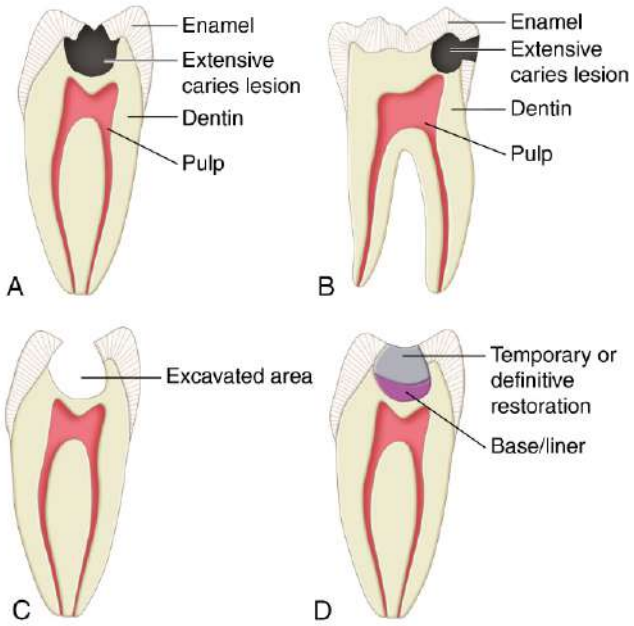



**Figure 2.44** Schematic representation of caries-control procedure. (A and B) Faciolingual (A) and mesiodistal (B) cross-sections of mandibular first molar show extensive pre-operative occlusal and proximal caries lesions. (C) Tooth after excavation of extensive caries. Note remaining unsupported enamel. (D) Temporary amalgam restoration inserted after appropriate liner or base.

complete caries removal (excavation) to hard dentin as a criteria for a successful clinical board examination.

III. The patient should be clearly informed of the risks and benefits of the SCR to soft dentin protocol and provide informed consent. If the patient is not willing to accept the risks, then the alternative, either complete caries removal with a higher risk of complications (endodontic therapy, post-operative sensitivity) or tooth extraction should be presented.

IV. A sealed tooth–restoration interface is critical for the success of the restorative procedure. When a proper seal is *not* obtained, or when/if the seal is compromised, marginal leakage and ingress of bacteria/fluids/nutrients will allow lesion progression. This is particularly critical when soft dentin remains under the restoration.

V. When excavating advanced (deep) dentin caries, if a caries-free DEJ cannot be obtained, a protective restoration (sedative filling, temporary restoration) should be used and tooth restorability should be reassessed. A permanent restoration should not be used unless a sound DEJ (or peripheral dentin margin) is reached (Figs. 2.44 and 2.45).

VI. In slowly advancing lesions, it is practical to remove the soft (infected) dentin until firm (demineralized, affected) dentin is reached. Despite being demineralized, with adequate marginal seal, the lesion may arrest and the affected dentin may remineralize. This allows for the most conservative tooth preparation without jeopardizing the restoration.

VII. The pulpal diagnoses outlined here rely on signs and symptoms of pulp pathology using the best diagnostic tools available. However, actual pulpal status is difficult to determine clinically—bacteria and toxins progressing ahead of caries may cause areas of undetectable and asymptomatic pulpal necrosis or irreversible pulpitis.

VIII. Use of calcium hydroxide (or other liner/base materials) after caries removal and before use of a protective or final restoration does not appear to improve outcomes.[144]

IX. Teeth that are restorable only by use of full cuspal coverage restoration generally are not appropriate for the SCR technique because of the difficulty of evaluating the tooth for possible failures such as continuing caries activity under the full coverage restoration. Another concern is the cost of rectifying failures.

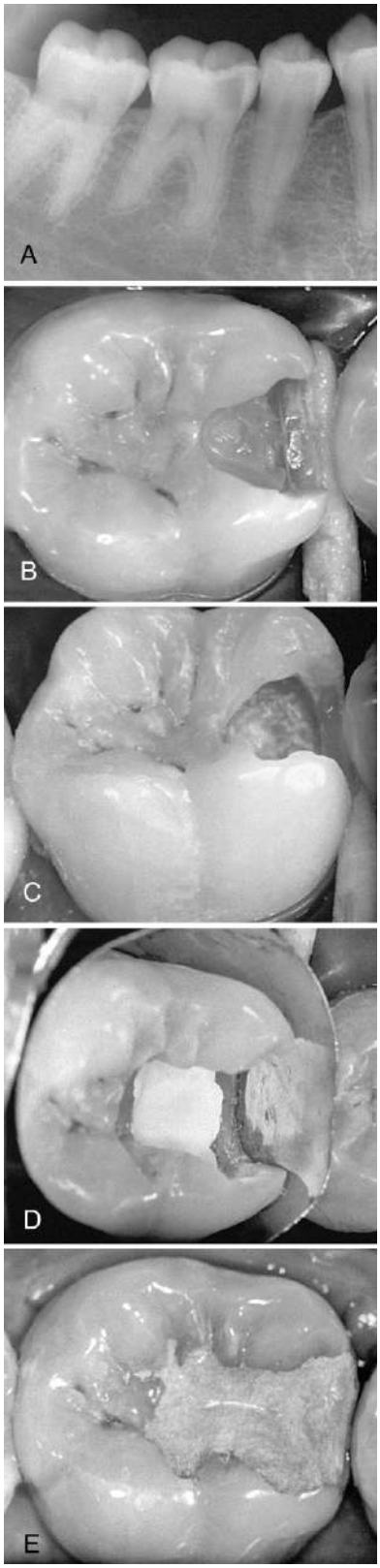



**Figure 2.45** (A) Preoperative clinical radiograph illustrating extensive caries lesion in the proximal and occlusal regions of the mandibular right first molar. Initial caries excavation of tooth. (B) Remaining caries requires further excavation. Also, note the wedge in place, protecting rubber dam and soft tissue; it has been lightly shaved by a bur. (C) Remaining unsupported enamel under mesiolingual cusp. (D) Tooth ready for placement of temporary restoration. Carious involvement required further extension than that seen in (B) and (C). Liner or base material has been applied to deepest excavated areas and matrix, appropriately wedged, has been placed. (E) Temporary restoration completed for caries-control procedure. Caries has been eliminated, the pulp adequately protected and inter-arch and intra-arch positions of tooth maintained by caries-control procedure.

**Table 2.16 Caries-control restoration**

| | |
|---|---|
| Initial treatment | Caries risk assessment<br>Education and motivation<br>Thorough evaluation and documentation of lesions<br>Temporization of all large cavitated lesions by caries-control restorations<br>Specific non-restorative, therapeutic treatment (Table 2.11)<br>Plaque control (Table 2.11, technique C)<br>Dietary control (Table 2.11, technique A) |
| Preliminary assessment | Gingival response as a marker of plaque biofilm control effectiveness<br>Arrest of cavitated lesions with SDF as indicated<br>Pulpal response of teeth with selective caries-excavation and restoration<br>Assessment of patient compliance with medications, oral hygiene and dietary control measures |
| Follow-up care | Careful clinical evaluation of teeth<br>Replacement of caries-control restorations with permanent restorations<br>Monitoring of plaque biofilm and MS levels<br>Further antimicrobial treatment and dietary reassessment as indicated by new cavitations, non-cavitated lesions, or high MS levels |

*MS*, Mutans streptococci.

X. This protocol recognizes that exposures will occur despite admonitions to leave soft and leathery (infected) dentin rather than expose the pulp. Whenever there is a pulp exposure, the treatment should be consistent with existing endodontic guidelines for vital pulp exposures.

XI. Advances in vital pulp therapy may allow successful endodontic therapy of pulp exposures without the need for root canal treatment. With the advent of *tricalcium silicates*, the prognosis of pulp capping (partial and full pulpotomy of permanent mature teeth) has tremendously improved. Recent observational and randomized trials, as well as systematic reviews and meta-analyses, have shown that these procedures have success for pulp vitality that exceeds 90%.[145,146] Complete caries removal with the option of vital pulp therapy should be one of the options offered to the patient during the informed consent process.

XII. The SCR protocol can be applied either on a single tooth being treated or to several teeth in a caries control phase treatment. High caries risk patients with multiple cavitated caries lesions can benefit from a treatment plan designed specifically to gain control over the rapid progression of their disease. The term *caries control* refers to a procedure in which multiple teeth with acute threatening caries are treated quickly by
   i. applying a medication (e.g. SDF) to arrest the caries lesions or
   ii. removing the infected tooth structure, medicating the pulp if necessary and restoring the defect(s) with a temporary or definite material.

The intent of caries-control procedures is immediate, corrective intervention of advanced caries lesions so as to prevent pulpal disease and avoid possible sequelae such as toothache, root canal therapy or more complex ultimate restorations.

This allows the caries process in these cavitated lesions to be essentially arrested and the overall bacterial load to be reduced. Caries-control procedures must be accompanied by other preventive measures (Table 2.16). Teeth rapidly treated by caries-control procedures subsequently may be treated by using routine restorative techniques if appropriate pulpal responses are obtained.

## Summary

Caries management efforts must be directed not at the tooth level (traditional or surgical treatment) but at the total-patient level (medical model of treatment). Restorative treatment does not cure the caries process. Instead, identifying and eliminating the causative factors for caries must be the primary focus, in addition to the restorative repair of damage caused by caries.

Much of the remainder of this textbook presents information on when and how to restore tooth defects. Many tooth defects are the result of caries activity. Only implementation of appropriate caries-preventive measures will lower the probability that caries lesions will recur. Tooth restorations are preventive in the sense that they remove numerous cariogenic organisms in the affected site and eliminate a protected habitat for other cariogenic bacteria; however, they primarily repair the tooth damage caused by caries and have only a limited impact on the patient's overall caries risk.

## Acknowledgements

The authors would like to thank Professor JM ('Bob') ten Cate PhD, University of Amsterdam, for his review of the Ecologic Basis of Dental Caries: The Role of the Biofilm section of this chapter.

## References


1. Keyes PH: Research in dental caries, *J Am Dent Assoc* 76(6):1357–1373, 1968.
2. Chaussain-Miller C, Fioretti F, Goldberg M, et al: The role of matrix metalloproteinases (MMPs) in human caries, *J Dent Res* 85(1):22–32, 2006.
3. Featherstone JD: The caries balance: the basis for caries management by risk assessment, *Oral Health Prev Dent* 2(Suppl 1):259–264, 2004.
4. Young DA, Kutsch VK, Whitehouse J: A clinician's guide to CAMBRA: a simple approach, *Compend Contin Educ Dent* 30(2):92–94, 2009 (96, 98, passim).
5. Marsh PD: Dental plaque as a biofilm and a microbial community—implications for health and disease, *BMC Oral Health* 6(Suppl 1):S14, 2006.
6. Welch JL, Rossetti BJ, Riekem CW, et al: Biogeography of a human oral microbiome at the micron scale, *Proc Natl Acad Sci USA* 113(6):E791–E800, 2016.
7. Ling Z, Kong J, Jia P, et al: Analysis of oral microbiota in children with dental caries by PCR-DGGE and barcoded pyrosequencing, *Microb Ecol* 60(3):677–690, 2010.
8. Hannig C, Hannig M: The oral cavity—a key system to understand substratum-dependent bioadhesion on solid surfaces in man, *Clin Oral Invest* 13(2):123–139, 2009.
9. Juhl M: Three-dimensional replicas of pit and fissure morphology in human teeth, *Scand J Dent Res* 91(2):90–95, 1983.
10. Takahashi N, Nyvad B: Ecological hypothesis of dentin and root caries, *Caries Res* 50:422–431, 2016.
11. Brown LR: Effects of selected caries preventive regimes on microbial changes following radiation-induced xerostomia in cancer patients, *Microbiol Abstr Spec Suppl* 1:275, 1976.
12. Dreizen SBL: Xerostomia and dental caries, *Microbiol Abstr Spec Suppl* 1263, 1976.
13. Mandel ID: Salivary factors in caries prediction, *Sp. Suppl. Microbiol. Abstr.* In Bibby BG, Shern RJ, editors: In *Proceedings "methods of caries prediction"*, ed 4, Arlington, VA, 1978, Information Retrieval, Inc., pp 147–162.
14. Arnold RR, Russell JE, Devine SM, et al: Antimicrobial activity of the secretory innate defense factors lactoferrin, lactoperoxidase, and lysozyme. In Guggenheim B, editor: *Cariology today*, Basel, 1984, Karger, pp 75–88.
15. Mandel I, Ellison SA: Naturally occurring defense mechanism in saliva. In Tanzer JM, editor: *Animal models in cariology (supplement to microbiology abstracts)*, Washington, DC, 1981, Information Retrieval.
16. van Houte J: Microbiological predictors of caries risk, *Adv Dent Res* 7(2):87–96, 1993.
17. Hay DI: Specific functional salivary proteins. In Guggenheim B, editor: *Cariology today*, Basel, 1984, Karger.
18. Parfitt GJ: The speed of development of the carious cavity, *Br Dent J* 100:204–207, 1956.
19. Ritter AV, Shugars DA, Bader JD: Root caries risk indicators: a systematic review of risk models, *Comm Dent Oral Epidemiol* 38(5):383–397, 2010.
20. Du M, Jiang H, Tai B, et al: Root caries patterns and risk factors of middle-aged and elderly people in China, *Comm Dent Oral Epidemiol* 37(3):260–266, 2009.
21. Saunders RH Jr, Meyerowitz C: Dental caries in older adults, *Dent Clin North Am* 49(2):293–308, 2005.
22. Berry TG, Summitt JB, Swift EJ Jr: Root caries, *Oper Dent* 29(6): 601–607, 2004.
23. Backer DO: Posteruptive changes in dental enamel, *J Dent Res* 45:503, 1966.
24. Silverstone LM: In vitro studies with special reference to the enamel surface and the enamel–resin interface. In Silverstone LM, Dogon IC, editors: In *Proceedings of an international symposium on the acid etch technique*, St Paul, MN, 1975, North Central.
25. Silverstone LM, Hicks MJ: The structure and ultrastructure of the carious lesion in human dentin, *Gerodontics* 1(4):185–193, 1985.
26. Boushell LW, Nagaola H, Yamauchi M: Increased matrix metalloproteinase-2 and bone sialoprotein response to human coronal caries, *Caries Res* 45:453–459, 2011.
27. Bjorndal L, Kidd EA: The treatment of deep dentine caries lesions, *Dent Update* 32(7):402–404, 2005 (407–410, 413).
28. Bjørndal L, Thylstrup A: A structural analysis of approximal enamel caries lesions and subjacent dentin reactions, *Eur J Oral Sci* 103(1):25–31, 1995.
29. Ekstrand KR, Ricketts DN, Kidd EA: Do occlusal carious lesions spread laterally at the enamel–dentin junction? A histolopathological study, *Clin Oral Investig* 2(1):15–20, 1998.
30. Pashley DH: Clinical correlations of dentin structure and function, *J Prosthet Dent* 66(6):777–781, 1991.
31. Ogawa K, Yamashita Y, Ichijo T, et al: The ultrastructure and hardness of the transparent layer of human carious dentin, *J Dent Res* 62(1):7–10, 1983.
32. Miyashita H, Worthington HV, Qualtrough A, et al: Pulp management for caries in adults: maintaining pulp vitality, *Cochrane Database Syst Rev* 2:CD004484, 2007.
33. Maltz M, Oliveira EF, Fontanella V, et al: Deep caries lesions after incomplete dentine caries removal: 40-month follow-up study, *Caries Res* 41(6):493–496, 2007.
34. Fusayama T: Two layers of carious dentin: diagnosis and treatment, *Oper Dent* 4(2):63–70, 1979.
35. Shay K: The evolving impact of aging America on dental practice, *J Contemp Dent Pract* 5(4):101–110, 2004.
36. Leake JL: Clinical decision-making for caries management in root surfaces, *J Dent Educ* 65(10):1147–1153, 2001.
37. Thomson WM: Dental caries experience in older people over time: what can the large cohort studies tell us? *Br Dent J* 196(2):89–92, 2004 (discussion 87).
38. Tanzer JM: Xylitol chewing gum and dental caries, *Int Dent J* 45(Suppl 1):65–76, 1995.
39. Berkey DB, Berg RG, Ettinger RL, et al: The old–old dental patient: the challenge of clinical decision-making, *J Am Dent Assoc* 127(3):321–332, 1996.
40. Surmont PA, Martens LC: Root surface caries: an update, *Clin Prev Dent* 11(3):14–20, 1989.
41. Yaacob M, Worthington HV, Deacon SA, et al: Powered versus manual toothbrushing for oral health, *Cochrane Database Syst Rev*(6), 2014.
42. Edgar WM: Saliva and dental health. Clinical implications of saliva: report of a consensus meeting, *Br Dent J* 169(3–4):96–98, 1990.
43. Hayes C: The effect of non-cariogenic sweeteners on the prevention of dental caries: a review of the evidence, *J Dent Educ* 65(10):1106–1109, 2001.
44. Haveman CW, Summitt JB, Burgess JO, et al: Three restorative materials and topical fluoride gel used in xerostomic patients: a clinical comparison, *J Am Dent Assoc* 134:177–184, 2003.
45. Beck JD, Kohout F, Hunt RJ: Identification of high caries risk adults: attitudes, social factors and diseases, *Int Dent J* 38(4):231–238, 1988.
46. Beck JD: Risk revisited, *Comm Dent Oral Epidemiol* 26(4):220–225, 1998.
47. Fontana M, Zero DT: Assessing patients' caries risk, *J Am Dent Assoc* 137(9):1231–1239, 2006.
48. Steinberg S: Understanding and managing dental caries: a medical approach, *Alpha Omegan* 100(3):127–134, 2007.
49. Steinberg S: Adding caries diagnosis to caries risk assessment: the next step in caries management by risk assessment

(CAMBRA), *Compend Contin Educ Dent* 30(8):522, 2009 (24–26, 28 passim).
50. Doméjean S, Featherstone JDB, White J: Validation of the CDA CAMBRA caries risk assessment: a six-year retrospective study, *J Calif Dent Assoc* 39:709–715, 2011.
51. Cheng J, Chaffee BW, Cheng NF, et al: Understanding treatment effect mechanisms of the CAMBRA randomized trial in reducing caries increment, *J Dent Res* 94(1):44–51, 2015.
52. Ismail AI, Pitts NB, Tellez M, et al: The International Caries Classification and Management System (ICCMS™): an example of a caries management pathway, *BMC Oral Health* 15(Suppl 1):S9, 2015.
53. Pitts NB, Ekstrand KR: ICDAS Foundation: International Caries Detection and Assessment System (ICDAS) and its International Caries Classification and Management System (ICCMS)—methods for staging of the caries process and enabling dentists to manage caries, *Comm Dent Oral Epidemiol* 41(1):e41–e52, 2013.
54. Ismail AI, Tellez M, Pitts NB, et al: Caries management pathways preserve dental tissues and promote oral health, *Comm Dent Oral Epidemiol* 41(1):e12–e40, 2013.
55. Campus G, Cagetti MG, Sale S, et al: Cariogram validity in schoolchildren: a two-year follow-up study, *Caries Res* 46:16–22, 2012.
56. Petersson GH, Isberg PE, Twetman S: Caries risk profiles in schoolchildren over 2 years assessed by Cariogram, *Int J Paediatr Dent* 20:341–346, 2010.
57. Domejean-Orliaguet S, Gansky SA, Featherstone JD: Caries risk assessment in an educational environment, *J Dent Educ* 70(12):1346–1354, 2006.
58. Hausen H: Caries prediction—state of the art, *Comm Dent Oral Epidemiol* 25(1):87–96, 1997.
59. Kidd EA: The use of diet analysis and advice in the management of dental caries in adult patients, *Oper Dent* 20(3):86–93, 1995.
60. Löe H: Human research model for the production and prevention of gingivitis, *J Dent Res* 50:256, 1971.
61. Krase B: *Caries risk*, Chicago, IL, 1985, Quintessence.
62. Klock B, Krasse B: Effect of caries-preventive measures in children with high numbers of *S. mutans* and lactobacilli, *Scand J Dent Res* 86(4):221–230, 1978.
63. Gorur A, Lyle DM, Schaudinn C, et al: Biofilm removal with a dental water jet, *Compend Contin Educ Dent* 30(Spec No. 1):1–6, 2009.
64. Brown LJ, Lazar V: The economic state of dentistry. Demand-side trends, *J Am Dent Assoc* 129(12):1685–1691, 1998.
65. HHS US Department of Health & Human Services: New assessments and actions on Fluoride. 2011. Available from: http://www.hhs.gov/news/press/2011pres/01/20110107a.html.
66. Svanberg M, Westergren G: Effect of SnF2, administered as mouthrinses or topically applied, on *Streptococcus mutans*, *Streptococcus sanguis* and lactobacilli in dental plaque and saliva, *Scand J Dent Res* 91(2):123–129, 1983.
67. Beltran-Aguilar ED, Goldstein JW, Lockwood SA: Fluoride varnishes. A review of their clinical use, cariostatic mechanism, efficacy and safety, *J Am Dent Assoc* 131(5):589–596, 2000.
68. Centers for Disease Control and Prevention: Recommendations for using fluoride to prevent and control dental caries in the United States, *MMWR Morb Mortal Wkly Rep* 50((RR-14) 1):30, 2001.
69. Newbrun E: Topical fluorides in caries prevention and management: a North American perspective, *J Dent Educ* 65(10):1078–1083, 2001.
70. Weintraub JA: Fluoride varnish for caries prevention: comparisons with other preventive agents and recommendations for a community-based protocol, *Spec Care Dent* 23(5):180–186, 2003.
71. Borutta A, Kunzel W, Rubsam F: The caries-protective efficacy of 2 fluoride varnishes in a 2-year controlled clinical trial [translation], *Dtsch Zahn Mund Kieferheilkd Zentralbl* 79(7):543–549, 1991.
72. Borutta A, Reuscher G, Hufnagl S, et al: Caries prevention with fluoride varnishes among preschool children [translation], *Gesundheitswesen* 68(11):731–734, 2006.
73. Castellano JB, Donly KJ: Potential remineralization of demineralized enamel after application of fluoride varnish, *Am J Dent* 17(6):462–464, 2004.
74. Haugejorden O, Nord A: Caries incidence after topical application of varnishes containing different concentrations of sodium fluoride: 3-year results, *Scand J Dent Res* 99(4):295–300, 1991.
75. Seppa L: Effects of a sodium fluoride solution and a varnish with different fluoride concentrations on enamel remineralization in vitro, *Scand J Dent Res* 96(4):304–309, 1988.
76. Alaluusua S, Kleemola-Kujala E, Gronroos L, et al: Salivary caries-related tests as predictors of future caries increment in teenagers. A three-year longitudinal study, *Oral Microbiol Immunol* 5(2):77–81, 1990.
77. Yamaga R, Nishino M, Yoshida S, et al: Diammine silver fluoride and its clinical application, *J Osaka Univ Dent Sch* 12:1–20, 1972.
78. Horst JA, Ellenikiotis H, Milgrom PL: UCSF protocol for caries arrest using silver diamine fluoride: rationale, indications and consent, *J Calif Dent Assoc* 44(1):16–28, 2016.
79. Mei ML, Ito L, Cao Y, et al: Inhibitory effect of silver diamine fluoride on dentine demineralisation and collagen degradation, *J Dent* 41(9):809–817, 2013.
80. Mei ML, Li QL, Chu CH, et al: Antibacterial effects of silver diamine fluoride on multi-species cariogenic biofilm on caries, *Ann Clin Microbiol Antimicrob* 26(12):4, 2013.
81. Markowitz K, Pashley DH: Discovering new treatments for sensitive teeth: the long path from biology to therapy, *J Oral Rehabil* 35(4):300–315, 2008.
82. Rosenblatt A, Stamford TC, Niederman R: Silver diamine fluoride: a caries "silver-fluoride bullet", *J Dent Res* 88(2):116–125, 2009.
83. Tan HP, Lo ECM, Dyson JE, et al: Randomized clinical trial on arresting dental root caries through silver diamine fluoride applications in community-dwelling elders, *Br Dent J* 221(7):409, 2016.
84. Yee R, Holmgren C, Mulder J, et al: Efficacy of silver diamine fluoride for arresting caries treatment, *J Dent Res* 88(7):644–647, 2009.
85. Hutter JW, Chan JT, Featherstone JD, et al: Professionally applied topical fluoride. Executive summary of evidence-based clinical recommendations, *J Am Dent Assoc* 137(8):1151–1159, 2006.
86. Simonsen RJ: Cost effectiveness of pit and fissure sealant at 10 years, *Quintessence Int* 20(2):75–82, 1989.
87. Tinanoff N, Douglass JM: Clinical decision making for caries management in children, *Pediatr Dent* 24(5):386–392, 2002.
88. Ahovuo-Saloranta A, Hiiri A, Nordblad A, et al: Pit and fissure sealants for preventing dental decay in the permanent teeth of children and adolescents, *Cochrane Database Syst Rev* 3, 2004, CD001830.
89. Beauchamp J, Caufield PW, Crall JJ, et al: Evidence-based clinical recommendations for the use of pit-and-fissure sealants: a report of the American Dental Association Council on Scientific Affairs, *J Am Dent Assoc* 139(3):257–268, 2008.
90. Bader JD, Shugars DA, Bonito AJ: Systematic reviews of selected dental caries diagnostic and management methods, *J Dent Educ* 65(10):960–968, 2001.
91. Simonsen RJ: Pit and fissure sealant: review of the literature, *Pediatr Dent* 24(5):393–414, 2002.
92. Wright JT, Tampi MP, Graham L, et al: Sealants for preventing and arresting pit-and-fissure occlusal caries in primary and permanent molars. A systematic review of randomized

controlled trials—a report of the American Dental Association and the American Academy of Pediatric Dentistry, *J Am Dent Assoc* 147(8):631–645, 2016 (e18).

93. Paris S, Meyer-Lueckel H: Masking of labial enamel white spot lesions by resin infiltration—a clinical report, *Quintessence Int* 40(9):713–718, 2009.
94. Paris S, Meyer-Lueckel H: Inhibition of caries progression by resin infiltration in situ, *Caries Res* 44(1):47–54, 2010.
95. Meyer-Lueckel H, Balbach A, Schikowsky C, et al: Pragmatic RCT on the efficacy of proximal caries infiltration, *J Dent Res* 95(5), 2016.
96. Sreebny LM, Schwartz SS: A reference guide to drugs and dry mouth—2nd edition, *Gerodontology* 14(1):33–47, 1997.
97. Swarn A, Ritter AV, Donovan T, et al: Caries risk evaluation: correlation between chair-side, laboratory and clinical tests, *J Dent Res* 89, 2010 (Special Issue B, USB of abstracts #4272).
98. Barkvoll P, Rolla G, Svendsen K: Interaction between chlorhexidine digluconate and sodium lauryl sulfate in vivo, *J Clin Periodontol* 16(9):593–595, 1989.
99. Emilson CG: Potential efficacy of chlorhexidine against mutans streptococci and human dental caries, *J Dent Res* 73(3):682–691, 1994.
100. Slot DE, Vaandrager NC, Van Loveren C, et al: The effect of chlorhexidine varnish on root caries: a systematic review, *Caries Res* 45(2):162–173, 2011.
101. Ashley P: Effectiveness of chlorhexidine varnish for preventing caries uncertain, *Evid Based Dent* 11(4):108, 2010.
102. Deshpande A, Jadad AR: The impact of polyol-containing chewing gums on dental caries: a systematic review of original randomized controlled trials and observational studies, *J Am Dent Assoc* 139(12):1602–1614, 2008.
103. Trahan L: Xylitol: a review of its action on mutans streptococci and dental plaque—its clinical significance, *Int Dent J* 45(Suppl 1):77–92, 1995.
104. Campus G, Cagetti MG, Sale S: Six months of high-dose xylitol in high-risk caries subjects—a 2-year randomised, clinical trial, *Clin Oral Invest* 17:785–791, 2013.
105. Bader JD, Vollmer WM, Shugars DA, et al: Results from the xylitol for adult caries trial (X-ACT), *J Am Dent Assoc* 144(1):21–30, 2013.
106. Ritter AV, Bader JD, Leo MC, et al: Tooth-surface-specific effects of xylitol: randomized trial results, *J Dent Res* 92(6):512–517, 2013.
107. Riley P, Moore D, Ahmed F, et al: Xylitol-containing products for preventing dental caries in children and adults, *Cochrane Database Syst Rev* 26(3), 2015.
108. Bueno-Silva B, Koo H, Falsetta ML, et al: Effect of neovestitol–vestitol containing Brazilian red propolis on accumulation of biofilm in vitro and development of dental caries in vivo, *J Bioadhesion Biofilm Res* 29:1233–1242, 2013.
109. Ástvaldsdóttir Á, Naimi-Akbar A, Davidson T, et al: Arginine and caries prevention: a systematic review, *Caries Res* 50(4):383–393, 2016.
110. Bader JD, Shugars DA, Bonito AJ: A systematic review of selected caries prevention and management methods, *Comm Dent Oral Epidemiol* 29:399–411, 2001.
111. Tung MS, Eichmiller FC: Amorphous calcium phosphates for tooth mineralization, *Compend Contin Educ Dent* 25(9 Suppl 1):9–13, 2004.
112. Cai F, Shen P, Morgan MV, et al: Remineralization of enamel subsurface lesions in situ by sugar-free lozenges containing casein phosphopeptide–amorphous calcium phosphate, *Aust Dent J* 48(4):240–243, 2003.
113. Morgan MV, Adams GG, Bailey DL, et al: The anticariogenic effect of sugar-free gum containing CPP–ACP nanocomplexes on approximal caries determined using digital bitewing radiography, *Caries Res* 42(3):171–184, 2008.
114. Reynolds EC: Remineralization of enamel subsurface lesions by casein phosphopeptide-stabilized calcium phosphate solutions, *J Dent Res* 76(9):1587–1595, 1997.
115. Reynolds EC, Cai F, Shen P, et al: Retention in plaque and remineralization of enamel lesions by various forms of calcium in a mouthrinse or sugar-free chewing gum, *J Dent Res* 82(3):206–211, 2003.
116. Yengopal V, Mickenautsch S: Caries preventive effect of casein phosphopeptide-amorphous calcium phosphate (CPP–ACP): a meta-analysis, *Acta Odontol Scand* 67(6):1–12, 2009.
117. Llena C, Forner L, Baca P: Anticariogenicity of casein phosphopeptide-amorphous calcium phosphate: a review of the literature, *J Contemp Dent Pract* 10(3):1–9, 2009.
118. Lehner T, Challacombe SJ, Caldwell J: An immunological investigation into the prevention of caries in deciduous teeth of rhesus monkeys, *Arch Oral Biol* 20(5–6):305–310, 1975.
119. Taubman MA, Smith DJ: Effects of local immunization with glucosyltransferase fractions from *Streptococcus mutans* on dental caries in rats and hamsters, *J Immunol* 118(2):710–720, 1977.
120. Innes NP, Frencken JE, Bjorndal L, et al: Managing carious lesions: consensus recommendations on terminology, *Adv Dent Res* 28(2):49–57, 2016.
121. Schwendicke F, Frencken JE, Bjorndal L, et al: Managing carious lesions: consensus recommendations on carious tissue removal, *Adv Dent Res* 28(2):58–67, 2016.
122. Bjorndal L: Buonocore Memorial Lecture. Dentin caries: progression and clinical management, *Oper Dent* 27(3):211–217, 2002.
123. Bjorndal L: Indirect pulp therapy and stepwise excavation, *Pediatr Dent* 30(3):225–229, 2008.
124. Ricketts DN, Pitts NB: Novel operative treatment options, *Monogr Oral Sci* 21:174–187, 2009.
125. Thompson V, Craig RG, Curro FA, et al: Treatment of deep carious lesions by complete excavation or partial removal: a critical review, *J Am Dent Assoc* 139(6):705–712, 2008.
126. Oen KT, Thompson VP, Vena D, et al: Attitudes and expectations of treating deep caries: a PEARL network survey, *Gen Dent* 55(3):197–203, 2007.
127. Pinheiro SL, Simionato MR, Imparato JC, et al: Antibacterial activity of glass–ionomer cement containing antibiotics on caries lesion microorganisms, *Am J Dent* 18(4):261–266, 2005.
128. Ricketts D: Management of the deep carious lesion and the vital pulp dentine complex, *Br Dent J* 191(11):606–610, 2001.
129. Ricketts DN, Kidd EA, Innes N, et al: Complete or ultraconservative removal of decayed tissue in unfilled teeth, *Cochrane Database Syst Rev* 3, 2006, CD003808.
130. Uribe S: Partial caries removal in symptomless teeth reduces the risk of pulp exposure, *Evid Based Dent* 7(4):94, 2006.
131. van Amerongen WE: Dental caries under glass ionomer restorations, *J Public Health Dent* 56(3 Spec No):150–154, 1996 (discussion 161–163).
132. Bjorndal L, Larsen T: Changes in the cultivable flora in deep carious lesions following a stepwise excavation procedure, *Caries Res* 34(6):502–508, 2000.
133. Maltz M, Garcia R, Jardim JJ, et al: Randomized trial of partial vs. stepwise caries removal: 3-year follow-up, *J Dent Res* 91(11): 1026–1031, 2012.
134. Alves LS, Fontanella V, Damo AC, et al: Qualitative and quantitative radiographic assessment of sealed carious dentin: a 10-year prospective study, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 109(1):135–141, 2010.
135. Lima KC, Coelho LT, Pinheiro IVA, et al: Microbiota of dentinal caries as assessed by reverse-capture checkerboard analysis, *Caries Res* 45:21–30, 2011.
136. Hayashi M, Fujitani M, Yamaki C, et al: Ways of enhancing pulp preservation by stepwise excavation—a systematic review, *J Dent* 39(2):95–107, 2011.

137. Banerjee A, Watson TF, Kidd EA: Dentine caries: take it or leave it? *Dent Update* 27(6):272–276, 2000.
138. Bjorndal L: Indirect pulp therapy and stepwise excavation, *J Endod* 34(7 Suppl):S29–S33, 2008.
139. Bjorndal L, Reit C, Bruun G, et al: Treatment of deep caries lesions in adults: randomized clinical trials comparing stepwise vs. direct complete excavation, and direct pulp capping vs. partial pulpotomy, *Eur J Oral Sci* 118(3):290–297, 2010.
140. Banerjee A, Kidd EA, Watson TF: In vitro validation of carious dentin removed using different excavation criteria, *Am J Dent* 16(4):228–230, 2003.
141. Hamama HH, Yiu CK, Burrow MF, et al: Chemical, morphological and microhardness changes of dentine after chemomechanical caries removal, *Aust Dent J* 58(3):283–292, 2013.
142. Hamama HH, Yiu CK, Burrow MF, et al: Systematic review and meta-analysis of randomized clinical trials on chemomechanical caries removal, *Oper Dent* 40(4):E167–E178, 2015.
143. Schwendicke F, Meyer-Lückel H, Dorfer C, et al: Failure of incompletely excavated teeth: a systematic review, *J Dent* 41(7):569–580, 2013.
144. Corralo DJ, Maltz M: Clinical and ultrastructural effects of different liners/restorative materials on deep carious dentin: a randomized clinical trial, *Caries Res* 47(3):243–250, 2013.
145. Alqaderi H, Lee CT, Borzangy S, et al: Coronal pulpotomy for cariously exposed permanent posterior teeth with closed apices: a systematic review and meta-analysis, *J Dent* 44:1–7, 2016.
146. Hoefler V, Nagaoka H, Miller CS: Long-term survival and vitality outcomes of permanent teeth following deep caries treatment with step-wise and partial-caries-removal: a systematic review, *J Dent* 54(11):25–32, 2016.
147. Winston AE, Bhaskar SN: Caries prevention in the 21st century, *J Am Dent Assoc* 129(11):1579–1587, 1998.
148. Burgess JO, Gallo JR: Treating root-surface caries, *Dent Clin North Am* 46(2):385–404, 2002 (vii–viii).
149. Haveman CW, Redding SW: Dental management and treatment of xerostomic patients, *Tex Dent J* 115(6):43–56, 1998.

## Futher Reading

150. Baum LJ: Dentinal pulp conditions in relation to caries lesions, *Int Dent J* 20:309–337, 1970.
151. Brannstrom M, Lind PO: Pulpal response to early dental caries, *J Dent Res* 44(5):1045–1050, 1965.
152. Hilton TJ: Keys to clinical success with pulp capping: a review of the literature, *Oper Dent* 34(5):615–625, 2009.
153. Kuboki Y, Liu CF, Fusayama T: Mechanism of differential staining in carious dentin, *J Dent Res* 62(6):713–714, 1983.
154. Riley JL 3rd, Gordan VV, Rindal DB, et al: General practitioners' use of caries-preventive agents in adult patients versus pediatric patients: findings from the dental practice-based research network, *J Am Dent Assoc* 141(6):679–687, 2010.
155. Riley JL 3rd, Gordan VV, Rindal DB, et al: Preferences for caries prevention agents in adult patients: findings from the dental practice-based research network, *Comm Dent Oral Epidemiol* 38(4):360–370, 2010.
156. American Dental Association: *Sc: key dental facts*, Chicago, IL, 2004, American Dental Association.
157. CDC: Populations receiving optimally fluoridated public drinking water—United States, 1992–2006, *MMWR Morb Mortal Wkly Rep* 57(27):737–741, 2008.
158. Chow LC, Takagi S, Vogel GL: Amorphous calcium phosphate: the contention of bone, *J Dent Res* 77(1):6, 1998 (author reply 7).
159. Swift EJ: Treatment options for the exposed vital pulp, *Pract Periodontics Aesthet Dent* 11:735–739, 1999.
160. Bjorndal L, Larsen T, Thylstrup A: A clinical and microbiological study of deep carious lesions during stepwise excavation using long treatment intervals, *Caries Res* 31(6):411–417, 1997.
161. Bjorndal L, Thylstrup A: A practice-based study on stepwise excavation of deep carious lesions in permanent teeth: a 1-year follow-up study, *Comm Dent Oral Epidemiol* 26(2):122–128, 1998.
162. Foley J, Evans D, Blackwell A: Partial caries removal and cariostatic materials in carious primary molar teeth: a randomised controlled clinical trial, *Br Dent J* 197(11):697–701, 2004 (discussion 689).
163. Innes NP, Evans DJ, Stirrups DR: The Hall Technique: a randomized controlled clinical trial of a novel method of managing carious primary molars in general dental practice: acceptability of the technique and outcomes at 23 months, *BMC Oral Health* 7:18, 2007.
164. Kidd EA: How "clean" must a cavity be before restoration? *Caries Res* 38(3):305–313, 2004.
165. Kidd EA, Fejerskov O: What constitutes dental caries? Histopathology of carious enamel and dentin related to the action of cariogenic biofilms, *J Dent Res* 83(Spec No C):C35–C38, 2004.
166. Leksell E, Ridell K, Cvek M, et al: Pulp exposure after stepwise versus direct complete excavation of deep carious lesions in young posterior permanent teeth, *Endod Dent Traumatol* 12(4):192–196, 1996.
167. Maltz M, de Oliveira EF, Fontanella V, et al: A clinical, microbiologic, and radiographic study of deep caries lesions after incomplete caries removal, *Quintessence Int* 33(2):151–159, 2002.
168. Mertz-Fairhurst EJ, Adair SM, Sams DR, et al: Cariostatic and ultraconservative sealed restorations: nine-year results among children and adults, *ASDC J Dent Child* 62(2):97–107, 1995.
169. Mertz-Fairhurst EJ, Curtis JW Jr, Ergle JW, et al: Ultraconservative and cariostatic sealed restorations: results at year 10, *J Am Dent Assoc* 129(1):55–66, 1998.
170. Mertz-Fairhurst EJ, Schuster GS, Fairhurst CW: Arresting caries by sealants: results of a clinical study, *J Am Dent Assoc* 112(2):194–197, 1986.
171. Mertz-Fairhurst EJ, Schuster GS, Williams JE, et al: Clinical progress of sealed and unsealed caries, Part II: standardized radiographs and clinical observations, *J Prosthet Dent* 42(6):633–637, 1979.
172. Mertz-Fairhurst EJ, Schuster GS, Williams JE, et al: Clinical progress of sealed and unsealed caries, Part I: depth changes and bacterial counts, *J Prosthet Dent* 42(5):521–526, 1979.
173. Mjor IA: Pulp–dentin biology in restorative dentistry. Part 7: the exposed pulp, *Quintessence Int* 33(2):113–135, 2002.
174. Takahashi N, Nyvad B: The role of bacteria in the caries process: ecological perspectives, *J Dent Res* 90(3):294–303, 2011.
175. Ferjeskov O, Nyvad B, Kidd E, editors: *Dental Caries: the disease and its clinical management*, ed 3, Oxford, 2015, Wiley-Blackwell.
176. Simón-Soro A, Tomás I, Cabrera-Rubio R, et al: Microbial geography of the oral

cavity, *J Dent Res* 92(7):616–621, 2013.

177. Hannig C, Hannig M, Attin T: Enzymes in the acquired enamel pellicle, *Eur J Oral Sci* 113(1):2–13, 2005.
178. Ekstrand KR, Martignon S, Ricketts DJN, et al: Detection and activity assessment of primary coronal caries lesions: a methodologic study, *Oper Dent* 32(3):225–235, 2007.
179. Nyvad B, Machiulskiene V, Baelum V: Reliability of a new caries diagnostic system differentiating between active and inactive caries lesions, *Caries Res* 33(4):252–260, 1999.
180. Zanini M, Hennequin M, Cousson PY: A review of criteria for the evaluation of pulpotomy outcomes in mature permanent teeth, *J Endod* 42(8):1167–1174, 2016.
181. Xiao J, Klein MI, Falsetta ML, et al: The exopolysaccharide matrix modulates the interaction between 3D architecture and virulence of a mixed-species oral biofilm, *PLoS Pathog* 8(4):e1002623, 2012.
182. Koo H, Falsetta ML, Klein MI: The exopolysaccharide matrix: a virulence determinant of cariogenic biofilm, *J Dent Res* 92(12):1065–1073, 2013.

Page left intentionally blank